

Sensation and Perception

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Sensation and Perception

Adam John Privitera

The topics of sensation and perception are among the oldest and most important in all of psychology. People are equipped with senses such as sight, hearing and taste that help us to take in the world around us. Amazingly, our senses have the ability to convert real-world information into electrical information that can be processed by the brain. The way we interpret this information-- our perceptions-- is what leads to our experiences of the world. In this module, you will learn about the biological processes of sensation and how these can be combined to create perceptions.

Learning Objectives

- Differentiate the processes of sensation and perception.
- Explain the basic principles of sensation and perception.
- Describe the function of each of our senses.
- Outline the anatomy of the sense organs and their projections to the nervous system.
- Apply knowledge of sensation and perception to real world examples.
- Explain the consequences of multimodal perception.

Introduction

"Once I was hiking at Cape Lookout State Park in Tillamook, Oregon. After passing through a vibrantly colored, pleasantly scented, temperate rainforest, I arrived at a cliff overlooking the Pacific Ocean."

I grabbed the cold metal railing near the edge and looked out at the sea. Below me, I could see a pod of sea lions swimming in the deep blue water. All around me I could smell the salt from the sea and the scent of wet, fallen leaves."

This description of a single memory highlights the way a person's senses are so important to our experience of the world around us.

Before discussing each of our extraordinary senses individually, it is necessary to cover some basic concepts that apply to all of them. It is probably best to start with one very important distinction that can often be confusing: the difference between sensation and perception. The *physical* process during which our sensory organs—those involved with hearing and taste, for example—respond to external stimuli is called **sensation**. Sensation happens when you eat noodles or feel the wind on your face or hear a car horn honking in the distance. During sensation, our sense organs are engaging in **transduction**, the conversion of one form of energy into another. Physical energy such as light or a sound wave is converted into a form of energy the brain can understand: electrical stimulation. After our brain receives the electrical signals, we make sense of all this stimulation and begin to appreciate the complex world around us. This *psychological* process—making sense of the stimuli—is called **perception**. It is during this process that you are able to *identify* a gas leak in your home or a song that reminds you of a specific afternoon spent with friends.



Our senses combine to create our perceptions of the world.
[Image: Adam John Privitera, CC BY-NC-SA 4.0, <https://goo.gl/H2QaA8>]

Regardless of whether we are talking about sight or taste or any of the individual senses, there are a number of basic principles that influence the way our sense organs work. The first of these influences is our ability to detect an external stimulus. Each sense organ—our eyes or tongue, for instance—requires a minimal amount of stimulation in order to detect a stimulus. This **absolute threshold** explains why you don't smell the perfume someone is wearing in a classroom unless they are somewhat close to you. Because absolute threshold changes throughout the day and based on what other stimuli you have recently experienced, researchers define absolute threshold as the minimum amount of stimulation needed to detect a stimulus 50% of the time.

The way we measure absolute thresholds is by using a method called **signal detection**. This process involves presenting stimuli of varying intensities to a research participant in order to determine the level at which he or she can reliably detect stimulation in a given sense. During one type of hearing test, for example, a person listens to increasingly louder tones (starting from silence). This type of test is called *the method of limits*, and it is an effort to determine the point, or threshold, at which a person begins to hear a stimulus (see Additional Resources for a video demonstration). In the example of louder tones, the method of limits test is using *ascending trials*. Some method of limits tests use *descending trials*, such as making a light grow dimmer until a person can no longer see it. Correctly indicating that a sound was heard is called a hit; failing to do so is called a miss. Additionally, indicating that a sound was heard when one wasn't played is called a *false alarm*, and correctly identifying when a sound wasn't played is a *correct rejection*.

Through these and other studies, we have been able to gain an understanding of just how remarkable our senses are. For example, the human eye is capable of detecting candlelight from 30 miles away in the dark. We are also capable of hearing the ticking of a watch in a quiet environment from 20 feet away. If you think that's amazing, I encourage you to read more about the extreme sensory capabilities of nonhuman animals; many animals possess what we would consider super-human abilities.

A similar principle to the absolute threshold discussed above underlies our ability to detect the difference between two stimuli of different intensities. The **differential threshold** (or difference threshold) or **just noticeable difference (JND)**, for each sense has been studied using similar methods to signal detection. To illustrate, find a friend and a few objects of known weight (you'll need objects that weigh 1, 2, 10 and 11 lbs.—or in metric terms: 1, 2, 5 and 5.5 kg). Have your friend hold the lightest object (1 lb. or 1 kg). Then, replace this object with the next heaviest and ask him or her to tell you which one weighs more. Reliably, your friend will say the second object every single time. It's extremely easy to tell the difference when something weighs double what another weighs! However, it is not so easy when the difference is a smaller percentage of the overall weight. It will be much harder for your friend to reliably tell the difference between 10 and 11 lbs. (or 5 versus 5.5 kg) than it is for 1 and 2 lbs. This is phenomenon is called **Weber's Law**, and it is the idea that bigger stimuli require larger differences to be noticed. As with the absolute threshold, your ability to notice differences varies throughout the day and based on what other stimuli you have recently experienced so the difference threshold is defined as the smallest difference detectable 50% of the time.

Crossing into the world of perception, it is clear that our experience influences how our brain processes things. You have tasted food that you like and food that you don't like. There are some bands you enjoy and others you can't stand. However, during the time you first eat

something or hear a band, you process those stimuli using bottom-up processing. This is when we build up to perception from the individual pieces. Sometimes, though, stimuli we've experienced in our past will influence how we process new ones. This is called top-down processing. The best way to illustrate these two concepts is with our ability to read. Read the following quote out loud:



Figure 1. An example of stimuli processing.

Notice anything odd while you were reading the text in the triangle? Did you notice the second “the”? If not, it’s likely because you were reading this from a top-down approach. Having a second “the” doesn’t make sense. We know this. Our brain knows this and doesn’t *expect* there to be a second one, so we have a tendency to skip right over it. In other words, your past experience has changed the way you perceive the writing in the triangle! A beginning reader—one who is using a bottom-up approach by carefully attending to each piece—would be less likely to make this error.

Finally, it should be noted that when we experience a sensory stimulus that doesn’t change, we stop paying attention to it. This is why we don’t feel the weight of our clothing, hear the hum of a projector in a lecture hall, or see all the tiny scratches on the lenses of our glasses. When a stimulus is constant and unchanging, we experience sensory adaptation. This occurs because if a stimulus does not change, our receptors quit responding to it. A great example of this occurs when we leave the radio on in our car after we park it at home for the night. When we listen to the radio on the way home from work the volume seems reasonable. However, the next morning when we start the car, we might be startled by how loud the radio is. We don’t remember it being that loud last night. What happened? We adapted to the constant stimulus (the radio volume) over the course of the previous day and increased the

volume at various times.

Now that we have introduced some basic sensory principles, let us take on each one of our fascinating senses individually.

Vision

How vision works

Vision is a tricky matter. When we see a pizza, a feather, or a hammer, we are actually seeing light bounce off that object and into our eye. Light enters the eye through the pupil, a tiny opening behind the cornea. The pupil regulates the amount of light entering the eye by contracting (getting smaller) in bright light and dilating (getting larger) in dimmer light. Once past the pupil, light passes through the lens, which focuses an image on a thin layer of cells in the back of the eye, called the retina.

Because we have two eyes in different locations, the image focused on each retina is from a slightly different angle (binocular disparity), providing us with our perception of 3D space (binocular vision). You can appreciate this by holding a pen in your hand, extending your arm in front of your face, and looking at the pen while closing each eye in turn. Pay attention to

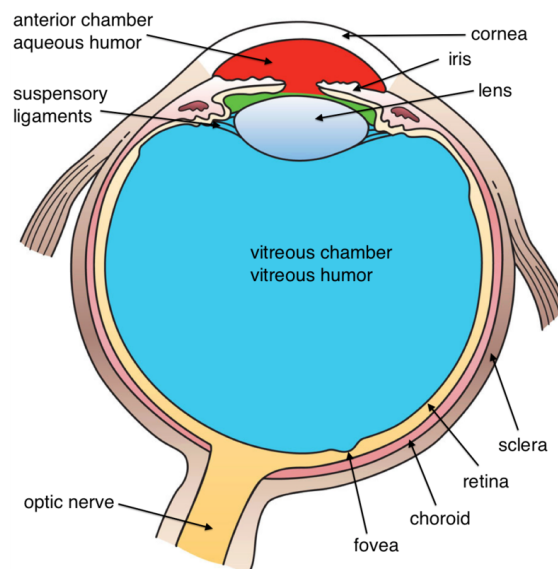


Figure 2. Diagram of the human eye. Notice the Retina, labeled here: this is the location of the Cones and Rods in the eye. [Image: Holly Fischer, <https://goo.gl/ozuG0Q>, CC BY 3.0, <https://goo.gl/TSlslq>]

the apparent position of the pen relative to objects in the background. Depending on which eye is open, the pen appears to jump back and forth! This is how video game manufacturers create the perception of 3D without special glasses; two slightly different images are presented on top of one another.

It is in the retina that light is transduced, or converted into electrical signals, by specialized cells called photoreceptors. The retina contains two main kinds of photoreceptors: rods and cones. Rods are primarily responsible for our ability to see in dim light conditions, such as during the night. Cones, on the other hand, provide us with the ability to see color and fine detail when the light is brighter. Rods and cones differ in their distribution across the retina, with the highest concentration of cones found in the fovea (the central region of focus), and rods dominating the periphery (see Figure 2). The difference in distribution can explain why looking directly at a dim star in the sky makes it seem to disappear; there aren't enough rods to process the dim light!

Next, the electrical signal is sent through a layer of cells in the retina, eventually traveling down the optic nerve. After passing through the thalamus, this signal makes it to the primary visual cortex, where information about light orientation and movement begin to come together (Hubel & Wiesel, 1962). Information is then sent to a variety of different areas of the cortex for more complex processing. Some of these cortical regions are fairly specialized—for example, for processing faces (fusiform face area) and body parts (extrastriate body area). Damage to these areas of the cortex can potentially result in a specific kind of agnosia, whereby a person loses the ability to perceive visual stimuli. A great example of this is illustrated in the writing of famous neurologist Dr. Oliver Sacks; he experienced *prosopagnosia*, the inability to recognize faces. These specialized regions for visual recognition comprise the ventral pathway (also called the “what” pathway). Other areas involved in processing location and movement make up the dorsal pathway (also called the “where” pathway). Together, these pathways process a large amount of information about visual stimuli (Goodale & Milner, 1992). Phenomena we often refer to as optical illusions provide misleading information to these “higher” areas of visual processing (see Additional Resources for websites containing amazing optical illusions).

Dark and light adaptation

Humans have the ability to adapt to changes in light conditions. As mentioned before, rods are primarily involved in our ability to see in dim light. They are the photoreceptors responsible for allowing us to see in a dark room. You might notice that this night vision ability takes around 10 minutes to turn on, a process called dark adaptation. This is because our rods

become bleached in normal light conditions and require time to recover. We experience the opposite effect when we leave a dark movie theatre and head out into the afternoon sun. During **light adaptation**, a large number of rods and cones are bleached at once, causing us to be blinded for a few seconds. Light adaptation happens almost instantly compared with dark adaptation. Interestingly, some people think pirates wore a patch over one eye in order to keep it adapted to the dark while the other was adapted to the light. If you want to turn on a light without losing your night vision, don't worry about wearing an eye patch, just use a red light; this wavelength doesn't bleach your rods.

Color vision



Figure 3. Stare at the center of the Canadian flag for fifteen seconds. Then, shift your eyes away to a white wall or blank piece of paper. You should see an "after image" in a different color scheme.

Our cones allow us to see details in normal light conditions, as well as color. We have cones that respond *preferentially*, not exclusively, for red, green and blue (Svaetichin, 1955). This **trichromatic theory** is not new; it dates back to the early 19th century (Young, 1802; Von Helmholtz, 1867). This theory, however, does not explain the odd effect that occurs when we look at a white wall after staring at a picture for around 30 seconds. Try this: stare at the image of

the flag in Figure 3 for 30 seconds and then immediately look at a sheet of white paper or a wall. According to the trichromatic theory of color vision, you should see white when you do that. Is that what you experienced? As you can see, the trichromatic theory doesn't explain the *afterimage* you just witnessed. This is where the **opponent-process theory** comes in (Hering, 1920). This theory states that our cones send information to *retinal ganglion cells* that respond to *pairs* of colors (red-green, blue-yellow, black-white). These specialized cells take information from the cones and compute the difference between the two colors—a process that explains why we cannot see reddish-green or bluish-yellow, as well as why we see afterimages. Color deficient vision can result from issues with the cones or retinal ganglion cells involved in color vision.

Hearing (Audition)

Some of the most well-known celebrities and top earners in the world are musicians. Our worship of musicians may seem silly when you consider that all they are doing is vibrating

the air a certain way to create sound waves, the physical stimulus for audition.

People are capable of getting a large amount of information from the basic qualities of sound waves. The *amplitude* (or intensity) of a sound wave codes for the loudness of a stimulus; higher amplitude sound waves result in louder sounds. The *pitch* of a stimulus is coded in the *frequency* of a sound wave; higher frequency sounds are higher pitched. We can also gauge the quality, or *timbre*, of a sound by the complexity of the sound wave. This allows us to tell the difference between bright and dull sounds as well as natural and synthesized instruments (Välimäki & Takala, 1996).

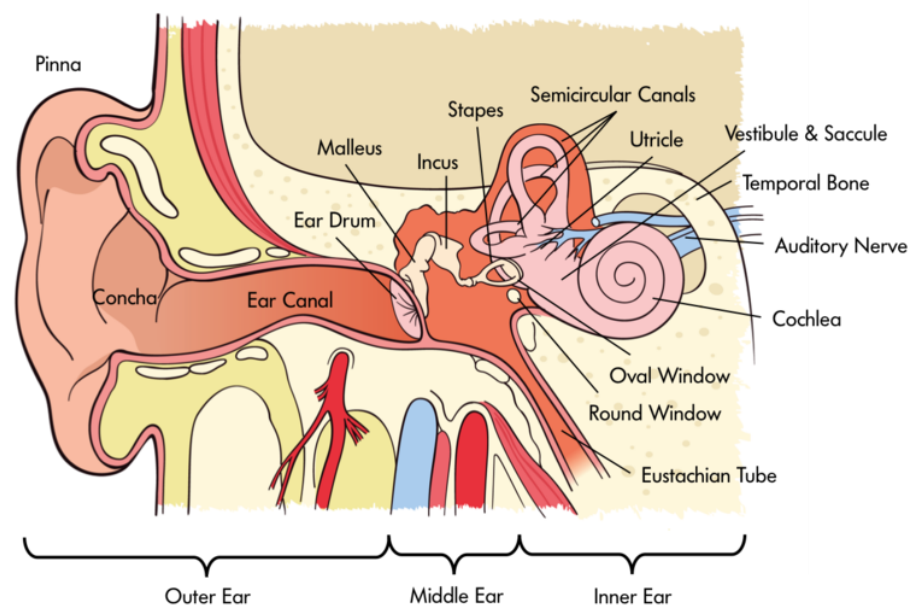


Figure 4. Diagram of the human ear. Notice the Cochlea labeled here: it is the location of the auditory Hair Cells that are tonotopically organized.

In order for us to sense sound waves from our environment they must reach our inner ear. Lucky for us, we have evolved tools that allow those waves to be funneled and amplified during this journey. Initially, sound waves are funneled by your pinna (the external part of your ear that you can actually see) into your auditory canal (the hole you stick Q-tips into despite the box advising against it). During their journey, sound waves eventually reach a thin, stretched membrane called the tympanic membrane (eardrum), which vibrates against the three smallest bones in the body—the malleus (hammer), the incus (anvil), and the stapes (stirrup)—collectively called the ossicles. Both the tympanic membrane and the ossicles amplify the sound waves before they enter the fluid-filled cochlea, a snail-shell-like bone structure containing auditory hair cells arranged on the basilar membrane (see Figure 4)

according to the frequency they respond to (called tonotopic organization). Depending on age, humans can normally detect sounds between 20 Hz and 20 kHz. It is inside the cochlea that sound waves are converted into an electrical message.

Because we have an ear on each side of our head, we are capable of localizing sound in 3D space pretty well (in the same way that having two eyes produces 3D vision). Have you ever dropped something on the floor without seeing where it went? Did you notice that you were somewhat capable of locating this object based on the sound it made when it hit the ground? We can reliably locate something based on which ear receives the sound first. What about the height of a sound? If both ears receive a sound at the same time, how are we capable of localizing sound vertically? Research in cats (Populin & Yin, 1998) and humans (Middlebrooks & Green, 1991) has pointed to differences in the quality of sound waves depending on vertical positioning.

After being processed by auditory hair cells, electrical signals are sent through the *cochlear nerve* (a division of the vestibulocochlear nerve) to the thalamus, and then the **primary auditory cortex** of the temporal lobe. Interestingly, the tonotopic organization of the cochlea is maintained in this area of the cortex (Merzenich, Knight, & Roth, 1975; Romani, Williamson, & Kaufman, 1982). However, the role of the primary auditory cortex in processing the wide range of features of sound is still being explored (Walker, Bizley, & Schnupp, 2011).

Balance and the vestibular system

The inner ear isn't only involved in hearing; it's also associated with our ability to balance and detect where we are in space. The **vestibular system** is comprised of three semicircular canals—fluid-filled bone structures containing cells that respond to changes in the head's orientation in space. Information from the vestibular system is sent through the vestibular nerve (the other division of the vestibulocochlear nerve) to muscles involved in the movement of our eyes, neck, and other parts of our body. This information allows us to maintain our gaze on an object while we are in motion. Disturbances in the vestibular system can result in issues with balance, including vertigo.

Touch

Who doesn't love the softness of an old t-shirt or the smoothness of a clean shave? Who actually enjoys having sand in their swimsuit? Our skin, the body's largest organ, provides us with all sorts of information, such as whether something is smooth or bumpy, hot or cold, or even if it's painful. **Somatosensation**—which includes our ability to sense touch, temperature

and pain—transduces physical stimuli, such as fuzzy velvet or scalding water, into electrical potentials that can be processed by the brain.

Tactile sensation

Tactile stimuli—those that are associated with texture—are transduced by special receptors in the skin called mechanoreceptors. Just like photoreceptors in the eye and auditory hair cells in the ear, these allow for the conversion of one kind of energy into a form the brain can understand.

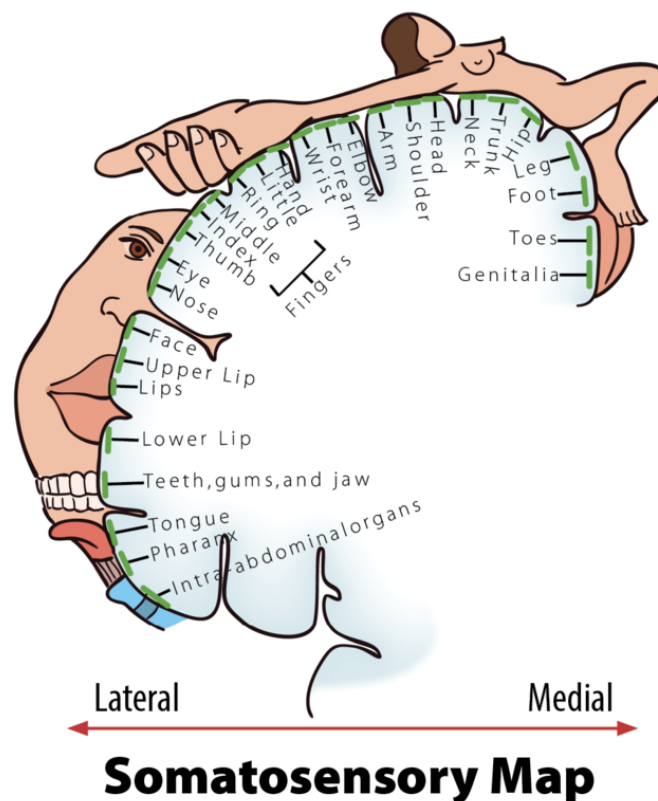


Figure 5. A drawing of the somatosensory cortex in the brain and the areas in the human body that correspond to it - they are drawn in proportion to the most sensitive or the most innervated parts of the body.

After tactile stimuli are converted by mechanoreceptors, information is sent through the thalamus to the primary somatosensory cortex for further processing. This region of the cortex is organized in a somatotopic map where different regions are sized based on the

sensitivity of specific parts on the opposite side of the body (Penfield & Rasmussen, 1950). Put simply, various areas of the skin, such as lips and fingertips, are more sensitive than others, such as shoulders or ankles. This sensitivity can be represented with the distorted proportions of the human body shown in Figure 5.

Pain

Most people, if asked, would love to get rid of pain (nociception), because the sensation is very unpleasant and doesn't appear to have obvious value. But the perception of pain is our body's way of sending us a signal that something is wrong and needs our attention. Without pain, how would we know when we are accidentally touching a hot stove, or that we should rest a strained arm after a hard workout?

Phantom limbs

Records of people experiencing phantom limbs after amputations have been around for centuries (Mitchell, 1871). As the name suggests, people with a phantom limb have the sensations such as itching seemingly coming from their missing limb. A phantom limb can also involve phantom limb pain, sometimes described as the muscles of the missing limb uncomfortably clenching. While the mechanisms underlying these phenomena are not fully understood, there is evidence to support that the damaged nerves from the amputation site are still sending information to the brain (Weinstein, 1998) and that the brain is reacting to this information (Ramachandran & Rogers-Ramachandran, 2000). There is an interesting treatment for the alleviation of phantom limb pain that works by tricking the brain, using a special mirror box to create a visual representation of the missing limb. The technique allows the patient to manipulate this representation into a more comfortable position (Ramachandran & Rogers-Ramachandran, 1996).

Smell and Taste: The Chemical Senses

The two most underappreciated senses can be lumped into the broad category of chemical senses. Both olfaction (smell) and gustation (taste) require the transduction of chemical stimuli into electrical potentials. I say these senses are underappreciated because most people would give up either one of these if they were forced to give up a sense. While this may not shock a lot of readers, take into consideration how much money people spend on the perfume industry annually (\$29 billion US Dollars). Many of us pay a lot more for a favorite brand of food because we prefer the taste. Clearly, we humans care about our chemical senses.

Olfaction (smell)

Unlike any of the other senses discussed so far, the receptors involved in our perception of both smell and taste bind directly with the stimuli they transduce. **Odorants** in our environment, very often mixtures of them, bind with olfactory receptors found in the **olfactory epithelium**. The binding of odorants to receptors is thought to be similar to how a lock and key operates, with different odorants binding to different specialized receptors based on their shape. However, the **shape theory of olfaction** isn't universally accepted and alternative theories exist, including one that argues that the vibrations of odorant molecules correspond to their subjective smells (Turin, 1996). Regardless of how odorants bind with receptors, the result is a pattern of neural activity. It is thought that our memories of these patterns of activity underlie our subjective experience of smell (Shepherd, 2005). Interestingly, because olfactory receptors send projections to the brain through the *cribriform plate* of the skull, head trauma has the potential to cause **anosmia**, due to the severing of these connections. If you are in a line of work where you constantly experience head trauma (e.g. professional boxer) and you develop anosmia, don't worry—your sense of smell will probably come back (Sumner, 1964).

Gustation (taste)

Taste works in a similar fashion to smell, only with receptors found in the taste buds of the tongue, called **taste receptor cells**. To clarify a common misconception, taste buds are not the bumps on your tongue (papillae), but are located in small divots around these bumps. These receptors also respond to chemicals from the outside environment, except these chemicals, called **tastants**, are contained in the foods we eat. The binding of these chemicals with taste receptor cells results in our perception of the five basic tastes: sweet, sour, bitter, salty and umami (savory)—although some scientists argue that there are more (Stewart et al., 2010). Researchers used to think these tastes formed the basis for a map-like organization of the tongue; there was even a clever rationale for the concept, about how the back of the tongue sensed



Ghost Pepper, also known as Bhut Jolokia is one of the hottest peppers in the world, it's 10 times hotter than a habanero, and 400 times hotter than tabasco sauce. What do you think would happen to your taste receptor cells if you took a bite out of this little guy? [Image: Richard Elzey, <https://goo.gl/suJHNq>, CC BY 2.0, <https://goo.gl/9uSnqN>]

bitter so we would know to spit out poisons, and the front of the tongue sensed sweet so we could identify high-energy foods. However, we now know that all areas of the tongue with taste receptor cells are capable of responding to every taste (Chandrashekar, Hoon, Ryba, & Zuker, 2006).

During the process of eating we are not limited to our sense of taste alone. While we are chewing, food odorants are forced back up to areas that contain olfactory receptors. This combination of taste and smell gives us the perception of **flavor**. If you have doubts about the interaction between these two senses, I encourage you to think back to consider how the flavors of your favorite foods are impacted when you have a cold; everything is pretty bland and boring, right?

Putting it all Together: Multimodal Perception

Though we have spent the majority of this module covering the senses individually, our real-world experience is most often multimodal, involving combinations of our senses into one perceptual experience. This should be clear after reading the description of walking through the forest at the beginning of the module; it was the combination of senses that allowed for that experience. It shouldn't shock you to find out that at some point information from each of our senses becomes integrated. Information from one sense has the potential to influence how we perceive information from another, a process called **multimodal perception**.

Interestingly, we actually respond more strongly to multimodal stimuli compared to the sum of each single modality together, an effect called the **superadditive effect of multisensory integration**. This can explain how you're still able to understand what friends are saying to you at a loud concert, as long as you are able to get visual cues from watching them speak. If you were having a quiet conversation at a café, you likely wouldn't need these additional cues. In fact, the **principle of inverse effectiveness** states that you are *less* likely to benefit from additional cues from other modalities if the initial unimodal stimulus is strong enough (Stein & Meredith, 1993).

Because we are able to process multimodal sensory stimuli, and the results of those processes are qualitatively different from those of unimodal stimuli, it's a fair assumption that the brain is doing something qualitatively different when they're being processed. There has been a growing body of evidence since the mid-90's on the neural correlates of multimodal perception. For example, neurons that respond to both visual and auditory stimuli have been identified in the *superior temporal sulcus* (Calvert, Hansen, Iversen, & Brammer, 2001). Additionally, multimodal "what" and "where" pathways have been proposed for auditory and

tactile stimuli (Renier et al., 2009). We aren't limited to reading about these regions of the brain and what they do; we can experience them with a few interesting examples (see Additional Resources for the "McGurk Effect," the "Double Flash Illusion," and the "Rubber Hand Illusion").

Conclusion

Our impressive sensory abilities allow us to experience the most enjoyable and most miserable experiences, as well as everything in between. Our eyes, ears, nose, tongue and skin provide an interface for the brain to interact with the world around us. While there is simplicity in covering each sensory modality independently, we are organisms that have evolved the ability to process multiple modalities as a unified experience.

Outside Resources

Audio: Auditory Demonstrations. CD published by the Acoustical Society of America (ASA).

You can listen to the demonstrations here

<http://www.auditory.org/democds.html>

Book: Ackerman, D. (1990). A natural history of the senses. Vintage.

<https://www.dianeackerman.com/a-natural-history-of-the-senses>

Book: Sacks, O. (1998). The man who mistook his wife for a hat: And other clinical tales. Simon and Schuster.

<http://www.oliversacks.com/books-by-oliver-sacks/man-mistook-wife-hat/>

Video: Acquired knowledge and its impact on our three-dimensional interpretation of the world - Anamorphic Illusions

<https://youtu.be/tBNHPk-Lnkk>

Video: Seeing Sound, Tasting Color

<https://www.youtube.com/watch?v=FTr1VnXKr4A>

Video: The Phantom Limb Phenomenon

<https://www.youtube.com/watch?v=1mHlv5ToMTM>

Web: A special ringtone that is only audible to younger people.

<https://www.youtube.com/watch?v=lrewnzQYrPI>

Web: Amazing library with visual phenomena and optical illusions, explained

<http://michaelbach.de/ot/index.html>

Web: An article on the discoveries in echolocation: the use of sound in locating people and things

<http://www.psychologicalscience.org/index.php/publications/observer/2015/december-15/u-sing-sound-to-get-around.html>

Web: An optical illusion demonstration the opponent-process theory of color vision.

<https://www.youtube.com/watch?v=qA2brNUo7WA>

Web: Anatomy of the eye

<https://www.aao.org/eye-health/anatomy/parts-of-eye>

Web: Animation showing tonotopic organization of the basilar membrane.

<https://www.youtube.com/watch?v=dyenMluFaUw>

Web: Best Illusion of the Year Contest website

<http://illusionoftheyear.com/>

Web: Demonstration of contrast gain adaptation

<https://michaelbach.de/ot/lum-contrastAdapt/>

Web: Demonstration of illusory contours and lateral inhibition. Mach bands

<http://michaelbach.de/ot/lum-MachBands/index.html>

Web: Demonstration of illusory contrast and lateral inhibition. The Hermann grid

http://michaelbach.de/ot/lum_herGrid/

Web: Demonstrations and illustrations of cochlear mechanics can be found here

<http://lab.rockefeller.edu/hudspeth/graphicalSimulations>

Web: Double Flash Illusion

<https://vimeo.com/39138252>

Web: Further information regarding what and where/how pathways

http://www.scholarpedia.org/article/What_and_where_pathways

Web: Great website with a large collection of optical illusions

<http://www.michaelbach.de/ot/>

Web: McGurk Effect Video

<https://www.youtube.com/watch?v=G-IN8vWm3m0>

Web: The Genetics of Taste

<http://www.smithsonianmag.com/arts-culture/the-genetics-of-taste-88797110/?no-ist>

Web: The Monell Chemical Sense Center website

<http://www.monell.org/>

Web: The Rubber Hand Illusion

<https://www.youtube.com/watch?v=sxwn1w7MJvk>

Web: The Tongue Map: Tasteless Myth Debunked

<http://www.livescience.com/7113-tongue-map-tasteless-myth-debunked.html>

Discussion Questions

1. There are a number of myths that exist about the sensory capabilities of infants. How would you design a study to determine what the true sensory capabilities of infants are?
2. A well-documented phenomenon experienced by millennials is the phantom vibration of a cell phone when no actual text message has been received. How can we use signal detection theory to explain this?
3. What physical features would an organism need in order to be really good at localizing sound in 3D space? Are there any organisms that currently excel in localizing sound? What features allow them to do this?
4. What issues would exist with visual recognition of an object if a research participant had his/her corpus callosum severed? What would you need to do in order to observe these deficits?

Vocabulary

Absolute threshold

The smallest amount of stimulation needed for detection by a sense.

Agnosia

Loss of the ability to perceive stimuli.

Anosmia

Loss of the ability to smell.

Audition

Ability to process auditory stimuli. Also called hearing.

Auditory canal

Tube running from the outer ear to the middle ear.

Auditory hair cells

Receptors in the cochlea that transduce sound into electrical potentials.

Binocular disparity

Difference in images processed by the left and right eyes.

Binocular vision

Our ability to perceive 3D and depth because of the difference between the images on each of our retinas.

Bottom-up processing

Building up to perceptual experience from individual pieces.

Chemical senses

Our ability to process the environmental stimuli of smell and taste.

Cochlea

Spiral bone structure in the inner ear containing auditory hair cells.

Cones

Photoreceptors of the retina sensitive to color. Located primarily in the fovea.

Dark adaptation

Adjustment of eye to low levels of light.

Differential threshold (or difference threshold)

The smallest difference needed in order to differentiate two stimuli. (See Just Noticeable Difference (JND))

Dorsal pathway

Pathway of visual processing. The “where” pathway.

Flavor

The combination of smell and taste.

Gustation

Ability to process gustatory stimuli. Also called taste.

Just noticeable difference (JND)

The smallest difference needed in order to differentiate two stimuli. (see Differential Threshold)

Light adaptation

Adjustment of eye to high levels of light.

Mechanoreceptors

Mechanical sensory receptors in the skin that response to tactile stimulation.

Multimodal perception

The effects that concurrent stimulation in more than one sensory modality has on the perception of events and objects in the world.

Nociception

Our ability to sense pain.

Odorants

Chemicals transduced by olfactory receptors.

Olfaction

Ability to process olfactory stimuli. Also called smell.

Olfactory epithelium

Organ containing olfactory receptors.

Opponent-process theory

Theory proposing color vision as influenced by cells responsive to pairs of colors.

Ossicles

A collection of three small bones in the middle ear that vibrate against the tympanic membrane.

Perception

The psychological process of interpreting sensory information.

Phantom limb

The perception that a missing limb still exists.

Phantom limb pain

Pain in a limb that no longer exists.

Pinna

Outermost portion of the ear.

Primary auditory cortex

Area of the cortex involved in processing auditory stimuli.

Primary somatosensory cortex

Area of the cortex involved in processing somatosensory stimuli.

Primary visual cortex

Area of the cortex involved in processing visual stimuli.

Principle of inverse effectiveness

The finding that, in general, for a multimodal stimulus, if the response to each unimodal component (on its own) is weak, then the opportunity for multisensory enhancement is very large. However, if one component—by itself—is sufficient to evoke a strong response, then the effect on the response gained by simultaneously processing the other components of the stimulus will be relatively small.

Retina

Cell layer in the back of the eye containing photoreceptors.

Rods

Photoreceptors of the retina sensitive to low levels of light. Located around the fovea.

Sensation

The physical processing of environmental stimuli by the sense organs.

Sensory adaptation

Decrease in sensitivity of a receptor to a stimulus after constant stimulation.

Shape theory of olfaction

Theory proposing that odorants of different size and shape correspond to different smells.

Signal detection

Method for studying the ability to correctly identify sensory stimuli.

Somatosensation

Ability to sense touch, pain and temperature.

Somatotopic map

Organization of the primary somatosensory cortex maintaining a representation of the arrangement of the body.

Sound waves

Changes in air pressure. The physical stimulus for audition.

Superadditive effect of multisensory integration

The finding that responses to multimodal stimuli are typically greater than the sum of the independent responses to each unimodal component if it were presented on its own.

Tastants

Chemicals transduced by taste receptor cells.

Taste receptor cells

Receptors that transduce gustatory information.

Top-down processing

Experience influencing the perception of stimuli.

Transduction

The conversion of one form of energy into another.

Trichromatic theory

Theory proposing color vision as influenced by three different cones responding preferentially to red, green and blue.

Tympanic membrane

Thin, stretched membrane in the middle ear that vibrates in response to sound. Also called the eardrum.

Ventral pathway

Pathway of visual processing. The “what” pathway.

Vestibular system

Parts of the inner ear involved in balance.

Weber's law

States that just noticeable difference is proportional to the magnitude of the initial stimulus.

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21

Vision

Simona Buetti & Alejandro Lleras

Vision is the sensory modality that transforms light into a psychological experience of the world around you, with minimal bodily effort. This module provides an overview of the most significant steps in this transformation and strategies that your brain uses to achieve this visual understanding of the environment.

Learning Objectives

- Describe how the eye transforms light information into neural energy.
- Understand what sorts of information the brain is interested in extracting from the environment and why it is useful.
- Describe how the visual system has adapted to deal with different lighting conditions.
- Understand the value of having two eyes.
- Understand why we have color vision.
- Understand the interdependence between vision and other brain functions.

What Is Vision?

Think about the spectacle of a starry night. You look up at the sky, and thousands of photons from distant stars come crashing into your retina, a light-sensitive structure at the back of your eyeball. These photons are millions of years old and have survived a trip across the universe, only to run into one of your photoreceptors. Tough luck: in one thousandth of a

second, this little bit of light energy becomes the fuel to a photochemical reaction known as **photoactivation**. The light energy becomes neural energy and triggers a cascade of neural activity that, a few hundredths of a second later, will result in your becoming aware of that distant star. You and the universe united by photons. That is the amazing power of vision. Light brings the world to you. Without moving, you know what's out there. You can recognize friends coming to meet you before you are able to hear them coming, ripe fruits from green ones on trees without having to taste them and before reaching out to grab them. You can also tell how quickly a ball is moving in your direction (Will it hit you? Can you hit it?).

How does all of that happen? First, light enters the eyeball through a tiny hole known as the pupil and, thanks to the refractive properties of your cornea and lens, this light signal gets projected sharply into the retina (see Outside Resources for links to a more detailed description of the eye structure). There, light is transduced into neural energy by about 200 million photoreceptor cells.

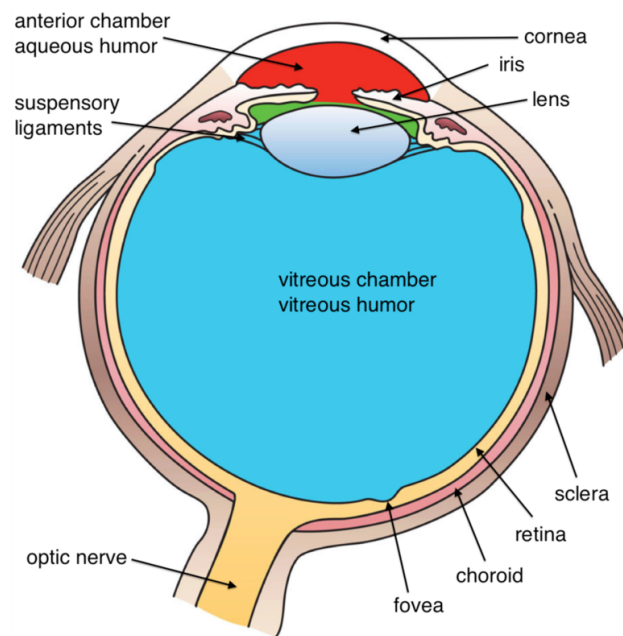


Diagram of the human eye [Image: Holly Fischer, <https://goo.gl/cMOQdh>, CC BY 3.0, <https://goo.gl/EHHVdU>]

This is where the information carried by the light about distant objects and colors starts being encoded by our brain. There are two different types of photoreceptors: **rods** and **cones**. The human eye contains more rods than cones. Rods give us sensitivity under dim lighting conditions and allow us to see at night. Cones allow us to see fine details in bright light and give us the sensation of color. Cones are tightly packed around the fovea (the central region

of the retina behind your pupil) and more sparsely elsewhere. Rods populate the periphery (the region surrounding the fovea) and are almost absent from the fovea.

But vision is far more complex than just catching photons. The information encoded by the photoreceptors undergoes a rapid and continuous set of ever more complex analysis so that, eventually, you can make sense of what's out there. At the fovea, visual information is encoded separately from tiny portions of the world (each about half the width of a human hair viewed at arm's length) so that eventually the brain can reconstruct in great detail fine visual differences from locations at which you are directly looking. This fine level of encoding requires lots of light and it is slow going (neurally speaking). In contrast, in the periphery, there is a different encoding strategy: detail is sacrificed in exchange for sensitivity. Information is summed across larger sections of the world. This aggregation occurs quickly and allows you to detect dim signals under very low levels of light, as well as detect sudden movements in your peripheral vision.

The Importance of Contrast

What happens next? Well, you might think that the eye would do something like record the amount of light at each location in the world and then send this information to the visual-processing areas of the brain (an astounding 30% of the cortex is influenced by visual signals!). But, in fact, that is not what eyes do. As soon as photoreceptors capture light, the nervous system gets busy analyzing *differences* in light, and it is these differences that get transmitted to the brain. The brain, it turns out, cares little about the overall amount of light coming from a specific part of the world, or in the scene overall. Rather, it wants to know: does the light coming from this one point differ from the light coming from the point next to it? Place your hand on the table in front of you. The contour of your hand is actually determined by the difference in light—the **contrast**—between the light coming from the skin in your hand and the light coming from the table underneath. To find the contour of your hand, we simply need to find the regions in the image where the difference in light between two adjacent points is maximal. Two points on your skin will reflect similar levels of light back to you, as will two points on the table. On the other hand, two points that fall on either side of the boundary contour between your hand and the table will reflect very different light.

The fact that the brain is interested in coding contrast in the world reveals something deeply important about the forces that drove the evolution of our brain: encoding the absolute amount of light in the world tells us little about what is out there. But if your brain can detect the sudden appearance of a *difference* in light somewhere in front of you, then it must be that something new is there. That contrast signal is *information*. That information may represent

something that you like (food, a friend) or something dangerous approaching (a tiger, a cliff). The rest of your visual system will work hard to determine what that thing is, but as quickly as 10ms after light enters your eyes, ganglion cells in your retinae have already encoded all the differences in light from the world in front of you.

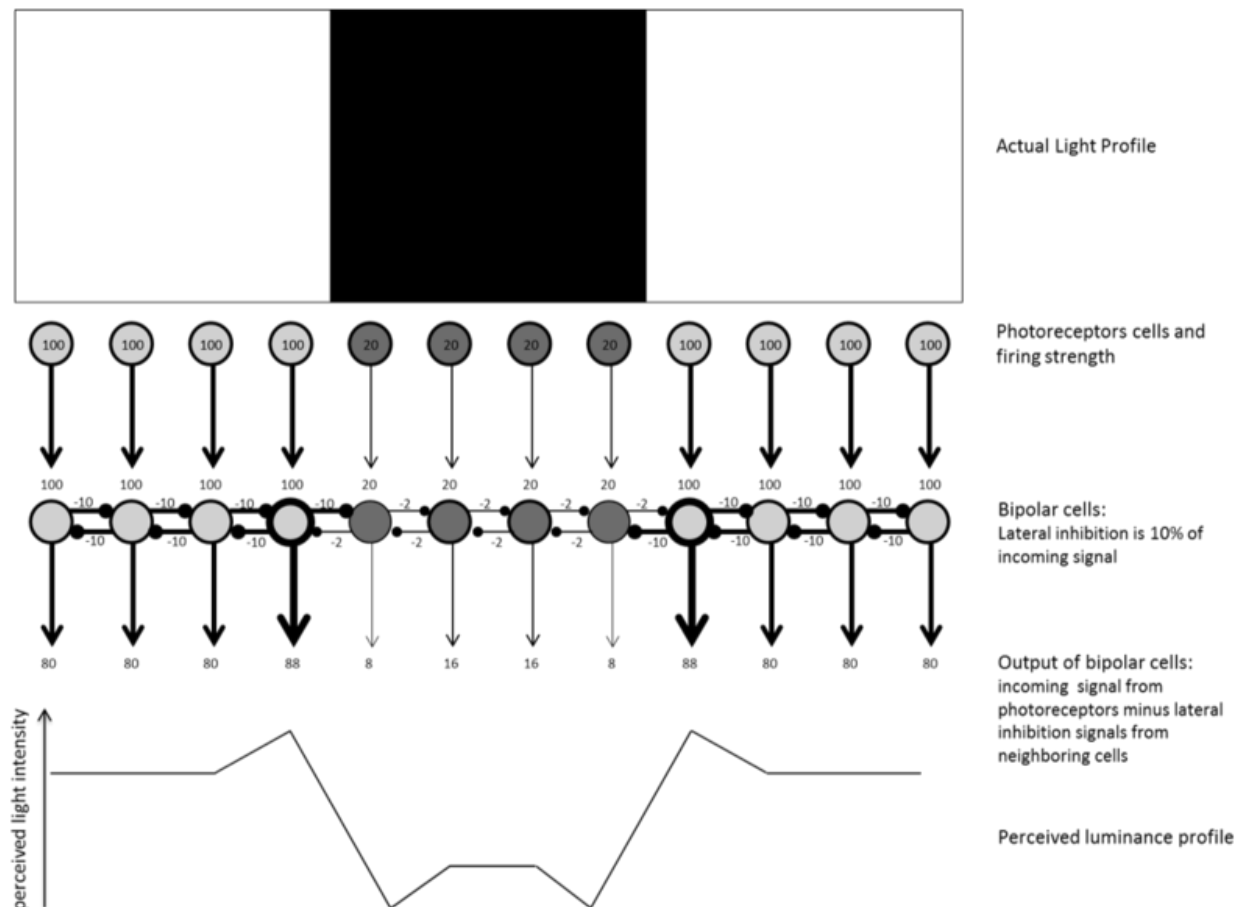


Figure 1. Illustration of Lateral Inhibition at work. The top of the figure shows a black stripe on a white background. The first row of circles illustrates photoreceptors responding in a graded fashion: the more light hits them, the more they fire. The numbers inside the circles represent how much these cells are firing, and the thickness of lines is also meant to illustrate the strength of neural firing. These photoreceptors activate the next layer of neurons in the retina: bipolar cells. These cells produce lateral inhibition signals, depicted by the horizontal lines that end with a small circle. The inhibition signals are proportional (here, 10% for ease) to the excitatory input they receive. Cells receiving 100 units will inhibit their neighbors by 10 units. Cells receiving 20 units will inhibit their neighbors by 2 units. The output of a bipolar cell will be determined by the input it receives minus all the lateral inhibition signals from its neighbors. As a result of the inhibition, notice how on the bright side of the edges, the firing rates are the highest (88) compared to nearby neurons just coding bright light (80). These higher values near the edge occur because these cells receive a comparatively small amount of inhibition from their "dark-side" neighbor (-2). Similarly, on the dark side of the edge, the firing rates are the lowest (8) of all the dark region (16) because these cells receive a comparatively large amount of inhibition from their "bright-side" neighbor (-10). Overall, the image is coded as a black stripe surrounded by brighter light, but now, thanks to lateral inhibition, all the edges in the image have been emphasized (enhanced), as illustrated by the perceived luminance profile at the bottom of the image.

Contrast is so important that your neurons go out of their way not only to encode differences in light but to exaggerate those differences for you, lest you miss them. Neurons achieve this via a process known as **lateral inhibition**. When a neuron is firing in response to light, it produces two signals: an output signal to pass on to the next level in vision, and a lateral signal to inhibit all neurons that are next to it. This makes sense on the assumption that nearby neurons are likely responding to the same light coming from nearby locations, so this information is somewhat redundant. The magnitude of the lateral inhibitory signal a neuron produces is proportional to the excitatory input that neuron receives: the more a neuron fires, the stronger the inhibition it produces. Figure 1 illustrates how lateral inhibition amplifies contrast signals at the edges of surfaces.

Sensitivity to Different Light Conditions



Rods and cones work in tandem to help you adjust when moving between extremes of dark and light. [Image: Pexels, CC0 Public Domain, <https://goo.gl/m25gce>]

Let's think for a moment about the range of conditions in which your visual system must operate day in and day out. When you take a walk outdoors on a sunny day, as many as billions of photons enter your eyeballs every second. In contrast, when you wake up in the middle of the night in a dark room, there might be as little as a few hundred photons per second entering your eyes. To deal with these extremes, the visual system relies on the different properties of the two types of photoreceptors. Rods are mostly responsible for processing light when photons are scarce (just a single photon can make a rod fire!), but it takes time to replenish the visual pigment that rods require for photoactivation. So, under bright conditions, rods are quickly bleached

(Stuart & Brige, 1996) and cannot keep up with the constant barrage of photons hitting them. That's when the cones become useful. Cones require more photons to fire and, more critically, their photopigments replenish much faster than rods' photopigments, allowing them to keep up when photons are abundant.

What happens when you abruptly change lighting conditions? Under bright light, your rods are bleached. When you move into a dark environment, it will take time (up to 30 minutes)

before they chemically recover (Hurley, 2002). Once they do, you will begin to see things around you that initially you could not. This phenomenon is called **dark adaptation**. When you go from dark to bright light (as you exit a tunnel on a highway, for instance), your rods will be bleached in a blaze and you will be blinded by the sudden light for about 1 second. However, your cones are ready to fire! Their firing will take over and you will quickly begin to see at this higher level of light.

A similar, but more subtle, adjustment occurs when the change in lighting is not so drastic. Think about your experience of reading a book at night in your bed compared to reading outdoors: the room may feel to you fairly well illuminated (enough so you can read) but the light bulbs in your room are not producing the billions of photons that you encounter outside. In both cases, you feel that your experience is that of a well-lit environment. You don't feel one experience as millions of times brighter than the other. This is because vision (as much of perception) is not proportional: seeing twice as many photons does not produce a sensation of seeing twice as bright a light. The visual system tunes into the current experience by favoring a range of contrast values that is most informative in that environment (Gardner et al., 2005). This is the concept of **contrast gain**: the visual system determines the mean contrast in a scene and represents values around that mean contrast best, while ignoring smaller contrast differences. (See the Outside Resources section for a demonstration.)

The Reconstruction Process

What happens once information leaves your eyes and enters the brain? Neurons project first into the thalamus, in a section known as the lateral geniculate nucleus. The information then splits and projects towards two different parts of the brain. Most of the computations regarding reflexive eye movements are computed in subcortical regions, the evolutionarily older part of the brain. Reflexive eye movements allow you to quickly orient your eyes towards areas of interest and to track objects as they move. The more complex computations, those that eventually allow you to have a visual experience of the world, all happen in the cortex, the evolutionarily newer region of the brain. The first stop in the cortex is at the **primary visual cortex** (also known as V1). Here, the "reconstruction" process begins in earnest: based on the contrast information arriving from the eyes, neurons will start computing information about color and simple lines, detecting various orientations and thicknesses. Small-scale motion signals are also computed (Hubel & Wiesel, 1962).

As information begins to flow towards other "higher" areas of the system, more complex computations are performed. For example, edges are assigned to the object to which they belong, backgrounds are separated from foregrounds, colors are assigned to surfaces, and

the global motion of objects is computed. Many of these computations occur in specialized brain areas. For instance, an area called MT processes global-motion information; the parahippocampal place area identifies locations and scenes; the fusiform face area specializes in identifying objects for which fine discriminations are required, like faces. There is even a brain region specialized in letter and word processing. These visual-recognition areas are located along the *ventral pathway* of the brain (also known as the **What pathway**). Other brain regions along the *dorsal pathway* (or **Where-and-How pathway**) will compute information about self- and object-motion, allowing you to interact with objects, navigate the environment, and avoid obstacles (Goodale and Milner, 1992).

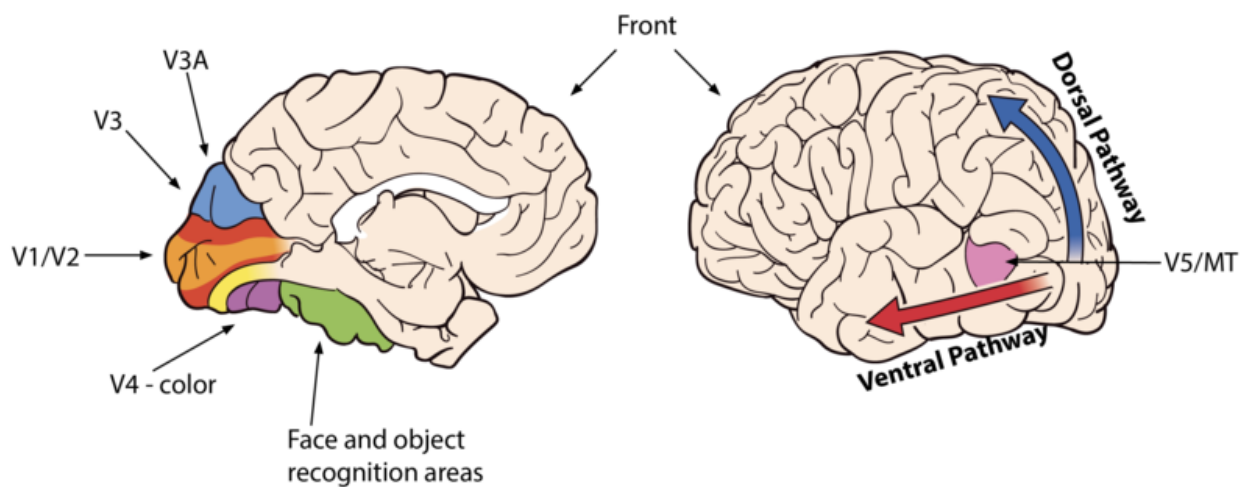


Figure 2: Areas of the brain

Now that you have a basic understanding of how your visual system works, you can ask yourself the question: why do you have two eyes? Everything that we discussed so far could be computed with information coming from a single eye. So why two? Looking at the animal kingdom gives us a clue. Animals who tend to be prey have eyes located on opposite sides of their skull. This allows them to detect predators whenever one appears anywhere around them. Humans, like most predators, have two eyes pointing in the same direction, encoding almost the exact scene twice. This redundancy gives us a **binocular advantage**: having two eyes not only provides you with two chances at catching a signal in front of you, but the minute difference in perspective that you get from each eye is used by your brain to reconstruct the sense of three-dimensional space. You can get an estimate of how far distant objects are from you, their size, and their volume. This is no easy feat: the signal in each eye is a two-dimensional projection of the world, like two separate pictures drawn upon your retinae. Yet, your brain

effortlessly provides you with a sense of depth by combining those two signals. This 3-D reconstruction process also relies heavily on all the knowledge you acquired through experience about spatial information. For instance, your visual system learns to interpret how the volume, distance, and size of objects change as they move closer or farther from you. (See the Outside Resources section for demonstrations.)

The Experience of Color

Perhaps one of the most beautiful aspects of vision is the richness of the color experience that it provides us. One of the challenges that we have as scientists is to understand why the human color experience is what it is. Perhaps you have heard that dogs only have 2 types of color photoreceptors, whereas humans have 3, chickens have 4, and mantis shrimp have 16. Why is there such variation across species? Scientists believe each species has evolved with different needs and uses color perception to signal information about food, reproduction, and health that are unique to their species. For example, humans have a specific sensitivity that allows you to detect slight changes in skin tone. You can tell when someone is embarrassed, aroused, or ill. Detecting these subtle signals is adaptive in a social species like ours.

How is color coded in the brain? The two leading theories of color perception were proposed in the mid-19th century, about 100 years before physiological evidence was found to corroborate them both (Svaetichin, 1956). **Trichromacy theory**, proposed by Young (1802) and Helmholtz (1867), proposed that the eye had three different types of color-sensitive cells based on the observation that any one color can be reproduced by combining lights from three lamps of different hue. If you can adjust separately the intensity of each light, at some point you will find the right combination of the three lights to match any color in the world. This principle is used today on TVs, computer screens, and any colored display. If you look closely enough at a pixel, you will find that it is composed of a blue, a red, and a green light, of varying intensities. Regarding the retina, humans have three types of cones: S-cones, M-cones, and L-cones (also known as blue, green, and red cones, respectively) that are sensitive to three different wavelengths of light.

Around the same time, Hering made a puzzling discovery: some colors are impossible to create. Whereas you can make yellowish greens, bluish reds, greenish blues, and reddish yellows by combining two colors, you can never make a reddish green or a bluish yellow. This observation led Hering (1892) to propose the **Opponent Process theory of color**: color is coded via three opponent channels (red-green, blue-yellow, and black-white). Within each channel, a comparison is constantly computed between the two elements in the pair. In other

words, colors are encoded as *differences* between two hues and not as simple combinations of hues. Again, what matters to the brain is *contrast*. When one element is stronger than the other, the stronger color is perceived and the weaker one is suppressed. You can experience this phenomenon by following the link below.

<http://nobaproject.com/assets/modules/module-visio...>

When both colors in a pair are present to equal extents, the color perception is canceled and we perceive a level of grey. This is why you cannot see a reddish green or a bluish yellow: they cancel each other out. By the way, if you are wondering where the yellow signal comes from, it turns out that it is computed by averaging the M- and L-cone signals. Are these colors uniquely human colors? Some think that they are: the red-green contrast, for example, is finely tuned to detect changes in human skin tone so you can tell when someone blushes or becomes pale. So, the next time you go out for a walk with your dog, look at the sunset and ask yourself, what color does my dog see? Probably none of the orange hues you do!

So now, you can ask yourself the question: do all humans experience color in the same way? Color-blind people, as you can imagine, do not see all the colors that the rest of us see, and this is due to the fact that they lack one (or more) cones in their retina. Incidentally, there are a few women who actually have four different sets of cones in their eyes, and recent research suggests that their experience of color can be (but not always is) richer than the one from three-coned people. A slightly different question, though, is whether all three-coned people have the same internal experiences of colors: is the red inside your head the same red inside your mom's head? That is an almost impossible question to answer that has been debated by philosophers for millennia, yet recent data suggests that there might in fact be cultural differences in the way we perceive color. As it turns out, not all cultures categorize colors in the same way, for example. And some groups "see" different shades of what we in the Western world would call the "same" color, as categorically different colors. The Berinmo tribe in New Guinea, for instance, appear to experience green shades that denote leaves that are alive as belonging to an entirely different color category than the sort of green shades that denote dying leaves. Russians, too, appear to experience light and dark shades of blue as different categories of colors, in a way that most Westerners do not. Further, current brain imaging research suggests that people's brains change (increase in white-matter volume) when they learn new color categories! These are intriguing and suggestive findings, for certain, that seem to indicate that our cultural environment may in fact have some (small) but definite impact on how people use and experience colors across the globe.

Integration with Other Modalities

Vision is not an encapsulated system. It interacts with and depends on other sensory modalities. For example, when you move your head in one direction, your eyes reflexively move in the opposite direction to compensate, allowing you to maintain your gaze on the object that you are looking at. This reflex is called the vestibulo-ocular reflex. It is achieved by integrating information from both the visual and the vestibular system (which knows about body motion and position). You can experience this compensation quite simply. First, while you keep your head still and your gaze looking straight ahead, wave your finger in front of you from side to side. Notice how the image of the finger appears blurry. Now, keep your finger steady and look at it while you move your head from side to side. Notice how your eyes reflexively move to compensate the movement of your head and how the image of the finger stays sharp and stable. Vision also interacts with your proprioceptive system, to help you find where all your body parts are, and with your auditory system, to help you understand the sounds people make when they speak. You can learn more about this in the Noba module about multimodal perception (<http://noba.to/cezw4qyn>).

Finally, vision is also often implicated in a blending-of-sensations phenomenon known as synesthesia. Synesthesia occurs when one sensory signal gives rise to two or more sensations. The most common type is *grapheme-color* synesthesia. About 1 in 200 individuals experience a sensation of color associated with specific letters, numbers, or words: the number 1 might always be seen as red, the number 2 as orange, etc. But the more fascinating forms of synesthesia blend sensations from entirely different sensory modalities, like taste and color or music and color: the taste of chicken might elicit a sensation of green, for example, and the timbre of violin a deep purple.

Concluding Remarks

We are at an exciting moment in our scientific understanding of vision. We have just begun to get a functional understanding of the visual system. It is not sufficiently evolved for us to recreate artificial visual systems (i.e., we still cannot make robots that “see” and understand light signals as we do), but we are getting there. Just recently, major breakthroughs in vision science have allowed researchers to significantly improve retinal prosthetics: photosensitive circuits that can be implanted on the back of the eyeball of blind people that connect to visual areas of the brain and have the capacity to partially restore a “visual experience” to these patients (Nirenberg & Pandarinath, 2012). And using functional magnetic brain imaging, we can now “decode” from your brain activity the images that you saw in your dreams while you were asleep (Horikawa, Tamaki, Miyawaki, & Kamitani, 2013)! Yet, there is still so much more to understand. Consider this: if vision is a construction process that takes time, whatever we see *now* is no longer what is front of us. Yet, humans can do amazing time-sensitive feats like

hitting a 90-mph fastball in a baseball game. It appears then that a fundamental function of vision is not just to know what is happening around you now, but actually to make an accurate inference about what you are about to see *next* (Enns & Lleras, 2008), so that you can keep up with the world. Understanding how this future-oriented, *predictive* function of vision is achieved in the brain is probably the next big challenge in this fascinating realm of research.

Outside Resources

Video: Acquired knowledge and its impact on our three-dimensional interpretation of the world - 3D Street Art

<https://youtu.be/bkJh3xxC6UU>

Video: Acquired knowledge and its impact on our three-dimensional interpretation of the world - Anamorphic Illusions

<http://www.youtube.com/watch?v=tBNHPk-Lnkk>

Video: Acquired knowledge and its impact on our three-dimensional interpretation of the world - Optical Illusion

<http://www.youtube.com/watch?v=YjmHofJ2da0&feature=related>

Web: Amazing library with visual phenomena and optical illusions, explained

<http://michaelbach.de/ot/index.html>

Web: Anatomy of the eye

<https://www.aao.org/eye-health/anatomy/parts-of-eye>

Web: Demonstration of contrast gain adaptation

<https://michaelbach.de/ot/lum-contrastAdapt/>

Web: Demonstration of illusory contours and lateral inhibition. Mach bands

<http://michaelbach.de/ot/lum-MachBands/index.html>

Web: Demonstration of illusory contrast and lateral inhibition. The Hermann grid

http://michaelbach.de/ot/lum_herGrid/

Web: Further information regarding what and where/how pathways

http://www.scholarpedia.org/article/What_and_where_pathways

Discussion Questions

1. When running in the dark, it is recommended that you never look straight at the ground. Why? What would be a better strategy to avoid obstacles?

2. The majority of ganglion cells in the eye specialize in detecting drops in the amount of light coming from a given location. That is, they increase their firing rate when they detect less light coming from a specific location. Why might the absence of light be more important than the presence of light? Why would it be evolutionarily advantageous to code this type of information?
3. There is a hole in each one of your eyeballs called the optic disk. This is where veins enter the eyeball and where neurons (the axons of the ganglion cells) exit the eyeball. Why do you not see two holes in the world all the time? Close one eye now. Why do you not see a hole in the world now? To “experience” a blind spot, follow the instructions in this website: http://michaelbach.de/ot/cog_blindSpot/index.html
4. Imagine you were given the task of testing the color-perception abilities of a newly discovered species of monkeys in the South Pacific. How would you go about it?
5. An important aspect of emotions is that we sense them in ourselves much in the same way as we sense other perceptions like vision. Can you think of an example where the concept of contrast gain can be used to understand people’s responses to emotional events?

Vocabulary

Binocular advantage

Benefits from having two eyes as opposed to a single eye.

Cones

Photoreceptors that operate in lighted environments and can encode fine visual details. There are three different kinds (S or blue, M or green and L or red) that are each sensitive to slightly different types of light. Combined, these three types of cones allow you to have color vision.

Contrast

Relative difference in the amount and type of light coming from two nearby locations.

Contrast gain

Process where the sensitivity of your visual system can be tuned to be most sensitive to the levels of contrast that are most prevalent in the environment.

Dark adaptation

Process that allows you to become sensitive to very small levels of light, so that you can actually see in the near-absence of light.

Lateral inhibition

A signal produced by a neuron aimed at suppressing the response of nearby neurons.

Opponent Process Theory

Theory of color vision that assumes there are four different basic colors, organized into two pairs (red/green and blue/yellow) and proposes that colors in the world are encoded in terms of the opponency (or difference) between the colors in each pair. There is an additional black/white pair responsible for coding light contrast.

Photoactivation

A photochemical reaction that occurs when light hits photoreceptors, producing a neural signal.

Primary visual cortex (V1)

Brain region located in the occipital cortex (toward the back of the head) responsible for processing basic visual information like the detection, thickness, and orientation of simple lines, color, and small-scale motion.

Rods

Photoreceptors that are very sensitive to light and are mostly responsible for night vision.

Synesthesia

The blending of two or more sensory experiences, or the automatic activation of a secondary (indirect) sensory experience due to certain aspects of the primary (direct) sensory stimulation.

Trichromacy theory

Theory that proposes that all of your color perception is fundamentally based on the combination of three (not two, not four) different color signals.

Vestibulo-ocular reflex

Coordination of motion information with visual information that allows you to maintain your gaze on an object while you move.

What pathway

Pathway of neural processing in the brain that is responsible for your ability to recognize what is around you.

Where-and-How pathway

Pathway of neural processing in the brain that is responsible for you knowing where things are in the world and how to interact with them.

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22

Hearing

Andrew J. Oxenham

Hearing allows us to perceive the world of acoustic vibrations all around us, and provides us with our most important channels of communication. This module reviews the basic mechanisms of hearing, beginning with the anatomy and physiology of the ear and a brief review of the auditory pathways up to the auditory cortex. An outline of the basic perceptual attributes of sound, including loudness, pitch, and timbre, is followed by a review of the principles of tonotopic organization, established in the cochlea. An overview of masking and frequency selectivity is followed by a review of the perception and neural mechanisms underlying spatial hearing. Finally, an overview is provided of auditory scene analysis, which tackles the important question of how the auditory system is able to make sense of the complex mixtures of sounds that are encountered in everyday acoustic environments.

Learning Objectives

- Describe the basic auditory attributes of sound.
- Describe the structure and general function of the auditory pathways from the outer ear to the auditory cortex.
- Discuss ways in which we are able to locate sounds in space.
- Describe various acoustic cues that contribute to our ability to perceptually segregate simultaneously arriving sounds.

Introduction

Hearing forms a crucial part of our everyday life. Most of our communication with others, via speech or music, reaches us through the ears. Indeed, a saying, often attributed to Helen Keller, is that blindness separates us from things, but deafness separates us from people. The ears respond to acoustic information, or sound—tiny and rapid variations in air pressure. Sound waves travel from the source and produce pressure variations in the listener's ear canals, causing the eardrums (or tympanic membranes) to vibrate. This module provides an overview of the events that follow, which convert these simple mechanical vibrations into our rich experience known as hearing, or auditory perception.



Hearing provides us with our most important connection to the people around us. [Image: Bindaas Madhavi, <https://goo.gl/Sv6TtR>, CC BY-NC-ND 2.0, <https://goo.gl/62XJAl>]

Perceptual Attributes of Sound

There are many ways to describe a sound, but the perceptual attributes of a sound can typically be divided into three main categories—namely, loudness, pitch, and timbre. Although all three refer to perception, and not to the physical sounds themselves, they are strongly related to various physical variables.

Loudness

The most direct physical correlate of loudness is sound intensity (or sound pressure) measured close to the eardrum. However, many other factors also influence the loudness of a sound, including its frequency content, its duration, and the context in which it is presented. Some of the earliest psychophysical studies of auditory perception, going back more than a century, were aimed at examining the relationships between perceived loudness, the physical sound intensity, and the just-noticeable differences in loudness (Fechner, 1860; Stevens, 1957). A great deal of time and effort has been spent refining various measurement methods. These methods involve techniques such as magnitude estimation, where a series of sounds (often sinusoids, or pure tones of single frequency) are presented sequentially at different sound levels, and subjects are asked to assign numbers to each tone, corresponding to the perceived

loudness. Other studies have examined how loudness changes as a function of the frequency of a tone, resulting in the international standard iso-loudness-level contours (ISO, 2003), which are used in many areas of industry to assess noise and annoyance issues. Such studies have led to the development of computational models that are designed to predict the loudness of arbitrary sounds (e.g., Moore, Glasberg, & Baer, 1997).

Pitch



Pitch is crucial to our perception and understanding of music and language. [Image: xroper7, <https://goo.gl/1E4sJY>, CC BY-NC 2.0, <https://goo.gl/tgFydH>]

Pitch plays a crucial role in acoustic communication. Pitch variations over time provide the basis of melody for most types of music; pitch contours in speech provide us with important prosodic information in non-tone languages, such as English, and help define the meaning of words in tone languages, such as Mandarin Chinese. Pitch is essentially the perceptual correlate of waveform periodicity, or repetition rate: The faster a waveform repeats over time, the higher is its perceived pitch. The most common pitch-evoking sounds are known as harmonic complex tones. They are complex because they consist of more than one frequency, and they are harmonic because the frequencies are all integer multiples of a common fundamental frequency (F_0). For instance, a harmonic

complex tone with a F_0 of 100 Hz would also contain energy at frequencies of 200, 300, 400 Hz, and so on. These higher frequencies are known as harmonics or overtones, and they also play an important role in determining the pitch of a sound. In fact, even if the energy at the F_0 is absent or masked, we generally still perceive the remaining sound to have a pitch corresponding to the F_0 . This phenomenon is known as the “pitch of the missing fundamental,” and it has played an important role in the formation of theories and models about pitch (de Cheveigné, 2005). We hear pitch with sufficient accuracy to perceive melodies over a range of F_0 s from about 30 Hz (Pressnitzer, Patterson, & Krumbholz, 2001) up to about 4–5 kHz (Attneave & Olson, 1971; Oxenham, Micheyl, Keebler, Loper, & Santurette, 2011). This range also corresponds quite well to the range covered by musical instruments; for instance, the modern grand piano has notes that extend from 27.5 Hz to 4,186 Hz. We are able to

discriminate changes in frequency above 5,000 Hz, but we are no longer very accurate in recognizing melodies or judging musical intervals.

Timbre

Timbre refers to the quality of sound, and is often described using words such as bright, dull, harsh, and hollow. Technically, timbre includes anything that allows us to distinguish two sounds that have the same loudness, pitch, and duration. For instance, a violin and a piano playing the same note sound very different, based on their sound quality or timbre.

An important aspect of timbre is the spectral content of a sound. Sounds with more high-frequency energy tend to sound brighter, tinnier, or harsher than sounds with more low-frequency content, which might be described as deep, rich, or dull. Other important aspects of timbre include the temporal envelope (or outline) of the sound, especially how it begins and ends. For instance, a piano has a rapid onset, or attack, produced by the hammer striking the string, whereas the attack of a clarinet note can be much more gradual. Artificially changing the onset of a piano note by, for instance, playing a recording backwards, can dramatically alter its character so that it is no longer recognizable as a piano note. In general, the overall spectral content and the temporal envelope can provide a good first approximation to any sound, but it turns out that subtle changes in the spectrum over time (or spectro-temporal variations) are crucial in creating plausible imitations of natural musical instruments (Risset & Wessel, 1999).

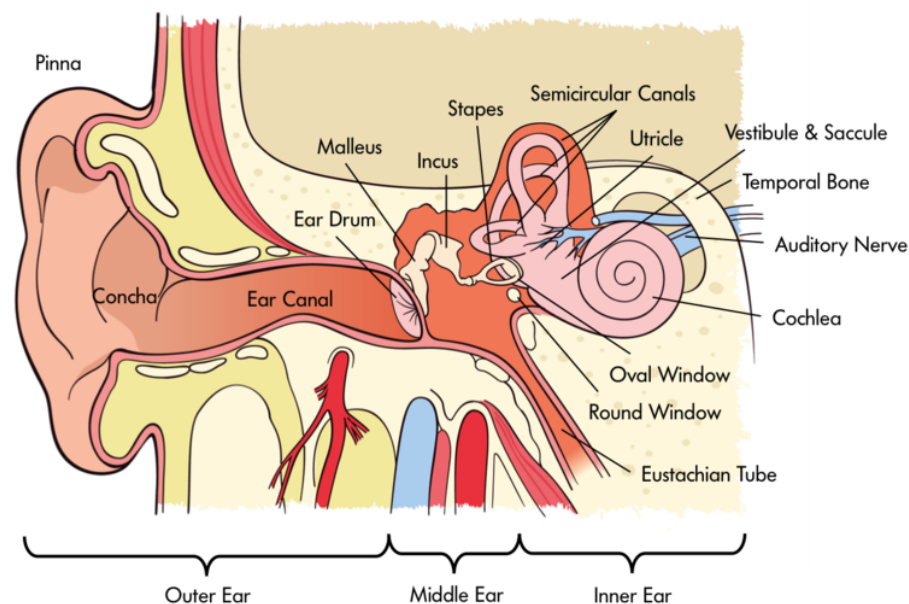


Figure 1: Diagram of the outer, middle, and inner ear.

An Overview of the Auditory System

Our auditory perception depends on how sound is processed through the ear. The ear can be divided into three main parts—the outer, middle, and inner ear (see Figure 1). The outer ear consists of the **pinna** (the visible part of the ear, with all its unique folds and bumps), the ear canal (or auditory meatus), and the **tympanic membrane**. Of course, most of us have two functioning ears, which turn out to be particularly useful when we are trying to figure out where a sound is coming from. As discussed below in the section on spatial hearing, our brain can compare the subtle differences in the signals at the two ears to localize sounds in space. However, this trick does not always help: for instance, a sound directly in front or directly behind you will not produce a difference between the ears. In these cases, the filtering produced by the pinnae helps us localize sounds and resolve potential front-back and up-down confusions. More generally, the folds and bumps of the pinna produce distinct peaks and dips in the frequency response that depend on the location of the sound source. The brain then learns to associate certain patterns of spectral peaks and dips with certain spatial locations. Interestingly, this learned association remains malleable, or plastic, even in adulthood. For instance, a study that altered the pinnae using molds found that people could learn to use their “new” ears accurately within a matter of a few weeks (Hofman, Van Riswick, & Van Opstal, 1998). Because of the small size of the pinna, these kinds of acoustic cues are only found at high frequencies, above about 2 kHz. At lower frequencies, the sound is basically unchanged whether it comes from above, in front, or below. The ear canal itself is a tube that helps to amplify sound in the region from about 1 to 4 kHz—a region particularly important for speech communication.

The middle ear consists of an air-filled cavity, which contains the middle-ear bones, known as the incus, malleus, and stapes, or anvil, hammer, and stirrup, because of their respective shapes. They have the distinction of being the smallest bones in the body. Their primary function is to transmit the vibrations from the tympanic membrane to the oval window of the cochlea and, via a form of lever action, to better match the impedance of the air surrounding the tympanic membrane with that of the fluid within the cochlea.

The inner ear includes the cochlea, encased in the temporal bone of the skull, in which the mechanical vibrations of sound are transduced into neural signals that are processed by the brain. The **cochlea** is a spiral-shaped structure that is filled with fluid. Along the length of the spiral runs the basilar membrane, which vibrates in response to the pressure differences produced by vibrations of the oval window. Sitting on the basilar membrane is the organ of Corti, which runs the entire length of the basilar membrane from the base (by the oval window) to the apex (the “tip” of the spiral). The organ of Corti includes three rows of outer hair cells

and one row of inner hair cells. The hair cells sense the vibrations by way of their tiny hairs, or stereocilia. The outer hair cells seem to function to mechanically amplify the sound-induced vibrations, whereas the inner hair cells form synapses with the auditory nerve and transduce those vibrations into action potentials, or neural spikes, which are transmitted along the auditory nerve to higher centers of the auditory pathways.

One of the most important principles of hearing—frequency analysis—is established in the cochlea. In a way, the action of the cochlea can be likened to that of a prism: the many frequencies that make up a complex sound are broken down into their constituent frequencies, with low frequencies creating maximal basilar-membrane vibrations near the apex of the cochlea and high frequencies creating maximal basilar-membrane vibrations nearer the base of the cochlea. This decomposition of sound into its constituent frequencies, and the frequency-to-place mapping, or “tonotopic” representation, is a major organizational principle of the auditory system, and is maintained in the neural representation of sounds all the way from the cochlea to the primary auditory cortex. The decomposition of sound into its constituent frequency components is part of what allows us to hear more than one sound at a time. In addition to representing frequency by place of excitation within the cochlea, frequencies are also represented by the timing of spikes within the auditory nerve. This property, known as “phase locking,” is crucial in comparing time-of-arrival differences of waveforms between the two ears (see the section on spatial hearing, below).

Unlike vision, where the primary visual cortex (or V1) is considered an early stage of processing, auditory signals go through many stages of processing before they reach the primary auditory cortex, located in the temporal lobe. Although we have a fairly good understanding of the electromechanical properties of the cochlea and its various structures, our understanding of the processing accomplished by higher stages of the auditory pathways remains somewhat sketchy. With the possible exception of spatial localization and neurons tuned to certain locations in space (Harper & McAlpine, 2004; Knudsen & Konishi, 1978), there is very little consensus on the how, what, and where of auditory feature extraction and representation. There is evidence for a “pitch center” in the auditory cortex from both human neuroimaging studies (e.g., Griffiths, Buchel, Frackowiak, & Patterson, 1998; Penagos, Melcher, & Oxenham, 2004) and single-unit physiology studies (Bendor & Wang, 2005), but even here there remain some questions regarding whether a single area of cortex is responsible for coding single features, such as pitch, or whether the code is more distributed (Walker, Bizley, King, & Schnupp, 2011).

Audibility, Masking, and Frequency Selectivity

Overall, the human cochlea provides us with hearing over a very wide range of frequencies. Young people with normal hearing are able to perceive sounds with frequencies ranging from about 20 Hz all the way up to 20 kHz. The range of intensities we can perceive is also impressive: the quietest sounds we can hear in the medium-frequency range (between about 1 and 4 kHz) have a sound intensity that is about a factor of 1,000,000,000,000 less intense than the loudest sound we can listen to without incurring rapid and permanent hearing loss. In part because of this enormous dynamic range, we tend to use a logarithmic scale, known as decibels (dB), to describe sound pressure or intensity. On this scale, 0 dB sound pressure level (SPL) is defined as 20 micro-Pascals (μPa), which corresponds roughly to the quietest perceptible sound level, and 120 dB SPL is considered dangerously loud.



When the frequency content of different sounds overlaps, masking occurs. Less intense sounds become difficult or impossible to hear because more intense sounds dominate and interfere. Crowded restaurants or busy city streets full of traffic are typical examples of places where certain sounds can "swamp" others. [Image: Peter van der Sluijs, <https://goo.gl/K8L4c0>, CC BY-SA 3.0, <https://goo.gl/eLCn2O>]

Masking is the process by which the presence of one sound makes another sound more difficult to hear. We all encounter masking in our everyday lives, when we fail to hear the phone ring while we are taking a shower, or when we struggle to follow a conversation in a noisy restaurant. In general, a more intense sound will mask a less intense sound, provided certain conditions are met. The most important condition is that the frequency content of the sounds overlap, such that the activity in the cochlea produced by a masking sound "swamps" that produced by the target sound. Another type of masking, known as "suppression," occurs when the response to the masker reduces the neural (and in some cases, the mechanical) response to the target sound. Because of the way that filtering in the cochlea functions, low-

frequency sounds are more likely to mask high frequencies than vice versa, particularly at high sound intensities. This asymmetric aspect of masking is known as the "upward spread of masking." The loss of sharp cochlear tuning that often accompanies cochlear damage leads to broader filtering and more masking—a physiological phenomenon that is likely to contribute to the difficulties experienced by people with hearing loss in noisy environments (Moore, 2007).

Although much masking can be explained in terms of interactions within the cochlea, there are other forms that cannot be accounted for so easily, and that can occur even when interactions within the cochlea are unlikely. These more central forms of masking come in different forms, but have often been categorized together under the term “informational masking” (Durlach et al., 2003; Watson & Kelly, 1978). Relatively little is known about the causes of informational masking, although most forms can be ascribed to a perceptual “fusion” of the masker and target sounds, or at least a failure to segregate the target from the masking sounds. Also relatively little is known about the physiological locus of informational masking, except that at least some forms seem to originate in the auditory cortex and not before (Gutschalk, Micheyl, & Oxenham, 2008).

Spatial Hearing

In contrast to vision, we have a 360° field of hearing. Our auditory acuity is, however, at least an order of magnitude poorer than vision in locating an object in space. Consequently, our auditory localization abilities are most useful in alerting us and allowing us to orient towards sources, with our visual sense generally providing the finer-grained analysis. Of course, there are differences between species, and some, such as barn owls and echolocating bats, have developed highly specialized sound localization systems.

Our ability to locate sound sources in space is an impressive feat of neural computation. The two main sources of information both come from a comparison of the sounds at the two ears. The first is based on interaural time differences (ITD) and relies on the fact that a sound source on the left will generate sound that will reach the left ear slightly before it reaches the right ear. Although sound is much slower than light, its speed still means that the time of arrival differences between the two ears is a fraction of a millisecond. The largest ITD we encounter in the real world (when sounds are directly to the left or right of us) are only a little over half a millisecond. With some practice, humans can learn to detect an ITD of between 10 and 20 μ s (i.e., 20 *millionths*



Humans are able to locate sound in space to determine whether the source is in front of us or behind us, or whether it is elevated or below us. [Photo: David Goehring, <https://goo.gl/UOLZpB>, CC BY 2.0, <https://goo.gl/BRvSA7>]

of a second) (Klump & Eady, 1956).

The second source of information is based in **interaural level differences (ILDs)**. At higher frequencies (higher than about 1 kHz), the head casts an acoustic “shadow,” so that when a sound is presented from the left, the sound level at the left ear is somewhat higher than the sound level at the right ear. At very high frequencies, the ILD can be as much as 20 dB, and we are sensitive to differences as small as 1 dB.

As mentioned briefly in the discussion of the outer ear, information regarding the elevation of a sound source, or whether it comes from in front or behind, is contained in high-frequency spectral details that result from the filtering effects of the pinnae.

In general, we are most sensitive to ITDs at low frequencies (below about 1.5 kHz). At higher frequencies we can still perceive changes in timing based on the slowly varying temporal envelope of the sound but not the temporal fine structure (Bernstein & Trahiotis, 2002; Smith, Delgutte, & Oxenham, 2002), perhaps because of a loss of neural phase-locking to the temporal fine structure at high frequencies. In contrast, ILDs are most useful at high frequencies, where the head shadow is greatest. This use of different acoustic cues in different frequency regions led to the classic and very early “duplex theory” of sound localization (Rayleigh, 1907). For everyday sounds with a broad frequency spectrum, it seems that our perception of spatial location is dominated by interaural time differences in the low-frequency temporal fine structure (Macpherson & Middlebrooks, 2002).

As with vision, our perception of distance depends to a large degree on context. If we hear someone shouting at a very low sound level, we infer that the shouter must be far away, based on our knowledge of the sound properties of shouting. In rooms and other enclosed locations, the reverberation can also provide information about distance: As a speaker moves further away, the direct sound level decreases but the sound level of the reverberation remains about the same; therefore, the ratio of direct-to-reverberant energy decreases (Zahorik & Wightman, 2001).

Auditory Scene Analysis

There is usually more than one sound source in the environment at any one time—imagine talking with a friend at a café, with some background music playing, the rattling of coffee mugs behind the counter, traffic outside, and a conversation going on at the table next to yours. All these sources produce sound waves that combine to form a single complex waveform at the eardrum, the shape of which may bear very little relationship to any of the waves produced

by the individual sound sources. Somehow the auditory system is able to break down, or decompose, these complex waveforms and allow us to make sense of our acoustic environment by forming separate auditory “objects” or “streams,” which we can follow as the sounds unfold over time (Bregman, 1990).

A number of heuristic principles have been formulated to describe how sound elements are grouped to form a single object or segregated to form multiple objects. Many of these originate from the early ideas proposed in vision by the so-called Gestalt psychologists, such as Max Wertheimer. According to these rules of thumb, sounds that are in close proximity, in time or frequency, tend to be grouped together. Also, sounds that begin and end at the same time tend to form a single auditory object. Interestingly, spatial location is not always a strong or reliable grouping cue, perhaps because the location information from individual frequency components is often ambiguous due to the effects of reverberation. Several studies have looked into the relative importance of different cues by “trading off” one cue against another. In some cases, this has led to the discovery of interesting auditory illusions, where melodies that are not present in the sounds presented to either ear emerge in the perception (Deutsch, 1979), or where a sound element is perceptually “lost” in competing perceptual organizations (Shinn-Cunningham, Lee, & Oxenham, 2007).

More recent attempts have used computational and neurally based approaches to uncover the mechanisms of auditory scene analysis (e.g., Elhilali, Ma, Micheyl, Oxenham, & Shamma, 2009), and the field of computational auditory scene analysis (CASA) has emerged in part as an effort to move towards more principled, and less heuristic, approaches to understanding



An infant with a cochlear implant. [Image: Bjorn Knetsch, <https://goo.gl/J2wCvj>, CC BY 2.0, <https://goo.gl/BRvSA7>]

the parsing and perception of complex auditory scenes (e.g., Wang & Brown, 2006). Solving this problem will not only provide us with a better understanding of human auditory perception, but may provide new approaches to “smart” hearing aids and cochlear implants, as well as automatic speech recognition systems that are more robust to background noise.

Conclusion

Hearing provides us with our most important connection to the people around us. The intricate physiology of the

auditory system transforms the tiny variations in air pressure that reach our ear into the vast array of auditory experiences that we perceive as speech, music, and sounds from the environment around us. We are only beginning to understand the basic principles of neural coding in higher stages of the auditory system, and how they relate to perception. However, even our rudimentary understanding has improved the lives of hundreds of thousands through devices such as cochlear implants, which re-create some of the ear's functions for people with profound hearing loss.

Outside Resources

Audio: Auditory Demonstrations from Richard Warren's lab at the University of Wisconsin, Milwaukee

<http://www4.uwm.edu/APL/demonstrations.html>

Audio: Auditory Demonstrations. CD published by the Acoustical Society of America (ASA). You can listen to the demonstrations here

<http://www.feilding.net/sfuad/musi3012-01/demos/audio/>

Web: Demonstrations and illustrations of cochlear mechanics can be found here

<http://lab.rockefeller.edu/hudspeth/graphicalSimulations>

Web: More demonstrations and illustrations of cochlear mechanics

<http://www.neurophys.wisc.edu/animations/>

Discussion Questions

1. Based on the available acoustic cues, how good do you think we are at judging whether a low-frequency sound is coming from in front of us or behind us? How might we solve this problem in the real world?
2. Outer hair cells contribute not only to amplification but also to the frequency tuning in the cochlea. What are some of the difficulties that might arise for people with cochlear hearing loss, due to these two factors? Why do hearing aids not solve all these problems?
3. Why do you think the auditory system has so many stages of processing before the signals reach the auditory cortex, compared to the visual system? Is there a difference in the speed of processing required?

Vocabulary

Cochlea

Snail-shell-shaped organ that transduces mechanical vibrations into neural signals.

Interaural differences

Differences (usually in time or intensity) between the two ears.

Pinna

Visible part of the outer ear.

Tympanic membrane

Ear drum, which separates the outer ear from the middle ear.

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Taste and Smell

Linda Bartoshuk & Derek Snyder

Humans are omnivores (able to survive on many different foods). The omnivore's dilemma is to identify foods that are healthy and avoid poisons. Taste and smell cooperate to solve this dilemma. Stimuli for both taste and smell are chemicals. Smell results from a biological system that essentially permits the brain to store rough sketches of the chemical structures of odor stimuli in the environment. Thus, people in very different parts of the world can learn to like odors (paired with calories) or dislike odors (paired with nausea) that they encounter in their worlds. Taste information is preselected (by the nature of the receptors) to be relevant to nutrition. No learning is required; we are born loving sweet and hating bitter. Taste inhibits a variety of other systems in the brain. Taste damage releases that inhibition, thus intensifying sensations like those evoked by fats in foods. Ear infections and tonsillectomies both can damage taste. Adults who have experienced these conditions experience intensified sensations from fats and enhanced palatability of high-fat foods. This may explain why individuals who have had ear infections or tonsillectomies tend to gain weight.

Learning Objectives

- Explain the salient properties of taste and smell that help solve the omnivore's dilemma.
- Distinguish between the way pleasure/displeasure is produced by smells and tastes.
- Explain how taste damage can have extensive unexpected consequences.

The Omnivore's Dilemma

Humans are **omnivores**. We can survive on a wide range of foods, unlike species, such as koalas, that have a highly specialized diet (for koalas, eucalyptus leaves). With our amazing dietary range comes a problem: the omnivore's dilemma (Pollan, 2006; Rozin & Rozin, 1981). To survive, we must identify healthy food and avoid poisons. The senses of taste and smell cooperate to give us this ability. Smell also has other important functions in lower animals (e.g., avoid predators, identify sexual partners), but these functions are less important in humans. This module will focus on the way taste and smell interact in humans to solve the omnivore's dilemma.



By just examining our teeth, it is apparent that we are animals who evolved to eat a wide range of foods. [Image: Justin Mclean, <https://goo.gl/ffk7ZV>, CC BY-NC 2.0, <https://goo.gl/VnKlK8>]

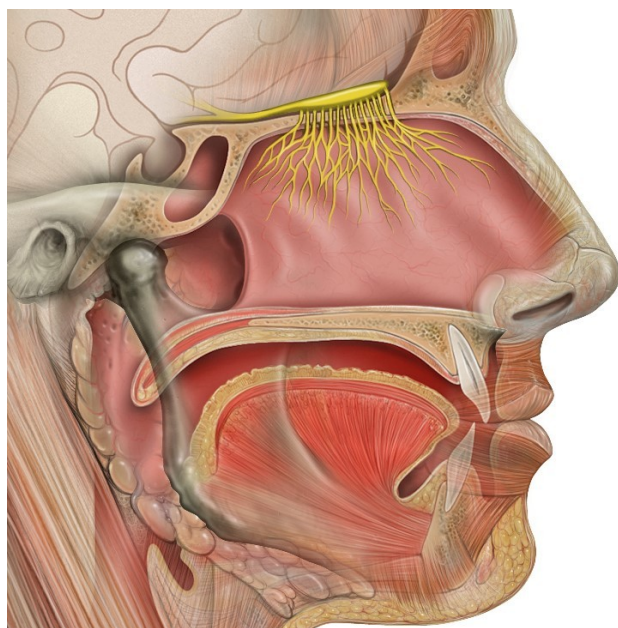
Taste and Smell Anatomy

Taste (**gustation**) and smell (**olfaction**) are both chemical senses; that is, the stimuli for these senses are chemicals. The more complex sense is olfaction. Olfactory receptors are complex proteins called G protein-coupled receptors (GPCRs). These structures are proteins that weave back and forth across the membranes of olfactory cells seven times, forming structures outside the cell that sense odorant molecules and structures inside the cell that activate the neural message ultimately conveyed to the brain by olfactory neurons. The structures that sense odorants can be thought of as tiny binding pockets with sites that respond to active parts of molecules (e.g., carbon chains). There are about 350 functional olfactory genes in humans; each gene expresses a particular kind of olfactory receptor. All olfactory receptors of a given kind project to structures called glomeruli (paired clusters of cells found on both sides of the brain). For a single molecule, the pattern of activation across the glomeruli paints

a picture of the chemical structure of the molecule. Thus, the olfactory system can identify a vast array of chemicals present in the environment. Most of the odors we encounter are actually mixtures of chemicals (e.g., bacon odor). The olfactory system creates an image for the mixture and stores it in memory just as it does for the odor of a single molecule (Shepherd, 2005).

Taste is simpler than olfaction. Bitter and sweet utilize GPCRs, just as olfaction does, but the number of different receptors is much smaller. For bitter, 25 receptors are tuned to different chemical structures (Meyerhof et al., 2010). Such a system allows us to sense many different poisons.

Sweet is even simpler. The primary sweet receptor is composed of two different G protein-coupled receptors; each of these two proteins ends in large structures reminiscent of Venus flytraps. This complex receptor has multiple sites that can bind different structures. The Venus flytrap endings open so that even some very large molecules can fit inside and stimulate the receptor.



Although smell plays a less integral role in our lives than it does for other animals, it is highly important in determining taste. In fact, if you plug your nose while eating chocolate, you will have a difficult time distinguishing it from any other kind of sweet. [Image: Patrick J. Lynch, <https://goo.gl/d5Clff>, CC BY 2.5, <https://goo.gl/0QtWcf>]

Bitter is inclusive (i.e., multiple receptors tuned to very different chemical structures feed into common neurons). Sweet is exclusive. There are many sugars with similar structures, but only three of these are particularly important to humans (sucrose, glucose, and fructose). Thus, our sweet receptor tunes out most sugars, leaving only the most important to stimulate the sweet receptor. However, the ability of the sweet receptor to respond to some non-sugars presents us with one of the great mysteries of taste. Several non-sugar molecules can stimulate the primary sweet receptor (e.g., saccharine, aspartame, cyclamate). These have given rise to the artificial sweetener industry, but their biological significance is unknown. What biological purpose is served by allowing these non-sugar molecules to stimulate the primary sweet receptor?

Some would have us believe that artificial sweeteners are a boon to those who want to lose weight. It seems like a no-brainer. Sugars have calories; saccharin does not. Theoretically, if we replace sugar with saccharin in our diets, we will lose weight. In fact, recent work showed that rats actually gained weight when saccharin was substituted for glucose (Swithers & Davidson, 2008). It turns out that substituting saccharin for sugar can increase appetite so more is eaten later. In addition, eating artificial sweeteners appears to alter metabolism, thus making losing weight even harder. So why did nature give us artificial sweeteners? We don't know.

One more mystery about sweet deserves comment. The discovery of the sweet receptor was met with great excitement because many investigators had searched for it for years. The fact that this complex receptor had multiple sites to which different molecules could bind explained why many different molecules taste sweet. However, this is actually a serious problem. No matter what molecule stimulates this receptor, the neural output from that receptor is the same. This would mean that the sweetness of all sweet substances would have to be the same. Yet artificial sweeteners do not taste exactly like sugar. The answer may lie in the fact that one of the two proteins that makes up the receptor can act alone, but only strong concentrations of sugar stimulate this isolated protein receptor. This permits the brain to distinguish between the sweetness of sugar and the sweetness of non-sugar molecules.

Salty and sour are the simplest tastes; these stimuli ionize (break into positively and negatively charged particles). The first event in the transduction series is the movement of the positively charged particle through channels in the taste cell membrane (Chaudhari & Roper, 2010).

Solving the omnivore's dilemma: Taste affect is hard-wired

The pleasure associated with sweet and salty and the displeasure associated with sour and bitter are hard-wired in the brain. Newborns love sweet (taste of mother's milk) and hate bitter (poisons) immediately. The receptors mediating salty taste are not mature at birth in humans, but when they are mature a few weeks after birth, the baby



Just how biologically ingrained is our love for sweets? Newborn infants immediately love sweet tastes, while the taste for salty foods takes longer to develop. [Image: shingleback, <https://goo.gl/fbUH3r>, CC BY 2.0, <https://goo.gl/BRvSA7>]

likes dilute salt (although more concentrated salt will evoke stinging sensations that will be avoided). Sour is generally disliked (protecting against tissue damage from acid?), but to the amazement of many parents, some young children appear to actually like the sour candies available today; this may be related to the breadth of their experience with fruits (Liem & Mennella, 2003). This hard-wired affect is the most salient characteristic of taste and this is why we classify only those taste qualities with hard-wired affect as “basic tastes.”

Another contribution to the omnivore’s dilemma: Olfactory affect is learned

The biological functions of olfaction depend on how odors enter our noses. Sniffing brings odorants through our nostrils. The odorants hit the turbinate bones and a puff of the odorized air rises to the top of the nasal cavity, where it goes through a narrow opening (the olfactory cleft) and arrives at the olfactory mucosa (the tissue that houses the olfactory receptors). Technically, this is called “orthonasal olfaction.” Orthonasal olfaction tells us about the world external to our bodies.

When we chew and swallow food, the odorants emitted by the food are forced up behind the palate (roof of the mouth) and enter our noses from the back; this is called “retronasal olfaction.” Ortho and retronasal olfaction involve the same odor molecules and the same olfactory receptors; however, the brain can tell the difference between the two and does not send the input to the same areas. Retronasal olfaction and taste project to some common areas where they are presumably integrated into flavor. Flavors tell us about the food we are eating.

If retronasal olfaction is paired with nausea, the food evoking the retronasal olfactory sensation becomes disliked. If retronasal olfaction is paired with situations the brain deems valuable (calories, sweet taste, pleasure from other sources, etc.), the food evoking that sensation becomes liked. These are called conditioned aversions and preferences (Rozin & Vollmecke, 1986).

Those who have experienced a conditioned aversion may have found that the dislike (even disgust) evoked when a flavor is paired with nausea can generalize to the smell of the food alone (orthonasal olfaction). Some years ago, Jeremy Wolfe and Linda Bartoshuk surveyed conditioned aversions among college students and staff that had resulted from consuming foods/beverages associated with nausea (Bartoshuk & Wolfe, 1990). In 29% of the aversions, subjects reported that even the smell of the food/beverage had become aversive. Other properties of food objects can become aversive as well. In one unusual case, an aversion to



There are few associations more powerful than the taste of food and an experience of sickness. In fact, many people will go their whole lives without ever trying a food again that once made them sick. [Image: James Palinsad, <https://goo.gl/r2Bph4>, CC BY-ND 2.0, <https://goo.gl/VnHMcZ>]

cheese crackers generalized to vanilla wafers apparently because the containers were similar. Conditioned aversions function to protect us from ingesting a food that our brains associate with illness. Conditioned preferences are harder to form, but they help us learn what is safe to eat.

Is the affect associated with olfaction ever hard-wired? Pheromones are said to be olfactory molecules that evoke specific behaviors. Googling “human pheromone” will take you to websites selling various sprays that are supposed to make one more sexually appealing. However, careful research does not support such claims in humans or any other mammals (Doty, 2010). For example, amniotic fluid was at one time believed to contain a pheromone

that attracted rat pups to their mother’s nipples so they could suckle. Early interest in identifying the molecule that acted as that pheromone gave way to understanding that the behavior was learned when a novel odorant, citral (which smells like lemons), was easily substituted for amniotic fluid (Pedersen, Williams, & Blass, 1982).

Central interactions: Key to understanding taste damage

The integration of retronasal olfaction and taste into flavor is not the only central interaction among the sensations evoked by foods. These integrations in most cases serve important biological functions, but occasionally they go awry and lead to clinical pathologies.

Taste is mediated by three cranial nerves; these are bilateral nerves, each of which innervates one side of the mouth. Since they do not connect in the peripheral nervous system, interactions across the midline must occur in the brain. Incidentally, studying interactions across the midline is a classic way to draw inferences about central interactions. Insights from studies of this type were very important to understanding central processes long before we had direct imaging of brain function.

Taste on the anterior two thirds of the tongue (the part you can stick out) is mediated by the chorda tympani nerve; taste on the posterior one third (the part that stays attached) is mediated by the glossopharyngeal nerve. Taste buds are tiny clusters of cells (like the segments of an orange) that are buried in the tissue of some papillae, the structures that give the tongue its bumpy appearance. Filiform papillae are the smallest and are distributed all over the tongue; they have no taste buds. In species like the cat, the filiform papillae are shaped like small spoons and help the cat hold liquids on the tongue while lapping (try lapping from a dish and you will see how hard it is without those special filiform papillae). Fungiform papillae (given this name because they resemble small button mushrooms) are larger circular structures on the anterior tongue (innervated by the chorda tympani). They contain about six taste buds each. Fungiform papillae can be seen with the naked eye, but swabbing blue food coloring on the tongue helps. The fungiform papillae do not stain as well as the rest of the tongue so they look like pink circles against a blue background. On some tongues, the spacing of fungiform papillae is like polka dots. Other tongues can have 10 times as many fungiform papillae, spaced so closely that there is little space between them. There is a connection between the density of fungiform papillae and the perception of taste. Those who experience the most intense taste sensations (we call them supertasters) tend to have the most fungiform papillae. Incidentally, this is a rare example in sensory processes of visible anatomical variation that correlates with function. We can look at the tongues of a variety of individuals and predict which of them will experience the most intense taste sensations.

The structures that house taste buds innervated by the glossopharyngeal nerve are called circumvallate papillae. They are relatively large structures arrayed in an inverted V shape across the back of the tongue. Each of them looks like a small island surrounded by a moat.

Taste nerves project to the brain, where they send inhibitory signals to one another. One of the biological consequences of this inhibition is taste constancy. Damage to one nerve reduces taste input but also reduces inhibition on the other nerves (Bartoshuk et al 2005). That release of inhibition intensifies the central neural



Approximately 25% of people are “non-tasters,” 50% are “medium tasters,” and another 25% are “super tasters.” The better the “taster” one is, the less he or she can typically tolerate spicy foods, salt, and the taste of alcohol. [Image: Rosie Rogers, <https://goo.gl/j7SepS>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

signals from the undamaged nerves, thereby maintaining whole mouth function. Interestingly, this release of inhibition can be so powerful that it actually increases whole mouth taste. The small effect of limited taste damage is one of the earliest clinical observations. In 1825, Brillat-Savarin described in his book *The Physiology of Taste* an interview with an ex-prisoner who had suffered a horrible punishment: amputation of his tongue. "This man, whom I met in Amsterdam, where he made his living by running errands, had had some education, and it was easy to communicate with him by writing. After I had observed that the forepart of his tongue has been cut off clear to the ligament, I asked him if he still found any flavor in what he ate, and if his sense of taste had survived the cruelty to which he had been subjected. He replied that ... he still possessed the ability to taste fairly well" (Brillat-Savarin, 1971, pg. 35). This injury damaged the chorda tympani but spared the glossopharyngeal nerve.

We now know that taste nerves not only inhibit one another but also inhibit other oral sensations. Thus, taste damage can intensify oral touch (fats) and oral burn (chilis). In fact, taste damage appears to be linked to pain in general. Consider an animal injured in the wild. If pain reduced eating, its chance of survival would be diminished. However, nature appears to have wired the brain such that taste input inhibits pain. Eating is reinforced and the animal's chances of survival increase.

Taste damage and weight gain

The effects of taste damage depend on the extent of damage. If only one taste nerve is damaged, then release of inhibition occurs. If the damage is extensive enough, function is lost with one possible exception. Preliminary data suggest that the more extensive the damage to taste, the greater the intensification of pain; this is obviously of clinical interest.

Damage to a single taste nerve can intensify oral touch (e.g., the creamy, viscous sensations evoked by fats). Perhaps most surprising, damage to a single taste nerve can intensify retronasal olfaction; this may occur as a secondary result from the intensification of whole mouth taste.

These sensory changes can alter the palatability of foods; in particular, high-fat foods can be rendered more palatable. Thus one of the first areas we examined was the possibility that mild taste damage could lead to increases in body mass index. Middle ear infections (otitis media) can damage the chorda tympani nerve; a tonsillectomy can damage the glossopharyngeal nerve. Head trauma damages both nerves, although it tends to take its greatest toll on the chorda tympani nerve. All of these clinical conditions increase body mass index in some individuals. More work is needed, but we suspect a link between the

intensification of fat sensations, enhancement of palatability of high-fat foods, and weight gain.

Outside Resources

Video: Inside the Psychologists Studio with Linda Bartoshuk

<http://www.youtube.com/watch?v=fO76jgepo74>

Video: Linda Bartoshuk at Nobel Conference 46

http://www.youtube.com/watch?v=l4tbg_yb4Ms

Video: Test your tongue: the science of taste

<http://www.youtube.com/watch?v=KvP7xGBzZNS>

Discussion Questions

1. In this module, we have defined “basic tastes” in terms of whether or not a sensation produces hard-wired affect. Can you think of any other definitions of basic tastes?
2. Do you think omnivores, herbivores, or carnivores have a better chance at survival?
3. Olfaction is mediated by one cranial nerve. Taste is mediated by three cranial nerves. Why do you think evolution gave more nerves to taste than to smell? What are the consequences of this?

Vocabulary

Conditioned aversions and preferences

Likes and dislikes developed through associations with pleasurable or unpleasurable sensations.

Gustation

The action of tasting; the ability to taste.

Olfaction

The sense of smell; the action of smelling; the ability to smell.

Omnivore

A person or animal that is able to survive by eating a wide range of foods from plant or animal origin.

Orthonasal olfaction

Perceiving scents/smells introduced via the nostrils.

Retronasal olfaction

Perceiving scents/smells introduced via the mouth/palate.

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Touch and Pain

Guro E. Løseth, Dan-Mikael Ellingson & Siri Leknes

The sensory systems of touch and pain provide us with information about our environment and our bodies that is often crucial for survival and well-being. Moreover, touch is a source of pleasure. In this module, we review how information about our environment and our bodies is coded in the periphery and interpreted by the brain as touch and pain sensations. We discuss how these experiences are often dramatically shaped by top-down factors like motivation, expectation, mood, fear, stress, and context. When well-functioning, these circuits promote survival and prepare us to make adaptive decisions. Pathological loss of touch can result in perceived disconnection from the body, and insensitivity to pain can be very dangerous, leading to maladaptive hazardous behavior. On the other hand, chronic pain conditions, in which these systems start signaling pain in response to innocuous touch or even in the absence of any observable sensory stimuli, have tremendous negative impact on the lives of the affected. Understanding how our sensory-processing mechanisms can be modulated psychologically and physiologically promises to help researchers and clinicians find new ways to alleviate the suffering of chronic-pain patients

Learning Objectives

- Describe the transduction of somatosensory signals: The properties of the receptor types as well as the difference in the properties of C-afferents and A-afferents and what functions these are thought to have.
- Describe the social touch hypothesis and the role of affective touch in development and bonding.
- Explain the motivation–decision model and descending modulation of pain, and give examples on how this circuitry can promote survival.
- Explain how expectations and context affect pain and touch experiences.

- Describe the concept of chronic pain and why treatment is so difficult.

Introduction

Imagine a life free of pain. How would it be—calm, fearless, serene? Would you feel invulnerable, invincible? Getting rid of **pain** is a popular quest—a quick search for “pain-free life” on Google returns well over 4 million hits—including links to various bestselling self-help guides promising a pain-free life in only 7 steps, 6 weeks, or 3 minutes. Pain management is a billion-dollar market, and involves much more than just pharmaceuticals. Surely a life with no pain would be a better one?



Figure 1A: Patient with HSAN-V genetic mutation affecting pain nerve growth. Severely affected 12-year-old boy with damages to his left knee and ankles. (Minde et al., 2004) [Used with permission]

Well, consider one of the “lucky few”: 12-year-old “Thomas” has never felt deep pain. Not even when a fracture made him walk around with one leg shorter than the other, so that the bones of his healthy leg were slowly crushed to destruction underneath the knee joint (see Figure 1A). For Thomas and other members of a large Swedish family, life without pain is a harsh reality because of a mutated gene that affects the growth of the nerves conducting deep pain. Most of those affected suffer from joint damage and frequent fractures to bones in their feet and hands; some end up in wheelchairs even before they reach puberty (Minde et al., 2004). It turns out pain—generally—serves us well.

Living without a sense of touch sounds less attractive than being free of pain—touch is a source of pleasure and essential to how we feel. Losing the sense of touch has severe implications—something patient G. L. experienced when an antibiotics treatment damaged the type of nerves that signal touch from her skin and the position of her joints and muscles. She reported feeling like she’d lost her physical self from her nose down, making her “disembodied”—like she no longer had any connection to the body attached to her head.

If she didn't look at her arms and legs they could just "wander off" without her knowing—initially she was unable to walk, and even after she relearned this skill she was so dependent on her visual attention that closing her eyes would cause her to land in a hopeless heap on the floor. Only light caresses like those from her children's hands can make her feel she has a body, but even these sensations remain vague and elusive (Olausson et al., 2002; Sacks, 1985).

Sensation

Cutaneous Senses of the Skin Connect the Brain to the Body and the Outside World

Touch and pain are aspects of the somatosensory system, which provides our brain with information about our own body (interoception) and properties of the immediate external world (exteroception) (Craig, 2002). We have somatosensory receptors located all over the body, from the surface of our skin to the depth of our joints. The information they send to the central nervous system is generally divided into four modalities: cutaneous senses (senses of the skin), proprioception (body position), kinesthesia (body movement), and nociception (pain, discomfort). We are going to focus on the cutaneous senses, which respond to tactile, thermal, and pruritic (itchy) stimuli, and events that cause tissue damage (and hence pain). In addition, there is growing evidence for a fifth modality specifically channeling *pleasant* touch (McGlone & Reilly, 2010).

Different Receptor Types Are Sensitive to Specific Stimuli

The skin can convey many sensations, such as the biting cold of a wind, the comfortable pressure of a hand holding yours, or the irritating itch from a woolen scarf. The different types of information activate specific receptors that convert the stimulation of the skin to electrical nerve impulses, a process called transduction. There are three main groups of receptors in our skin: *mechanoreceptors*, responding to mechanical stimuli, such as stroking, stretching, or vibration of the skin; *thermoreceptors*, responding to cold or hot temperatures; and *chemoreceptors*, responding to certain types of chemicals either applied externally or released within the skin (such as histamine from an inflammation). For an overview of the different receptor types and their properties, see Box 1. The experience of *pain* usually starts with activation of nociceptors—receptors that fire specifically to potentially tissue-damaging stimuli. Most of the nociceptors are subtypes of either chemoreceptors or mechanoreceptors. When tissue is damaged or inflamed, certain chemical substances are released from the cells,

and these substances activate the chemosensitive nociceptors. Mechanoreceptive nociceptors have a high threshold for activation—they respond to mechanical stimulation that is so intense it might damage the tissue.

Box 1. Categories of low-threshold mechanoreceptors*

Identity of receptor	Size of receptor*	Type of skin where found	Speed of adaptation*	Adequate stimulus*
Merkel's disks	Small, sharp borders	Glabrous*	Slow	Pressure
Meissner's corpuscles	Small, sharp borders	Glabrous	Rapid	Indentation
Ruffini corpuscles	Large, diffuse borders	Hairy + glabrous	Slow	Stretching
Pacinian corpuscles	Large, diffuse borders	Hairy + glabrous	Rapid	Vibration

* **Adequate stimulus:** The type of stimulus that the receptor is specialized to receive and respond to.

* **Glabrous Skin:** The hairless skin found on our palms and the soles of our feet. This skin has a higher density of receptors of a more complex range, which reflects the fact that we use these areas of our body to actively explore our surroundings and to discriminate tactile properties of objects we're interacting with.

* **Low-threshold mechanoreceptors:** Mechanoreceptors that respond to stimulus that is so light it doesn't threaten to damage the tissue around it. High-threshold mechanoreceptors respond to stimulation of higher intensity, and are a type of nociceptor.

* **Receptive field:** The space of skin or tissue in which stimulation will elicit a response in the receptor. Smaller receptive fields make the receptor more sensitive to details.

* **Speed adaptation:** Slowly adapting mechanoreceptors continue to fire action potentials during sustained stimulation. Rapidly adapting mechanoreceptors fire action potentials in response to stimulus onset and offset (i.e. to stimuli changes), and help detect stimulus movement on the skin.

Action Potentials in the Receptor Cells Travel as Nerve Impulses with Different Speeds

When you step on a pin, this activates a host of mechanoreceptors, many of which are nociceptors. You may have noticed that the sensation changes over time. First you feel a sharp

stab that propels you to remove your foot, and only then you feel a wave of more aching pain. The sharp stab is signaled via fast-conducting **A-fibers**, which project to the **somatosensory cortex**. This part of the cortex is **somatotopically organized**—that is, the sensory signals are represented according to where in the body they stem from (see Illustration, Figure 2). The unpleasant ache you feel after the sharp pin stab is a separate, simultaneous signal sent from the nociceptors in your foot via thin **C-pain or A δ -fibers** to the insular cortex and other brain regions involved in processing of emotion and interoception (see Figure 3a for a schematic representation of this pathway). The experience of stepping on a pin is, in other words, composed by two separate signals: one discriminatory signal allowing us to localize the touch stimulus and distinguish whether it's a blunt or a sharp stab; and one affective signal that lets us know that stepping on the pin is bad. It is common to divide pain into sensory-discriminatory and affective-motivational aspects (Auvray, Myin, & Spence, 2010). This

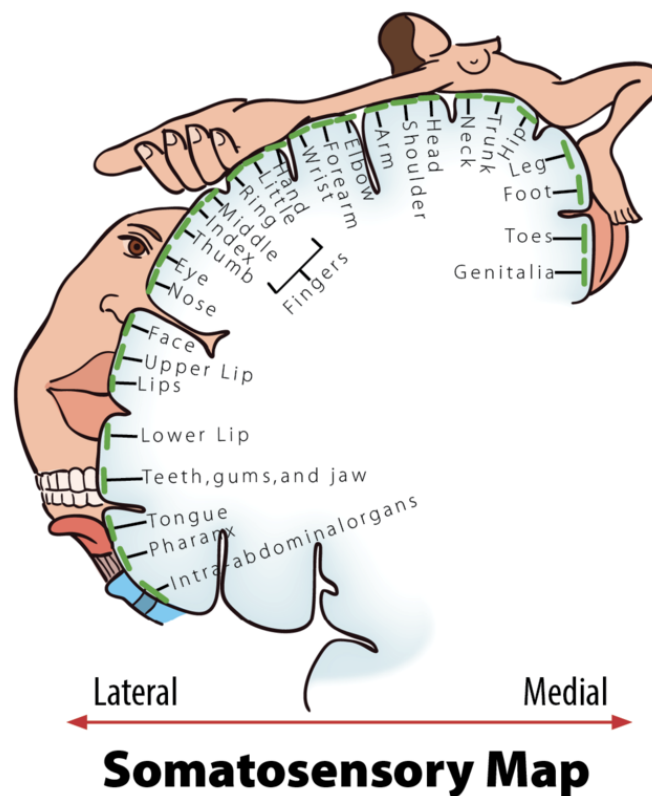


Figure 2: Somatosensory map: Body parts are represented in specific locations on the somatosensory cortex. Representations map out somatotopically, with the feet located medially and shoulders and arms laterally to the interhemispheric fissure. Facial structures are represented in a different location to the scalp and head; the face oriented upside down with the forehead pointing towards the shoulders.

distinction corresponds, at least partly, to how this information travels from the peripheral to the central nervous system and how it is processed in the brain (Price, 2000).

Affective Aspects of Touch Are Important for Development and Relationships

Touch senses are not just there for discrimination or detection of potentially painful events, as Harlow and Suomi (1970) demonstrated in a series of heartbreaking experiments where baby monkeys were taken from their mothers. The infant monkeys could choose between two artificial surrogate mothers—one “warm” mother without food but with a furry, soft cover; and one cold, steel mother with food. The monkey babies spent most of their time clinging to the soft mother, and only briefly moved over to the hard, steel mother to feed, indicating that touch is of “overpowering importance” to the infant (Harlow & Suomi, 1970, p. 161). Gentle touch is central for creating and maintaining social relationships in primates; they groom each other by stroking the fur and removing parasites—an activity important not only for their individual well-being but also for group cohesion (Dunbar, 2010; Keverne, Martensz, & Tuite, 1989). Although people don’t groom each other in the same way, gentle touch is important for us, too.

The sense of touch is the first to develop while one is in the womb, and human infants crave touch from the moment they’re born. From studies of human orphans, we know that touch is also crucial for human development. In Romanian orphanages where the babies were fed but not given regular attention or physical contact, the children suffered cognitive and neurodevelopmental delay (Simons & Land, 1987). Physical contact helps a crying baby calm down, and the soothing touch a mother gives to her child is thought to reduce the levels of stress hormones such as cortisol. High levels of cortisol have negative effects on neural development, and they can even lead to cell loss (Feldman, Singer, & Zagoory, 2010; Fleming, O’Day, & Kraemer, 1999; Pechtel & Pizzagalli, 2011). Thus, stress reduction through hugs and caresses might be important not only for children’s well-being, but also for the development of the infant brain.

The skin senses are similar across species, likely reflecting the evolutionary advantage of being able to tell what is touching you, where it’s happening, and whether or not it’s likely to cause tissue damage. An intriguing line of touch research suggests that humans, cats, and other animals have a special, evolutionarily preserved system that promotes gentle touch because it carries social and emotional significance. On a peripheral level, this system consists of a subtype of C-fibers that responds not to painful stimuli, but rather to gentle stroking touch—called C-tactile fibers. The firing rate of the C-tactile fibers correlates closely with how

pleasant the stroking feels—suggesting they are coding specifically for the gentle caresses typical of social affiliative touch (Löken, Wessberg, Morrison, McGlone, & Olausson, 2009). This finding has led to the **social touch hypothesis**, which proposes that C-tactile fibers form a system for touch perception that supports social bonding (Morrison, Löken, & Olausson, 2010; Olausson, Wessberg, Morrison, McGlone, & Vallbo, 2010). The discovery of the C-tactile system suggests that touch is organized in a similar way to pain; fast-conducting A-fibers contribute to sensory-discriminatory aspects, while thin C-fibers contribute to affective-motivational aspects (Löken, Wessberg, Morrison, McGlone, & Olausson, 2009). However, while these “hard-wired” afferent systems often provide us with accurate information about our environment and our bodies, how we experience touch or pain depends very much on top-down sources like motivation, expectation, mood, fear, and stress.

Modulation

Pain Is Necessary for Survival, but Our Brain Can Stop It if It Needs To

In April 2003, the climber Aron Ralston found himself at the floor of Blue John Canyon in Utah, forced to make an appalling choice: face a slow but certain death—or amputate his right arm. Five days earlier he fell down the canyon—since then he had been stuck with his right arm trapped between an 800-lb boulder and the steep sandstone wall. Weak from lack of food and water and close to giving up, it occurred to him like an epiphany that if he broke the two bones in his forearm he could manage to cut off the rest with his pocket knife. The thought of freeing himself and surviving made him so excited he spent the next 40 minutes completely engrossed in the task: first snapping his bones using his body as a lever, then sticking his fingers into the arm, pinching bundles of muscle fibers and severing them one by one, before cutting the blue arteries and the pale “noodle-like” nerves. The pain was unimportant. Only cutting through the thick white main nerve made him stop for a minute—the flood of pain, he describes, was like thrusting his entire arm “into a cauldron of magma.” Finally free, he rappelled down a cliff and walked another 7 miles until he was rescued by some hikers (Ralston, 2010). How is it possible to do something so excruciatingly painful to yourself, and still manage to walk, talk, and think rationally afterwards? The answer lies within the brain, where signals from the body are interpreted. When we perceive somatosensory and nociceptive signals from the body, the experience is highly subjective and malleable by motivation, attention, emotion, and context.

The Motivation–Decision Model and Descending Modulation of Pain

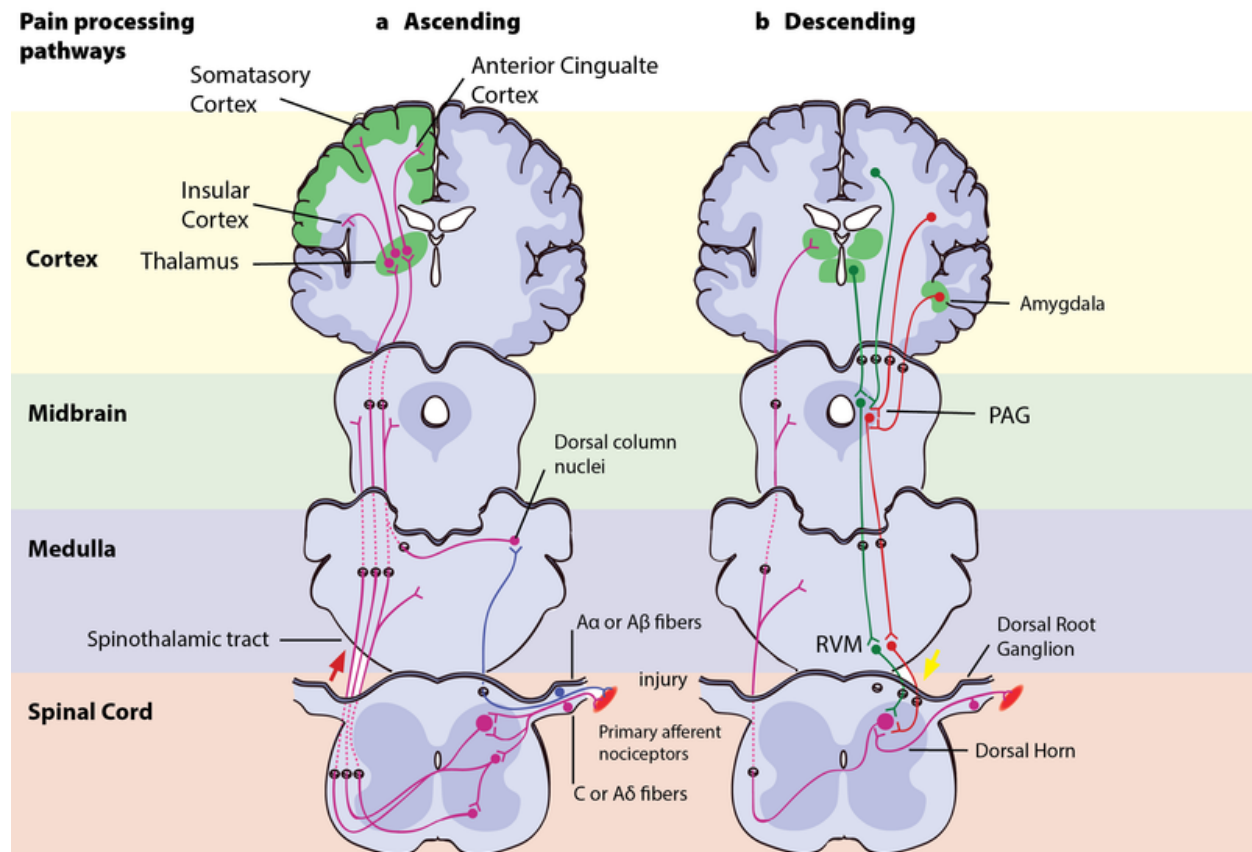


Figure 3: Pain processing pathways Left - Ascending pain pathways: An injury is signaled simultaneously via fast-conducting Aα or Aβ-fibers and slow-conducting C-pain or Aδ-fibers. The fast A-fibers signal pressure, stretching and other tissue movements to the somatosensory cortex via the dorsal column nuclei. The C-pain and Aδ-fibers send pain information from nociceptors in the tissue or skin, and transmit these signals to second order neurons in the dorsal horn of the spinal cord. The second order neurons then cross over to the opposite side, where they form the ascending spinothalamic tract. This tract projects signals to nuclei in the medulla and midbrain on the way up to the thalamus (T). The thalamus relays the information to the somatosensory and insular cortex, as well as cortical regions mediating different aspects of the pain experience such as affective responses in the cingulate cortex.

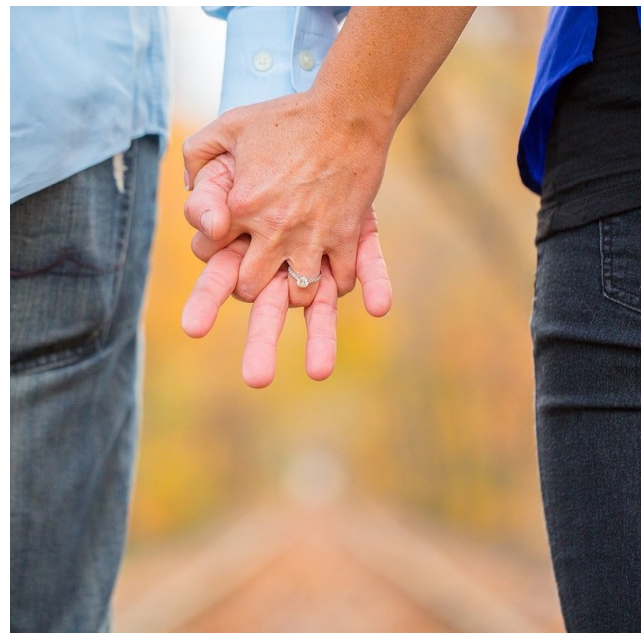
Right - Descending pain modulation pathways: Information from the environment and certain motivational states can activate this top-down pathway. Several areas in the limbic forebrain including the anterior cingulate and insular cortex, nuclei in the amygdala and the hypothalamus (H), project to the midbrain periaqueductal grey (PAG), which then modulates ascending pain transmission from the afferent pain system indirectly through the rostral ventromedial medulla (RVM) in the brainstem. This modulating system produces analgesia by the release of endogenous opioids, and uses ON- and OFF-cells to exert either inhibitory (green) or facilitatory (red) control of nociceptive signals at the spinal dorsal horn.

According to the *motivation-decision model*, the brain automatically and continuously evaluates the pros and cons of any situation—weighing impending threats and available rewards (Fields, 2004, 2006). Anything more important for survival than avoiding the pain activates the brain's **descending pain modulatory system**—a top-down system involving several parts of the brain and brainstem, which inhibits nociceptive signaling so that the more important actions can

be attended to (Figure 3b). In Aron's extreme case, his actions were likely based on such an unconscious decision process—taking into account his homeostatic state (his hunger, thirst, the inflammation and decay of his crushed hand slowly affecting the rest of his body), the sensory input available (the sweet smell of his dissolving skin, the silence around him indicating his solitude), and his knowledge about the threats facing him (death, or excruciating pain that won't kill him) versus the potential rewards (survival, seeing his family again). Aron's story illustrates the evolutionary advantage to being able to shut off pain: The descending pain modulatory system allows us to go through with potentially life-saving actions. However, when one has reached safety or obtained the reward, healing is more important. The very same descending system can then “crank up” nociception from the body to promote healing and motivate us to avoid potentially painful actions. To facilitate or inhibit nociceptive signals from the body, the descending pain modulatory system uses a set of ON- or OFF-cells in the brainstem, which regulates how much of the nociceptive signal reaches the brain. The descending system is dependent on opioid signaling, and analgesics like morphine relieve pain via this circuit (Petrovic, Kalso, Petersson, & Ingvar, 2002).

The Analgesic Power of Reward

Thinking about the good things, like his loved ones and the life ahead of him, was probably pivotal to Aron's survival. The promise of a reward can be enough to relieve pain. Expecting pain relief (getting less pain is often the best possible outcome if you're in pain, i.e., it is a reward) from a medical treatment contributes to the placebo effect—where pain relief is due at least partly to your brain's descending modulation circuit, and such relief depends on the brain's own opioid system (Eippert et al., 2009; Eippert, Finsterbusch, Bingel, & Buchel, 2009; Levine, Gordon, & Fields, 1978). Eating tasty food, listening to good music, or feeling pleasant touch on your skin also decreases pain in both animals and humans, presumably through the same mechanism in the brain (Leknes & Tracey, 2008). In a now classic experiment, Dum and Herz



Social rewards, like holding the hands or just seeing the picture of a loved one, can reduce sensations of pain. [Image: rogerl01, CC0 Public Domain, <https://goo.gl/m25gce>]

(1984) either fed rats normal rat food or let them feast on highly rewarding chocolate-covered candy (rats love sweets) while standing on a metal plate until they learned exactly what to expect when placed there. When the plate was heated up to a noxious/painful level, the rats that expected candy endured the temperature for twice as long as the rats expecting normal chow. Moreover, this effect was completely abolished when the rats' opioid (endorphin) system was blocked with a drug, indicating that the analgesic effect of reward anticipation was caused by endorphin release.

For Aron the climber, both the stress from knowing that death was impending and the anticipation of the reward it would be to survive probably flooded his brain with endorphins, contributing to the wave of excitement and euphoria he experienced while he carried out the amputation “like a five-year-old unleashed on his Christmas presents” (Ralston, 2010). This altered his experience of the pain from the extreme tissue damage he was causing and enabled him to focus on freeing himself. Our brain, it turns out, can modulate the perception of how unpleasant pain is, while still retaining the ability to experience the intensity of the sensation (Rainville, Duncan, Price, Carrier, & Bushnell, 1997; Rainville, Fine, Bushnell, & Duncan, 1992). Social rewards, like holding the hand of your boyfriend or girlfriend, have pain-reducing effects. Even looking at a picture of him/her can have similar effects—in fact, seeing a picture of a person we feel close to not only reduces subjective pain ratings, but also the activity in pain-related brain areas (Eisenberger et al., 2011). The most common things to do when wanting to help someone through a painful experience—being present and holding the person's hand—thus seems to have a measurably positive effect.

When Touch Becomes Painful or Pain Becomes Chronic

Chances are you've been sunburned a few times in your life and have experienced how even the lightest pat on the back or the softest clothes can feel painful on your over-sensitive skin. This condition, where innocuous touch gives a burning, tender sensation, is similar to a chronic condition called allodynia—where neuronal disease or injury makes touch that is normally pleasant feel unpleasantly painful. In allodynia, neuronal injury in the spinal dorsal horn causes A β -afferents, which are activated by non-nociceptive touch, to access nociceptive pathways (Liljencrantz et al., 2013). The result is that even gentle touch is interpreted by the brain as painful. While an acute pain response to noxious stimuli has a vital protective function, allodynia and other chronic pain conditions constitute a tremendous source of unnecessary suffering that affects millions of people. Approximately 100 million Americans suffer from chronic pain, and annual economic cost associated is estimated to be \$560–\$635 billion (Committee on Advancing Pain Research, Care, & Institute of Medicine, 2011). Chronic pain conditions are highly diverse, and they can involve changes on peripheral, spinal, central, and

psychological levels. The mechanisms are far from fully understood, and developing appropriate treatment remains a huge challenge for pain researchers.

Chronic pain conditions often begin with an injury to a peripheral nerve or the tissue surrounding it, releasing hormones and inflammatory molecules that *sensitize* nociceptors. This makes the nerve and neighboring afferents more excitable, so that also uninjured nerves become hyperexcitable and contribute to the persistence of pain. An injury might also make neurons fire nonstop regardless of external stimuli, providing near-constant input to the pain system. **Sensitization** can also happen in the brain and in the descending modulatory system of the brainstem (Zambreanu, Wise, Brooks, Iannetti, & Tracey, 2005). Exactly on which levels the pain perception is altered in chronic pain patients can be extremely difficult to pinpoint, making treatment an often exhausting process of trial and error. Suffering from chronic pain has dramatic impacts on the lives of the afflicted. Being in pain over a longer time can lead to depression, anxiety (fear or anticipation of future pain), and immobilization, all of which may in turn exacerbate pain (Wiech & Tracey, 2009). Negative emotion and attention to pain can increase sensitization to pain, possibly by keeping the descending pain modulatory system in facilitation mode. Distraction is therefore a commonly used technique in hospitals where patients have to undergo painful treatments like changing bandages on large burns. For chronic pain patients, however, diverting attention is not a long-term solution. Positive factors

like social support can reduce the risk of chronic pain after an injury, and so they can help to adjust to bodily change as a result of injury. We have already talked about how having a hand to hold might alleviate suffering. Chronic pain treatment should target these emotional and social factors as well as the physiological.



How powerful is the mind really? Well, the next time you have dental work done, know that some people opt for no Novocain, using only the power of their minds to overcome the pain. [Image: HypnoArt, CC0 Public Domain, <https://goo.gl/m25gce>]

The Power of the Mind

The context of pain and touch has a great impact on how we interpret it. Just imagine how different it would feel to Aron if someone amputated his hand against his will and for no discernible reason. Prolonged pain from injuries can be easier to bear if the incident causing them provides a positive context—like a war

wound that testifies to a soldier's courage and commitment—or **phantom pain** from a hand that was cut off to enable life to carry on. The relative meaning of pain is illustrated by a recent experiment, where the same moderately painful heat was administered to participants in two different contexts—one control context where the alternative was a nonpainful heat; and another where the alternative was an intensely painful heat. In the control context, where the moderate heat was the least preferable outcome, it was (unsurprisingly) rated as painful. In the other context it was the best possible outcome, and here the exact same moderately painful heat was actually rated as *pleasant*—because it meant the intensely painful heat had been avoided. This somewhat surprising change in perception—where pain becomes pleasant because it represents relief from something worse—highlights the importance of the meaning individuals ascribe to their pain, which can have decisive effects in pain treatment (Leknes et al., 2013). In the case of touch, knowing who or what is stroking your skin can make all the difference—try thinking about slugs the next time someone strokes your skin if you want an illustration of this point. In a recent study, a group of heterosexual males were told that they were about to receive sensual caresses on the leg by either a male experimenter or by an attractive female experimenter (Gazzola et al., 2012). The study participants could not see who was touching them. Although it was always the female experimenter who performed the caress, the heterosexual males rated the otherwise pleasant sensual caresses as clearly unpleasant when they believed the male experimenter did it. Moreover, brain responses to the “male touch” in somatosensory cortex were reduced, exemplifying how top-down regulation of touch resembles top-down pain inhibition.

Pain and pleasure not only share modulatory systems—another common attribute is that we don't need to be on the receiving end of it ourselves in order to experience it. How did you feel when you read about Aron cutting through his own tissue, or “Thomas” destroying his own bones unknowingly? Did you cringe? It's quite likely that some of your brain areas processing affective aspects of pain were active even though the nociceptors in your skin and deep tissue were not firing. Pain can be experienced vicariously, as can itch, pleasurable touch, and other sensations. Tania Singer and her colleagues found in an fMRI study that some of the same brain areas that were active when participants felt pain on their own skin (anterior cingulate and insula) were also active when they were given a signal that a loved one was feeling the pain. Those who were most “empathetic” also showed the largest brain responses (Singer et al., 2004). A similar effect has been found for pleasurable touch: The posterior insula of participants watching videos of someone else's arm being gently stroked shows the same activation as if they were receiving the touch themselves (Morrison, Bjornsdotter, & Olausson, 2011).

Summary

Sensory experiences connect us to the people around us, to the rest of the world, and to our own bodies. Pleasant or unpleasant, they're part of being human. In this module, we have seen how being able to inhibit pain responses is central to our survival—and in cases like that of climber Aron Ralston, that ability can allow us to do extreme things. We have also seen how important the ability to feel pain is to our health—illustrated by young “Thomas,” who keeps injuring himself because he simply doesn’t notice pain. While “Thomas” has to learn to avoid harmful activities without the sensory input that normally guides us, G. L. has had to learn how to keep approaching and move about in a world she can hardly feel at all, with a body that is practically disconnected from her awareness. Too little sensation or too much of it leads to no good, no matter how pleasant or unpleasant the sensation usually feels. As long as we have nervous systems that function normally, we are able to adjust the volume of the sensory signals and our behavioral reactions according to the context we’re in. When it comes to sensory signals like touch and pain, we are interpreters, not measuring instruments. The quest for understanding how our sensory-processing mechanisms can be modulated, psychologically and physiologically, promises to help researchers and clinicians find new ways to alleviate distress from chronic pain.

Outside Resources

Book: Butler, D. S., Moseley, G. L., & Sunyata. (2003). Explain pain (p. 19). Australia: Noigroup.

Book: Kringelbach, M. L., & Berridge, K. C. (Eds.). (2010). Pleasures of the brain (p. 343). Oxford, UK: Oxford University Press.

Book: Ralston, A. (2004). Between a rock and a hard place: The basis of the motion picture 127 Hours. New York, NY: Atria.

Book: Sacks, O. (1998). The man who mistook his wife for a hat: And other clinical tales. New York, NY: Simon & Schuster.

Video: BBC Documentary series “Human Senses,” Episode 3: Touch and Vision

http://watchdocumentary.org/watch/human-senses-episode-03-touch-and-vision-video_f3e33c14a.html

Video: BBC Documentary “Pleasure and Pain with Michael Mosley”

<http://www.bbc.co.uk/programmes/b00y377q>

Video: TEDxAdelaide - Lorimer Moseley – “Why Things Hurt”

<http://www.youtube.com/watch?v=gwd-wLdlHjs>

Video: Trailer for the film 127 Hours, directed by Danny Boyle and released in 2010

<http://www.youtube.com/watch?v=OlhLOWTnVoQ>

Web: Homepage for the International Association for the Study of Pain

<http://www.iasp-pain.org>

Web: Proceedings of the National Academy of Sciences Colloquium "The Neurobiology of Pain"

<http://www.pnas.org/content/96/14.toc#COLLOQUIUM>

Web: Stanford School of Medicine Pain Management Center

<http://paincenter.stanford.edu/>

Website resource aiming to communicate “advances and issues in the clinical sciences as they relate to the role of the brain and mind in chronic pain disorders,” led by Dr. Lorimer Moseley

<http://www.bodyinmind.org/>

Discussion Questions

1. Your friend has had an accident and there is a chance the injury might cause pain over a prolonged period. How would you support your friend? What would you say and do to ease the pain, and why do you think it would work?
2. We have learned that touch and pain sensation in many aspects do not reflect “objectively” the outside world or the body state. Rather, these experiences are shaped by various top-down influences, and they can even occur without any peripheral activation. This is similar to the way other sensory systems work, e.g., the visual or auditory systems, and seems to reflect a general way the brain process sensory events. Why do you think the brain interprets the incoming sensory information instead of giving a one-to-one readout the way a thermometer and other measuring instruments would? Imagine you instead had “direct unbiased access” between stimuli and sensation. What would be the advantages and disadvantages of this?
3. Feelings of pain or touch are subjective—they have a particular quality that you perceive subjectively. How can we know whether the pain you feel is similar to the pain I feel? Is it possible that modern scientists can objectively measure such subjective feelings?

Vocabulary

A-fibers

Fast-conducting sensory nerves with myelinated axons. Larger diameter and thicker myelin sheaths increases conduction speed. A β -fibers conduct touch signals from low-threshold mechanoreceptors with a velocity of 80 m/s and a diameter of 10 μ m; A δ -fibers have a diameter of 2.5 μ m and conduct cold, noxious, and thermal signals at 12 m/s. The third and fastest conducting A-fiber is the A α , which conducts proprioceptive information with a velocity of 120 m/s and a diameter of 20 μ m.

Allodynia

Pain due to a stimulus that does not normally provoke pain, e.g., when a light, stroking touch feels painful.

Analgesia

Pain relief.

C-fibers

C-fibers: Slow-conducting unmyelinated thin sensory afferents with a diameter of 1 μ m and a conduction velocity of approximately 1 m/s. C-pain fibers convey noxious, thermal, and heat signals; C-tactile fibers convey gentle touch, light stroking.

Chronic pain

Persistent or recurrent pain, beyond usual course of acute illness or injury; sometimes present without observable tissue damage or clear cause.

C-pain or A δ -fibers

C-pain fibers convey noxious, thermal, and heat signals

C-tactile fibers

C-tactile fibers convey gentle touch, light stroking

Cutaneous senses

The senses of the skin: tactile, thermal, pruritic (itchy), painful, and pleasant.

Descending pain modulatory system

A top-down pain-modulating system able to inhibit or facilitate pain. The pathway produces analgesia by the release of endogenous opioids. Several brain structures and nuclei are part

of this circuit, such as the frontal lobe areas of the anterior cingulate cortex, orbitofrontal cortex, and insular cortex; and nuclei in the amygdala and the hypothalamus, which all project to a structure in the midbrain called the periaqueductal grey (PAG). The PAG then controls ascending pain transmission from the afferent pain system indirectly through the rostral ventromedial medulla (RVM) in the brainstem, which uses ON- and OFF-cells to inhibit or facilitate nociceptive signals at the spinal dorsal horn.

Endorphin

An endogenous morphine-like peptide that binds to the opioid receptors in the brain and body; synthesized in the body's nervous system.

Exteroception

The sense of the external world, of all stimulation originating from outside our own bodies.

Interoception

The sense of the physiological state of the body. Hunger, thirst, temperature, pain, and other sensations relevant to homeostasis. Visceral input such as heart rate, blood pressure, and digestive activity give rise to an experience of the body's internal states and physiological reactions to external stimulation. This experience has been described as a representation of "the material me," and it is hypothesized to be the foundation of subjective feelings, emotion, and self-awareness.

Nociception

The neural process of encoding noxious stimuli, the sensory input from nociceptors. Not necessarily painful, and crucially not necessary for the experience of pain.

Nociceptors

High-threshold sensory receptors of the peripheral somatosensory nervous system that are capable of transducing and encoding noxious stimuli. Nociceptors send information about actual or impending tissue damage to the brain. These signals can often lead to pain, but nociception and pain are not the same.

Noxious stimulus

A stimulus that is damaging or threatens damage to normal tissues.

ocial touch hypothesis

Proposes that social touch is a distinct domain of touch. C-tactile afferents form a special pathway that distinguishes social touch from other types of touch by selectively firing in response to touch of social-affective relevance; thus sending affective information parallel to

the discriminatory information from the A β -fibers. In this way, the socially relevant touch stands out from the rest as having special positive emotional value and is processed further in affect-related brain areas such as the insula.

Pain

Defined as “an unpleasant sensory and emotional experience associated with actual or potential tissue damage, or described in terms of such damage,” according to the International Association for the Study of Pain.

Phantom pain

Pain that appears to originate in an amputated limb.

Placebo effect

Effects from a treatment that are not caused by the physical properties of a treatment but by the meaning ascribed to it. These effects reflect the brain's own activation of modulatory systems, which is triggered by positive expectation or desire for a successful treatment. Placebo analgesia is the most well-studied placebo effect and has been shown to depend, to a large degree, on opioid mechanisms. Placebo analgesia can be reversed by the pharmacological blocking of opioid receptors. The word “placebo” is probably derived from the Latin word “placebit” (“it will please”).

Sensitization

Increased responsiveness of nociceptive neurons to their normal input and/or recruitment of a response to normally subthreshold inputs. Clinically, sensitization may only be inferred indirectly from phenomena such as hyperalgesia or allodynia. Sensitization can occur in the central nervous system (central sensitization) or in the periphery (peripheral sensitization).

Social touch hypothesis

Proposes that social touch is a distinct domain of touch. C-tactile afferents form a special pathway that distinguishes social touch from other types of touch by selectively firing in response to touch of social-affective relevance; thus sending affective information parallel to the discriminatory information from the A β -fibers. In this way, the socially relevant touch stands out from the rest as having special positive emotional value and is processed further in affect-related brain areas such as the insula.

Somatosensory cortex

Consists of primary sensory cortex (S1) in the postcentral gyrus in the parietal lobes and secondary somatosensory cortex (S2), which is defined functionally and found in the upper bank of the lateral sulcus, called the parietal operculum. Somatosensory cortex also includes

parts of the insular cortex.

Somatotopically organized

When the parts of the body that are represented in a particular brain region are organized topographically according to their physical location in the body (see Figure 2 illustration).

Spinothalamic tract

Runs through the spinal cord's lateral column up to the thalamus. C-fibers enter the dorsal horn of the spinal cord and form a synapse with a neuron that then crosses over to the lateral column and becomes part of the spinothalamic tract.

Transduction

The mechanisms that convert stimuli into electrical signals that can be transmitted and processed by the nervous system. Physical or chemical stimulation creates action potentials in a receptor cell in the peripheral nervous system, which is then conducted along the axon to the central nervous system.

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25

The Vestibular System

Dora Angelaki & J. David Dickman

The vestibular system functions to detect head motion and position relative to gravity and is primarily involved in the fine control of visual gaze, posture, orthostasis, spatial orientation, and navigation. Vestibular signals are highly processed in many regions of the brain and are involved in many essential functions. In this module, we provide an overview of how the vestibular system works and how vestibular signals are used to guide behavior.

Learning Objectives

- Define the basic structures of the vestibular receptor system.
- Describe the neuroanatomy of the vestibuloocular, vestibulospinal, and vestibulo-thalamo-cortical pathways.
- Describe the vestibular commissural system.
- Describe the different multisensory cortical areas for motion perception.

Introduction

Remember the dizzy feeling you got as a child after you jumped off the merry-go-round or spun around like a top? These feelings result from activation of the vestibular system, which detects our movements through space but is not a conscious sense like vision or hearing. In fact, most vestibular functions are imperceptible, but vestibular-related sensations such as motion sickness can pop up rapidly when riding on a roller coaster, having a bumpy plane ride, or a sailing a boat in rough seas. However, these sensations are really side effects and

the vestibular system is actually extremely important for everyday activities, with vestibular signals being involved in much of the brain's information processing that controls such fundamental functions as balance, posture, gaze stabilization, spatial orientation, and navigation, to name a few. In many regions of the brain, vestibular information is combined with signals from the other senses as well as with motor information to give rise to motion perception, body awareness, and behavioral control. Here, we will explore the workings of the vestibular system and consider some of the integrated computations the brain performs using vestibular signals to guide our common behavior.



Although you may be most aware of the vestibular system in situations where you experience motion sickness, at other times its working in the background to provide you with essential functions like balance and stable vision. [Image: Leo Reynolds, <https://goo.gl/N8D94w>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

Structure of the vestibular receptors

The vestibular receptors lie in the inner ear next to the auditory cochlea. They detect rotational motion (head turns), linear motion (translations), and tilts of the head relative to gravity and transduce these motions into neural signals that can be sent to the brain. There are five vestibular receptors in each ear (Hearing module, Figure 1- <http://noba.to/jry3cu78>), including three semicircular canals (horizontal, anterior, and posterior) that transduce rotational angular accelerations and two otolith receptors (utricle and saccule) that transduce

linear accelerations (Lindeman, 1969). Together, the semicircular canals and otolith organs can respond to head motion and maintained static head position relative to gravity in all directions in 3D space.

These receptors are contained in a series of interconnected fluid filled tubes that are protected by a dense, overlying bone (Iurato, 1967). Each of the three semicircular canals lies in a plane that is orthogonal to the other two. The horizontal semicircular canal lies in a roughly horizontal head plane, whereas the anterior and posterior semicircular canals lie vertically in the head (Blanks, Curthoys, Bennett, & Markham, 1985). The semicircular canal receptor cells, termed **hair cells**, are located only in the middle of the circular tubes in a special epithelium, covered by a gelatinous membrane that stretches across the tube to form a fluid-tight seal like the skin of a drum (Figures 1A and 1B). Hair cells are so named due to an array of nearly 100 staggered-height **stereocilia** (like a church pipe organ) that protrude from the top of the cell into the overlying gelatin membrane (Wersäll, 1956). The shortest stereocilia are at one end of the cell and the tallest at the other (Lindeman, 1969). When the head is rotated, the fluid in the semicircular canals lags behind the head motion and pushes on the gelatin membrane, which bends the stereocilia.

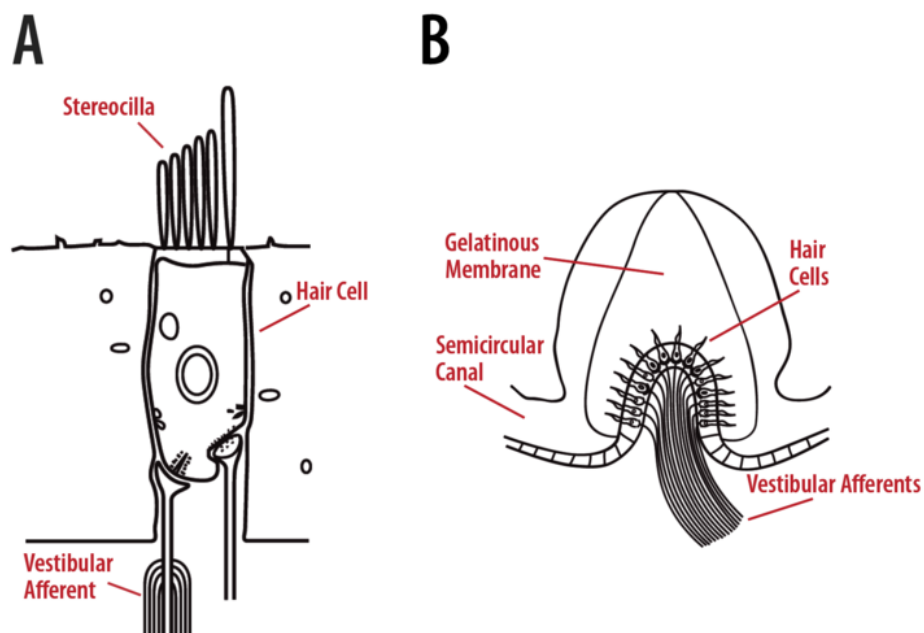


Figure 1. Receptor hair cells and semicircular canal structure. A) Vestibular hair cell in the receptor epithelium with stereocilia in the apical surface of the cell. Innervating afferent and efferent neurons make synaptic contacts with the basal surface of the cell. B) Semicircular canal structure showing the fluid duct, the hair cell stereocilia embedded in a gelatinous membrane on top of the hair cells, and the innervating afferent fibers.

As shown in Figure 2, when the head moves toward the receptor hair cells (e.g., left head turns for the left horizontal semicircular canal), the stereocilia are bent toward the tallest end and special **mechanically gated ion channels** in the tips of the cilia open, which excites (**depolarizes**) the cell (Shotwell, Jacobs, & Hudspeth, 1981). Head motion in the opposite direction causes bending toward the smallest stereocilia, which closes the channels and inhibits (**hyperpolarizes**) the cell. The left and right ear semicircular canals have opposite polarity, so for example, when you turn your head to the left, the receptors in the left horizontal semicircular canal will be excited while right ear horizontal canal receptors will be inhibited (Figure 3). The same relationship is true for the vertical semicircular canals. Vestibular **afferent nerve fibers** innervate the base of the hair cell and increase or decrease their neural firing rate as the receptor cell is excited or inhibited (Dickman and Correia, 1989), respectively, and then carry these signals regarding head rotational motion to the brain as part of the **vestibulocochlear nerve** (Cranial nerve VIII). They enter the brainstem and terminate in the ipsilateral vestibular nuclei, cerebellum, and reticular formation (Carleton & Carpenter, 1984; Dickman & Fang, 1996). The primary vestibular hair cell and afferent **neurotransmitters** are **glutamate** and **aspartate**. Due to the mechanical properties of the vestibular receptor system,

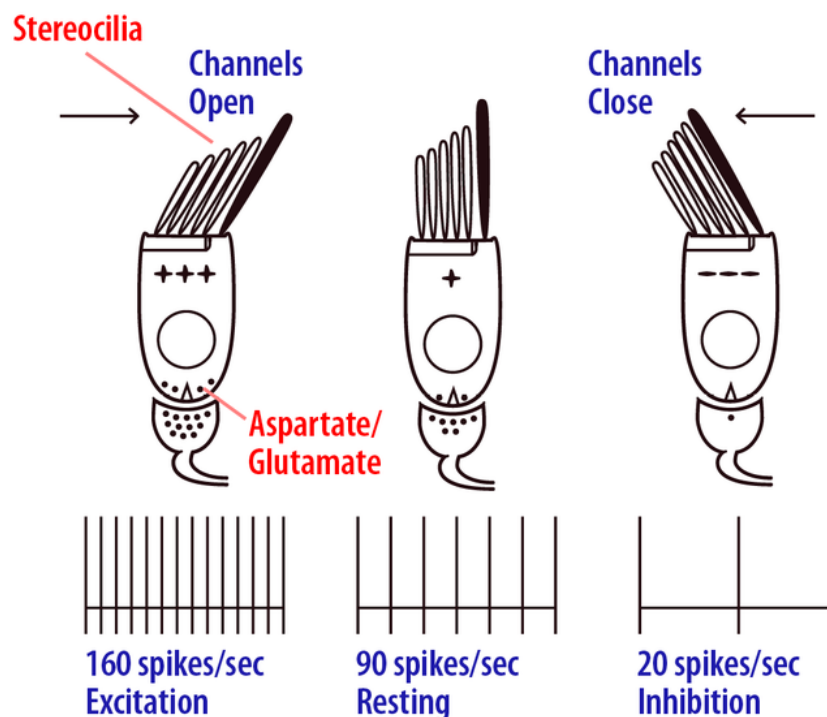


Figure 2. Vestibular receptor cell directional selectivity. Middle) At rest, hair cells release some neurotransmitter, producing a high, spontaneous firing rate in the innervating afferent fibers. Left) When the stereocilia are displaced toward the kinocilium, the cell is depolarized and the afferent firing rate is increased. Right) When the stereocilia are displaced away from the kinocilium, the cell is hyperpolarized and the afferent firing rate is decreased

rotational accelerations of the head are integrated into velocity signals (Van Egmond, Groen, & Jongkess, 1949) that are then encoded by semicircular canal afferents (Fernandez & Goldberg, 1971). **Detection thresholds** for rotational motion have shown that afferents can discriminate differences in head velocity on the order of 2 deg/sec, but also are sensitive to a broad range of natural head movements up to high head speeds in the hundreds of deg/sec (as you might experience when you make a fast head turn toward a loud sound, or are performing gymnastics; Sadeghi, Chacron, Taylor, & Cullen, 2007; Yu, Dickman, & Angelaki, 2012).

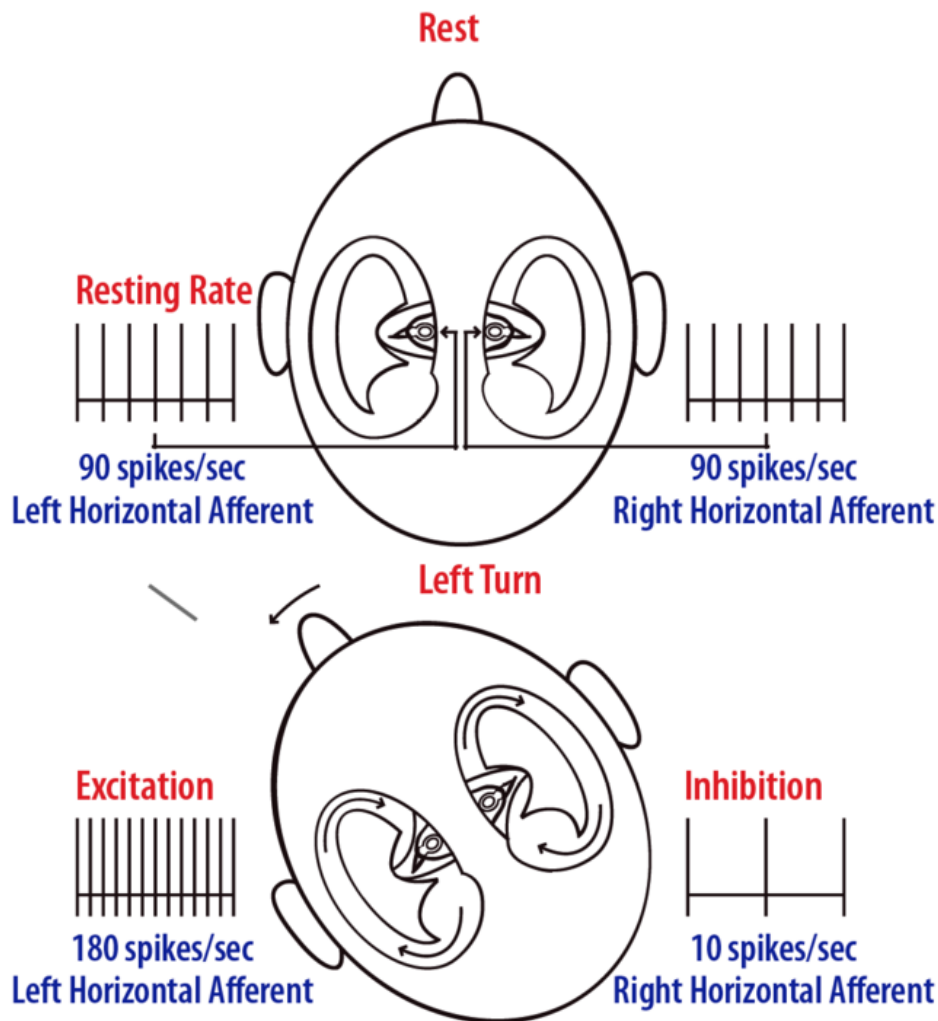


Figure 3. Receptor hair cells in the otolith organs. Receptor cells have stereocilia embedded in the gelatinous membrane, which is covered by thousands of calcium carbonate otoconia. Receptor cells are polarized in opposite directions relative to a central location and are innervated by VIIIth nerve afferent fibers.

Otolith receptors are sensitive to linear accelerations and tilts of the head relative to gravity (Fernandez & Goldberg, 1976a). The utricle otolith receptor lies parallel to the horizontal semicircular canal and the saccule receptor lies vertical in the head (Hearing module, Figure 1- <http://noba.to/jry3cu78>). As shown in Figure 4, a special otolith epithelium contains receptor hair cells whose stereocilia extend into a gelatin membrane that is covered by a layer of calcium carbonate crystals, termed **otoconia**, like rocks piled up to form a jetty (Lindeman, 1969). Otoconia are not affected by fluid movements but instead are displaced by linear accelerations, including translations (e.g., forward/backward or upward/downward motions) or changes in head position relative to gravity. These linear accelerations produce displacements of the otoconia (due to their high mass), much like rocks rolling down a hill or your coffee cup falling off the car dashboard when you push the gas pedal. Movements of the otoconia bend the hair cell stereocilia and open/close channels in a similar way to that described for the semicircular canals. However, otolith hair cells are polarized such that the tallest stereocilia are pointing toward the center of the utricle and away from the center in the saccule, which effectively splits the receptors into two opposing groups (Flock, 1964; Lindeman, 1969). In this way, some hair cells are excited and some inhibited for each linear motion force or head tilt experienced, with the population of receptors and their innervating afferents being **directionally tuned** to all motions or head tilts in 3D space (Fernandez & Goldberg, 1976b).

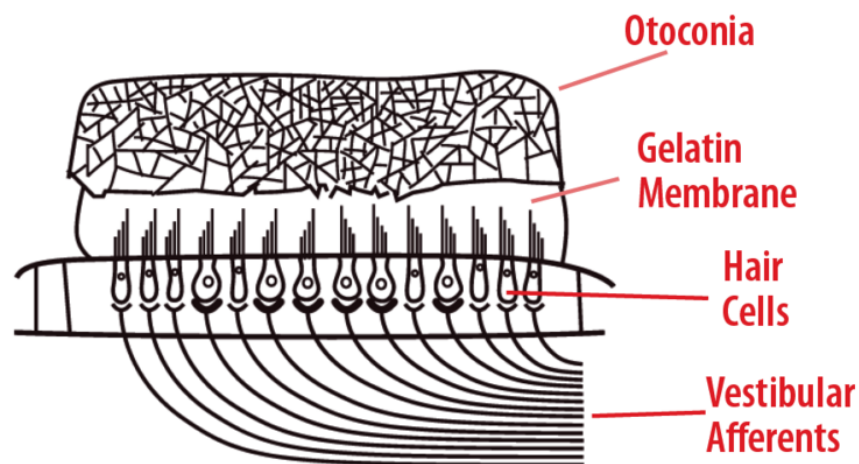


Figure 4. Vestibular response to horizontal plane head rotation. A) When the head is stationary, afferent fibers on both the sides of the head have equivalent firing so there is no sense of motion. B) When the head turns to the left, all of the left horizontal semicircular canal hair cells are excited and afferent fibers increase their firing rate. Conversely, right horizontal canal afferents decrease their firing rate

All vestibular hair cells and afferents receive connections from vestibular efferents, which are fibers projecting from the brain out to the vestibular receptor organs, whose function is not well understood. It is thought that efferents control the sensitivity of the receptor (Boyle, Carey, & Highstein, 1991). The primary efferents neurotransmitter is acetylcholine (Anniko & Arnold, 1991).

The vestibular nuclei

The vestibular nuclei comprise a large set of neural elements in the brainstem that receive motion and other multisensory signals, then regulate movement responses and sensory experience. Many vestibular nuclei neurons have reciprocal connections with the cerebellum that form important regulatory mechanisms for the control of eye movements, head movements, and posture. There are four major vestibular nuclei that lie in the rostral medulla and caudal pons of the brainstem; all receive direct input from vestibular afferents (Brodal, 1984; Precht & Shimazu, 1965). Many of these nuclei neurons receive convergent motion information from the opposite ear through an inhibitory commissural pathway that uses gamma-aminobutyric acid (GABA) as a neurotransmitter (Kasahara & Uchino, 1974; Shimazu & Precht, 1966). The commissural pathway is highly organized such that cells receiving horizontal excitatory canal signals from the ipsilateral ear will also receive contralateral inhibitory horizontal canal signals from the opposite ear. This fact gives rise to a “push-pull” vestibular function, whereby directional sensitivity to head movement is coded by opposing receptor signals. Because vestibular nuclei neurons receive information from bilateral inner ear receptors and because they maintain a high spontaneous firing rate (nearly 100 impulses/sec), they are thought to act to “compare” the relative discharge rates of left vs. right canal afferent firing activity. For example, during a leftward head turn, left brainstem nuclei neurons receive high firing-rate information from the left horizontal canal and low firing-rate information from the right horizontal canal. The comparison of activity is interpreted as a left head turn. Similar nuclei neuron responses exist when the head is pitched or rolled, with the vertical semicircular canals being stimulated by the rotational motion in their sensitivity planes. However, the opposing push-pull response from the vertical canals occurs with the anterior semicircular canal in one ear and the co-planar posterior semicircular canal of the opposite ear. Damage or disease that interrupts inner ear signal information from one side of the head can change the normal resting activity in the VIIIth nerve afferent fibers and will be interpreted by the brain as a head rotation, even though the head is stationary. These effects often lead to illusions of spinning or rotating that can be quite upsetting and may produce nausea or vomiting. However, over time the commissural fibers provide for vestibular compensation, a process by which the loss of unilateral vestibular receptor function is partially restored centrally and behavioral responses, such as the vestibuloocular

reflex (VOR) and postural responses, mostly recover (Beraneck et al., 2003; Fetter & Zee, 1988;; Newlands, Hesse, Haque, & Angelaki, 2001; Newlands & Perachio, 1990).

In addition to the commissural pathway, many vestibular nuclei neurons receive proprioceptive signals from the spinal cord regarding muscle movement and position, visual signals regarding spatial motion, other multisensory (e.g., trigeminal) signals, and higher order signals from the cortex. It is thought that the cortical inputs regulate fine gaze and postural control, as well as suppress the normal compensatory reflexes during motion in order to elicit volitional movements. Of special significance are convergent signals from the semicircular canal and otolith afferents that allow central vestibular neurons to compute specific properties of head motion (Dickman & Angelaki, 2002). For example, Einstein (1907) showed that linear accelerations are equivalent whether they arise from translational motion or from tilts of the head relative to gravity. The otolith receptors cannot discriminate between the two, so how is it that we can tell the difference between when we are translating forward and tilting backward, where the linear acceleration signaled by the otolith afferents is the same? Vestibular nuclei and cerebellar neurons use convergent signals from both the semicircular canals and the otolith receptors to discriminate between tilt and translation, and as a result, some cells encode head tilt (Zhou, 2006) while other cells encode translational motion (Angelaki, Shaikh, Green, & Dickman, 2004).

Vestibuloocular system

The vestibular system is responsible for controlling gaze stability during motion (Crane & Demer, 1997). For example, if we want to read the sign in a store window while walking by, we must maintain foveal fixation on the words while compensating for the combined rotational and translational head movements incurred during our stride. The vestibular system regulates compensatory eye, neck, spinal, and limb movements in order to maintain gaze (Keshner & Peterson, 1995). One of the major components contributing to gaze stability is the VOR, which produces reflexive eye movements that are equal in magnitude and opposite in direction to the perceived head motion in 3D space (Wilson et al., 1995). The VOR is so accurate and fast that it allows people to maintain visual fixation on objects of interest while experiencing demanding motion conditions, such as running, skiing, playing tennis, and driving. In fact, gaze stabilization in humans has been shown to be completely compensatory (essentially perfect) for most natural behaviors. To produce the VOR, vestibular neurons must control each of the six pairs of eye muscles in unison through a specific set of connections to the oculomotor nuclei (Ezure & Graf, 1984). The anterior and posterior semicircular canals along with the saccule control vertical and torsional (turning of the eye around the line of sight) eye movements, while the horizontal canals and the utricle control horizontal eye

movements.

To understand how the VOR works, let's take the example of the compensatory response for a leftward head turn while reading the words on a computer screen. The basic pathway consists of horizontal semicircular canal afferents that project to specific neurons in the vestibular nuclei. These nuclei cells, in turn, send an excitatory signal to the contralateral **abducens nucleus**, which projects through the sixth cranial nerve to innervate the **lateral rectus muscle** (Figure 5). Some abducens neurons send an excitatory projection back across

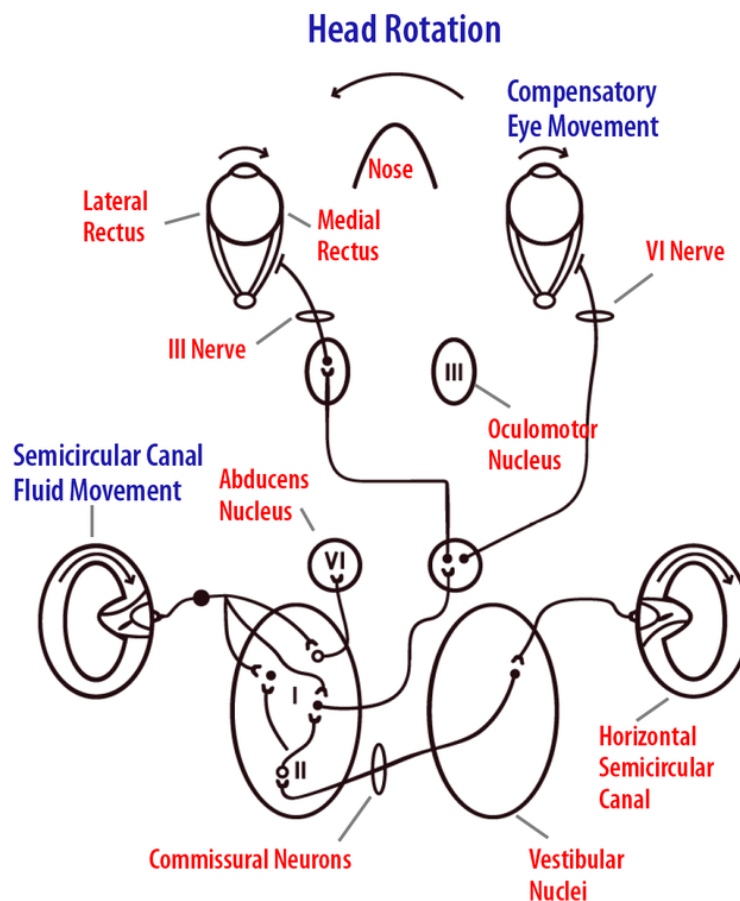


Figure 5. Vestibuloocular reflex. During a leftward head turn, the left horizontal semicircular canal receptors are excited, while the right ear receptors are inhibited. The left excitatory signals excite vestibular nuclei neurons. These cells project across the brain to excite motor neurons in the right abducens nucleus (VI) that excite the lateral rectus muscle of the right eye and to cells in the oculomotor nucleus (III) that excite the medial rectus muscle of the left eye. This moves both eyes to the right to exactly match the leftward head movement and stabilize visual gaze upon a target of interest. The right ear inhibitory signals cross to neurons in the left vestibular nucleus that decrease their firing rate. These cells are inhibitory and decrease their firing rate to further increase the response of rightward motor eye muscle cells.

the midline to a subdivision of cells in the ipsilateral **oculomotor nucleus**, which, in turn, projects through the third cranial nerve to innervate the right (ipsilateral) medial rectus muscle. When a leftward head turn is made, the left horizontal canal vestibular afferents will increase their firing rate and consequently increase the activity of vestibular nuclei neurons projecting to the opposite (contralateral) right abducens nucleus. The abducens neurons produce contraction of the right lateral rectus and, through a separate cell projection to the left oculomotor nucleus, excite the left medial rectus muscles. In addition, matching bilateral inhibitory connections relax the left lateral rectus and right medial rectus eye muscles. The resulting rightward eye movement for both eyes stabilizes the object of interest upon the retina for greatest visual acuity.

During linear translations, a different type of VOR also occurs (Paige & Tomko, 1991). For example, sideways motion to the left results in a horizontal rightward eye movement to maintain visual stability on an object of interest. In a similar manner, vertical up-down head movements (such as occur while walking or running) elicit oppositely directed vertical eye movements (Angelaki, McHenry, & Hess, 2000). For these reflexes, the amplitude of the translational VOR depends on viewing distance. This is due to the fact that the **vergence angle** (i.e., the angle between the lines of sight for each eye) varies as a function of the inverse of the distance to the viewed visual object (Schwarz, Busetini, & Miles, 1989). Visual objects that are far away (2 meters or more) require no vergence angle, but as the visual objects get closer (e.g., when holding your finger close to your nose), a large vergence angle is needed. During translational motion, the eyes will change their vergence angle as the visual object moves from close to farther away (or vice versa). These responses are a result of activation of the otolith receptors, with connections to the oculomotor nuclei similar to those described above for the rotational vestibuloocular reflex. With tilts of the head, the resulting eye movement is termed **torsion**, and consists of a rotational eye movement around the line of sight that is in the direction opposite to the head tilt. As mentioned above, there are major reciprocal connections between the vestibular nuclei and the cerebellum. It has been well established that these connections are crucial for adaptive motor learning in the vestibuloocular reflex (Lisberger, Pavelko, & Broussard, 1994).

Vestibulo-spinal network

There are two vestibular descending pathways that regulate body muscle responses to motion and gravity, consisting of the **lateral vestibulo-spinal tract** (LVST) and the **medial vestibulo-spinal tract** (MVST). Reflexive control of head and neck muscles arises through the neurons in the medial vestibulospinal tract (MVST). These neurons comprise the rapid vestibulocollic reflex (VCR) that serves to stabilize the head in space and participates in gaze control (Peterson,

Goldber, Bilotto, & Fuller, 1985). The MVST neurons receive input from vestibular receptors and the cerebellum, and somatosensory information from the spinal cord. MVST neurons carry both excitatory and inhibitory signals to innervate neck flexor and extensor motor neurons in the spinal cord. For example, if one trips over a crack in the pavement while walking, MVST neurons will receive downward and forward linear acceleration signals from the otolith receptors and forward rotation acceleration signals from the vertical semicircular canals. The VCR will compensate by providing excitatory signals to the dorsal neck flexor muscles and inhibitory signals to the ventral neck extensor muscles, which moves the head upward and opposite to the falling motion to protect it from impact.



Similar to how the gyroscope in your cell phone can detect which angle it's at to adjust the screen, so do the MSVT neurons inform your body which direction it's headed in order to make the appropriate bodily adjustments. [Image: Kleman Gellek, <https://goo.gl/DR9rpR>, CC BY 4.0, <https://goo.gl/QuGXFp>]

The LVST comprises a topographic organization of vestibular nuclei cells that receive substantial input from the cerebellum, proprioceptive inputs from the spinal cord, and convergent afferent signals from vestibular receptors. LVST fibers project ipsilaterally to many levels of motor neurons in the cord to provide coordination of different muscle groups for postural control (Shinoda, Sugiuchi, Futami, Ando, & Kawasaki, 1994). LVST neurons contain either acetylcholine or glutamate as a neurotransmitter and exert an excitatory influence upon extensor muscle motor neurons. For example, LVST fibers produce extension of the contralateral axial and limb musculature when the body is tilted sideways. These actions serve to stabilize the body's center of gravity in order to preserve upright posture.

Vestibulo-autonomic control

Some vestibular nucleus neurons send projections to the reticular formation, dorsal pontine nuclei, and nucleus of the solitary tract. These connections regulate breathing and circulation through compensatory vestibular autonomic responses that stabilize respiration and blood pressure during body motion and changes relative to gravity. They may also be important for induction of motion sickness and emesis.

Vestibular signals in the thalamus and cortex

The cognitive perception of motion, spatial orientation, and navigation through space arises through multisensory information from vestibular, visual, and somatosensory signals in the thalamus and cortex (Figure 6A). Vestibular nuclei neurons project bilaterally to the several thalamic regions. Neurons in the ventral posterior group respond to either vestibular signals alone, or to vestibular plus somatosensory signals, and projects to primary somatosensory cortex (area 3a, 2v), somatosensory association cortex, posterior parietal cortex (areas 5 and 7), and the insula of the temporal cortex (Marlinski & McCrea, 2008; Meng, May, Dickman, & Angelaki, 2007). The posterior nuclear group (PO), near the medial geniculate body, receives both vestibular and auditory signals as well as inputs from the superior colliculus and spinal cord, indicating an integration of multiple sensory signals. Some anterior pulvinar neurons also respond to motion stimuli and project to cortical area 3a, the posterior insula, and the temporo-parietal cortex (PIVC). In humans, electrical stimulation of the thalamic areas produces sensations of movement and sometimes dizziness.

Area 2v cells respond to motion, and electrical stimulation of this area in humans produces sensations of moving, spinning, or dizziness. Area 3a lies at the base of the central sulcus adjacent to the motor cortex and is thought to be involved in integrative motor control of the head and body (Guldin, Akbarian, & Grusser, 1992). Neurons in the PIVC are multisensory, responding to body motion, somatosensory, proprioceptive, and visual motion stimuli (Chen, DeAngelis, & Angelaki, 2011; Grusser, Pause, & Schreiter, 1982). PIVC and areas 3a and 2v are heavily interconnected. Vestibular neurons also have been observed in the posterior parietal cortex; in area 7, in the ventral intraparietal area (VIP), the medial intraparietal area (MIP), and the medial superior temporal area (MST). VIP contains multimodal neurons involved in spatial coding. MIP and MST neurons respond to body motion through space by multisensory integration of visual motion and vestibular signals (Gu, DeAngelis, & Angelaki, 2007) and many MST cells are directly involved in heading perception (Gu, Watkins, Angelaki, & DeAngelis, 2006). Lesions of the parietal cortical areas can result in confusions in spatial awareness. Finally, areas involved with the control of saccades and pursuit eye movements, including area 6, area 8, and the superior frontal gyrus, receive vestibular signals (Fukushima, Sato, Fukushima, Shinmei, & Kaneko, 2000). How these different cortical regions contribute to our perception of motion and spatial orientation is still not well understood.

Spatial orientation and navigation

Our ability to know where we are and to navigate different spatial locations is essential for survival. It is believed that a cognitive map of our environment is created through exploration

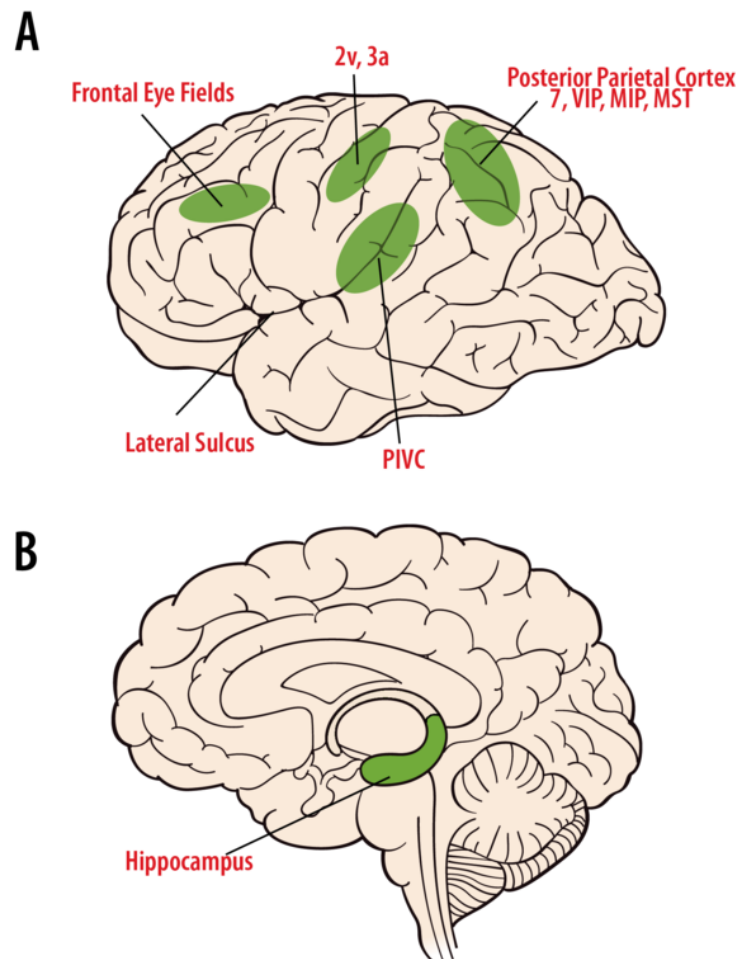


Figure 6. Cortical regions of the brain known to be involved with vestibular processing. A) The frontal eye fields control eye movements and receive vestibular motion information. Areas 2v and 3a are somatosensory areas that map body location and movement signals. Area PIVC responds to body and head motion information. The posterior parietal cortex is involved with motion perception and responds to both visual and vestibular motion cues. B) The hippocampus and parahippocampal regions are involved with spatial orientation and navigation functions. All receive vestibular signals regarding body and head motion.

and then used for spatial orientation and navigation, such as driving to the store, or walking through a dark house (McNaughton, Battaglia, Jensen, Moser, & Moser, 2006). Cells in the limbic system and the hippocampus that contribute to these functions have been identified, including place cells, grid cells, and head direction cells (Figure 6B). Place cells in the hippocampus encode specific locations in the environment (O'Keefe, 1976). Grid cells in the entorhinal cortex encode spatial maps in a tessellated pattern (Hafting, Fyhn, Molden, Moser, & Moser, 2005). Head direction cells in the anterior-dorsal thalamus encode heading direction, independent of spatial location (Taube, 1995). It is thought that these cell types work together

to provide for spatial orientation, spatial memory, and our ability to navigate. Both place cells and head direction cells depend upon a functioning vestibular system to maintain their directional and orientation information (Stackman, Clark, & Taube, 2002). The pathway by which vestibular signals reach the navigation network is not well understood; however, damage to the vestibular system, hippocampus, and dorsal thalamus regions often disrupts our ability to orient in familiar environments, navigate from place to place, or even to find our way home.

Motion sickness



A common treatment of motion sickness is Dramamine, which helps to reduce the sensitivity of input from your vestibular system to the rest of your body. [Image: Mike Baird, <https://goo.gl/zfeqqr>, CC BY 2.0, <https://goo.gl/BRvSA7>]

Although a number of conditions can produce motion sickness, it is generally thought that it is evoked from a mismatch in sensory cues between vestibular, visual, and proprioceptive signals (Yates, Miller, & Lucot, 1998). For example, reading a book in a car on a winding road can produce motion sickness, whereby the accelerations experienced by the vestibular system do not match the visual input. However, if one looks out the window at the scenery going by during the same travel, no sickness occurs because the visual and vestibular cues are in alignment. Sea sickness, a form of motion sickness, appears to be a special case and arises from unusual vertical oscillatory and roll motion. Human studies have found that low frequency oscillations of 0.2 Hz and large amplitudes (such as found in large seas during a storm) are most likely to cause motion sickness, with higher frequencies offering little problems.

Summary

Here, we have seen that the vestibular system transduces and encodes signals about head motion and position with respect to gravity, information that is then used by the brain for many essential functions and behaviors. We actually understand a great deal regarding vestibular contributions to fundamental reflexes, such as compensatory eye movements and

balance during motion. More recent progress has been made toward understanding how vestibular signals combine with other sensory cues, such as vision, in the thalamus and cortex to give rise to motion perception. However, there are many complex cognitive abilities that we know require vestibular information to function, such as spatial orientation and navigation behaviors, but these systems are only just beginning to be investigated. Future research regarding vestibular system function will likely be geared to seeking answers to questions regarding how the brain copes with vestibular signal loss. In fact, according to the National Institutes of Health, nearly 35% of Americans over the age of 40 (69 million people) have reported chronic vestibular-related problems. It is therefore of significant importance to human health to better understand how vestibular cues contribute to common brain functions and how better treatment options for vestibular dysfunction can be realized.

Outside Resources

Animated Video of the Vestibular System

<http://sites.sinauer.com/neuroscience5e/animations14.01.html>

Discussion Questions

1. If a person sustains loss of the vestibular receptors in one ear due to disease or trauma, what symptoms would the person suffer? Would the symptoms be permanent?
2. Often motion sickness is relieved when a person looks at far distance objects, such as things located on the far horizon. Why does far distance viewing help in motion sickness while close distance view (like reading a map or book) make it worse?
3. Vestibular signals combine with visual signals in certain areas of cortex and assist in motion perception. What types of cues does the visual system provide for self motion through space? What types of vestibular signals would be consistent with rotational versus translational motion?

Vocabulary

Abducens nucleus

A group of excitatory motor neurons in the medial brainstem that send projections through the VIth cranial nerve to control the ipsilateral lateral rectus muscle. In addition, abducens interneurons send an excitatory projection across the midline to a subdivision of cells in the ipsilateral oculomotor nucleus, which project through the IIIrd cranial nerve to innervate the ipsilateral medial rectus muscle.

Acetylcholine

An organic compound neurotransmitter consisting of acetic acid and choline. Depending upon the receptor type, acetylcholine can have excitatory, inhibitory, or modulatory effects.

Afferent nerve fibers

Single neurons that innervate the receptor hair cells and carry vestibular signals to the brain as part of the vestibulocochlear nerve (cranial nerve VIII).

Aspartate

An excitatory amino acid neurotransmitter that is widely used by vestibular receptors, afferents, and many neurons in the brain.

Compensatory reflexes

A stabilizing motor reflex that occurs in response to a perceived movement, such as the vestibuloocular reflex, or the postural responses that occur during running or skiing.

Depolarized

When receptor hair cells have mechanically gated channels open, the cell increases its membrane voltage, which produces a release of neurotransmitter to excite the innervating nerve fiber.

Detection thresholds

The smallest amount of head motion that can be reliably reported by an observer.

Directional tuning

The preferred direction of motion that hair cells and afferents exhibit where a peak excitatory response occurs and the least preferred direction where no response occurs. Cells are said to be “tuned” for a best and worst direction of motion, with in-between motion directions eliciting a lesser but observable response.

Gamma-aminobutyric acid

A major inhibitory neurotransmitter in the vestibular commissural system.

Gaze stability

A combination of eye, neck, and head responses that are all coordinated to maintain visual fixation (fovea) upon a point of interest.

Glutamate

An excitatory amino acid neurotransmitter that is widely used by vestibular receptors, afferents, and many neurons in the brain.

Hair cells

The receptor cells of the vestibular system. They are termed hair cells due to the many hairlike cilia that extend from the apical surface of the cell into the gelatin membrane. Mechanical gated ion channels in the tips of the cilia open and close as the cilia bend to cause membrane voltage changes in the hair cell that are proportional to the intensity and direction of motion.

Hyperpolarizes

When receptor hair cells have mechanically gated channels close, the cell decreases its membrane voltage, which produces less release of neurotransmitters to inhibit the innervating nerve fiber.

Lateral rectus muscle

An eye muscle that turns outward in the horizontal plane.

Lateral vestibulo-spinal tract

Vestibular neurons that project to all levels of the spinal cord on the ipsilateral side to control posture and balance movements.

Mechanically gated ion channels

Ion channels located in the tips of the stereocilia on the receptor cells that open/close as the cilia bend toward the tallest/smallest cilia, respectively. These channels are permeable to potassium ions, which are abundant in the fluid bathing the top of the hair cells.

Medial vestibulo-spinal tract

Vestibular nucleus neurons project bilaterally to cervical spinal motor neurons for head and neck movement control. The tract principally functions in gaze direction and stability during motion.

Neurotransmitters

A chemical compound used to send signals from a receptor cell to a neuron, or from one neuron to another. Neurotransmitters can be excitatory, inhibitory, or modulatory and are packaged in small vesicles that are released from the end terminals of cells.

Oculomotor nuclei

Includes three neuronal groups in the brainstem, the abducens nucleus, the oculomotor nucleus, and the trochlear nucleus, whose cells send motor commands to the six pairs of eye muscles.

Oculomotor nucleus

A group of cells in the middle brainstem that contain subgroups of neurons that project to the medial rectus, inferior oblique, inferior rectus, and superior rectus muscles of the eyes through the 3rd cranial nerve.

Otoconia

Small calcium carbonate particles that are packed in a layer on top of the gelatin membrane that covers the otolith receptor hair cell stereocilia.

Otolith receptors

Two inner ear vestibular receptors (utricle and saccule) that transduce linear accelerations and head tilt relative to gravity into neural signals that are then transferred to the brain.

Proprioceptive

Sensory information regarding muscle position and movement arising from receptors in the muscles, tendons, and joints.

Semicircular canals

A set of three inner ear vestibular receptors (horizontal, anterior, posterior) that transduce head rotational accelerations into head rotational velocity signals that are then transferred to the brain. There are three semicircular canals in each ear, with the major planes of each canal being orthogonal to each other.

Stereocilia

Hairlike projections from the top of the receptor hair cells. The stereocilia are arranged in ascending height and when displaced toward the tallest cilia, the mechanical gated channels open and the cell is excited (depolarized). When the stereocilia are displaced toward the smallest cilia, the channels close and the cell is inhibited (hyperpolarized).

Torsion

A rotational eye movement around the line of sight that consists of a clockwise or counterclockwise direction.

Vergence angle

The angle between the line of sight for the two eyes. Low vergence angles indicate far-viewing objects, whereas large angles indicate viewing of near objects.

Vestibular compensation

Following injury to one side of vestibular receptors or the vestibulocochlear nerve, the central vestibular nuclei neurons gradually recover much of their function through plasticity mechanisms. The recovery is never complete, however, and extreme motion environments can lead to dizziness, nausea, problems with balance, and spatial memory.

Vestibular efferents

Nerve fibers originating from a nucleus in the brainstem that project from the brain to innervate the vestibular receptor hair cells and afferent nerve terminals. Efferents have a modulatory role on their targets, which is not well understood.

Vestibular system

Consists of a set of motion and gravity detection receptors in the inner ear, a set of primary nuclei in the brainstem, and a network of pathways carrying motion and gravity signals to many regions of the brain.

Vestibulocochlear nerve

The VIIIth cranial nerve that carries fibers innervating the vestibular receptors and the cochlea.

Vestibuloocular reflex

Eye movements produced by the vestibular brainstem that are equal in magnitude and opposite in direction to head motion. The VOR functions to maintain visual stability on a point of interest and is nearly perfect for all natural head movements.

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26

Time and Culture

Robert V. Levine

There are profound cultural differences in how people think about, measure, and use their time. This module describes some major dimensions of time that are most prone to cultural variation.

Learning Objectives

- Understand how cultures differ in the views of time and the importance of these differences for social behavior.
- Explore major components of social time.
- Use these concepts to better understand the hidden dimensions of culture.

Introduction

It is said that “time is money” in industrialized economies. Workers are paid by the hour, lawyers charge by the minute, and advertising is sold by the second (US\$3.3 million for a 30-second commercial, or a little over \$110,000 per second, for the 2012 Super Bowl). Remarkably, the civilized mind has reduced time—the most obscure and abstract of all intangibles—to the most objective of all quantities: money. With time and things on the same value scale, we can establish how many of our working hours equal the price of a product in a store.

This way of thinking about time is not universal, however. Beliefs about time remain profoundly different from culture to culture. Research shows that cultural differences in time can be as

vast as those between languages. In one particularly telling study of the roots of culture shock, Spradley and Phillips asked a group of returning Peace Corps volunteers to rank 33 items concerning the amount of cultural adjustment each had required of them. The list included a wide range of items familiar to fearful travelers, such as “the type of food eaten,” the “personal cleanliness of most people,” “the number of people of your own race,” and “the general standard of living.” But aside from mastering the foreign language, the two greatest difficulties for the Peace Corps volunteers were concerned with social time: “the general pace of life,” followed by one of its most significant components, “how punctual most people are” (Spradley & Phillips, 1972).



The idea that “time is money” may be good for business but is it good for society in general? What other ways do people around the world conceptualize and experience time? [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

Half a century ago anthropologist Edward Hall described cultural rules of **social time** as the “**silent language**” (Hall, 1983). These informal patterns of time “are seldom, if ever, made explicit. They exist in the air around us. They are either familiar and comfortable or unfamiliar and wrong.” The world over, children simply pick up their society’s conceptions of early and late, of waiting and rushing, of the past, the present, and the future, as they mature. No dictionary clearly defines these rules of time for them or for strangers who stumble over the maddening incongruities between the time sense they bring with them and the one they face in a new land.

Cultures may differ on many aspects of social time—its value, meaning, how it should be divided, allocated, and measured. The following dimensions are particularly prone to different cultural, as well as individual, interpretations:

Work Versus Leisure

There are cultural differences in the value placed on work, on leisure, and upon the balance between the two. Although some balance is universal, the preferred formulas differ both across cultures and between individuals in each culture. The differences are marked even within highly industrialized countries, The United States and Japan are famous for long work

hours, as exemplified by the terms “workaholic” and “*karoshi*” (“death by overwork”) (Levine, 1997). European nations tend to also emphasize work, with many differences among countries, but generally put greater emphasis on preserving nonwork time than do people in the United States and Japan (Levine, 2012).

Time spent within the workplace also varies across cultures. People tend to spend more of their work time on-task in some cultures and more of that time socializing—informal chatting, having tea or coffee with others, etc.—in other cultures. Studies have found wide cultural variation in answers to the question: “In the companies for which you have worked, what percent of time do people typically spend on tasks that are part of their job description.” For example, people working in companies in large cities in the United States tend to report in the range of “80 percent task time, 20 percent social time.” On the other hand, people working in companies in India, Nepal, Indonesia, Malaysia, and some Latin American countries tend to give answers closer to “50 percent task time, 50 percent social time” (Brislin and Kim, 2003).

Sequence

Each culture sets rules concerning the appropriate sequence of tasks and activities. Is it work before play, or vice versa? Do people take all of their sleep at night, or is there a siesta in the midafternoon? Is one expected to have coffee or tea and socialize, and for how long, before getting down to serious business? There are also customs about sequences over the long run. For example, how long is the socially accepted period of childhood, if it exists at all, and when is it time to assume the responsibilities of an adult?

Clock and Event Time

The most fundamental difference in timekeeping throughout history has been between people operating by the clock and those who measure time by social events (Lauer, 1981). This profound difference in thinking about time continues to divide cultures today. Under **clock time**, the hour on the timepiece governs the beginning and ending of activities. Under event time, scheduling is determined by the flow of the activity. Events begin and end when, by mutual consensus, participants “feel” the time is right (Levine, 1997).

In event-time societies, modes of time-reckoning tend to express social experience. Sometimes activities occur in finely coordinated sequences, but without observing the clock. For example, anthropologists have described how participants at an Indian wake move from gathering time to prayer time, singing time, intermission, and mealtime. They move by consensual feeling—when “the time feels right”—but with no apparent concern for the time

on the clock.

Many countries exhort event time as a philosophy of life. In East Africa, there is a popular adage that “Even the time takes its time.” In Trinidad, it is commonly said that “Any time is Trinidad time” (Birth, 1999). In the United States and much of Europe, by contrast, the right way to measure time is assumed to be by the clock. This is especially true when it comes to work hours. Time is money, and any time not focused on-task is seen as wasted time.

Even the language of time may be more or less event-oriented. The Kachin people of North Burma, for example, have no single word equivalent of “time.” They use the word *ahkying* to refer to the “time” of the clock, *na* to a long “time,” *tawng* to a short “time,” *ta* to springtime, and *asak* to the “time” of a person’s life. Whereas, clock time cultures treat time as an objective entity—it is a noun in English—the Kachin words for time are treated more like adverbs (Levine, 1997).



Cultures differ greatly in their adherence to “clock time”. While some cultures pride themselves on managing time down to the second (the trains run on time!), others have a more flexible view of when is the right time for an event to begin or end. [Image: John Tenniel, CC0 Public Domain, <https://goo.gl/m25gce>]

These different ways of time-keeping can often lead to cultural misunderstandings. Individuals operating on clock time are careful to be punctual and expect the same of others. Those on event time are more spontaneous in beginning and ending events and, as a result, tend to be less punctual and more understanding when others are less punctual.

There are also differences within cultures—on both the individual and situational levels. To take just one example, some workers may prosper under clearly defined schedules while others may prefer to complete their work on their own schedules. Similarly, some jobs (for example, financial traders) demand clock-time precision while others (for example, some creative arts) thrive on the spontaneity of event-time scheduling. Levine (2012) argues for fluency in both approaches and to recognize when either is more beneficial.

Calendars

Many cultures use social activities to define their calendars rather than the other way around. The calendars of the Nuer people from the Upper Nile in the Sudan, for example, are based on the seasonal changes in their environment. They know that the month of *kur* is occurring *because* they are building their fishing dams and cattle camps. When they break camp and return to their villages, they know it must now be the month of *dwat*.

Most societies have some type of week, but it is not always seven days long. The Muysca of Colombia had a three-day week. The Incas of Peru had a 10-day week. Often the length of the week reflects cycles of activities, rather than the other way around. For many, the market is the main activity requiring group coordination. The Khasi people hold their markets every eighth day. Consequently, they have made their week eight days long and named the days of the week after the places where the main markets occur (Levine, 2005).

Polychronic and Monochronic Time

Industrial/organizational psychologists emphasize the significance of monochronic versus polychronic work patterns (Bluedorn, 2002). People and organizations in clock-time cultures are more likely to emphasize **monochronic (M-time)** approaches, meaning they like to focus on one activity at a time. People in event time cultures, on the other hand, tend to emphasize **polychronic (P-time)** approaches, meaning they prefer to do several things at once. These labels were originally developed by Hall (1983). M-time people like to work from start to finish

in linear sequence: The first task is begun and completed before turning to another, which is then begun and completed. In polychronic time, however, one project goes on until there is an inclination or inspiration to turn to another, which may lead to an idea for another, then back to first, with intermittent and unpredictable pauses and reassumptions of one task or another. Progress on P-time occurs a little at a time on each task.



What should dictate the conclusion of a good conversation, other time commitments or the flow of the conversation itself? P-time cultures prefer the latter. [Image: mikedogh, <https://goo.gl/AjWjke>, CC BY-SA 2.0, <https://goo.gl/osSgSv>]

P-time cultures are characterized by a strong involvement with people. They emphasize the completion of human transactions rather than keeping to schedules. For example, two P-time

individuals who are deep in conversation will typically choose to arrive late for their next appointment rather than cut into the flow of their discussion. Both would be insulted, in fact, if their partner were to abruptly terminate the conversation before it came to a spontaneous conclusion.

Levine (2012) argues for the value of shifting between each approach depending on the characteristics of the individuals and the situations involved. In a corporation, for example, some positions may require tight scheduling of time (e.g., accountants during tax time). On the other hand, employees in research and development may be most productive when less tightly controlled.

Silence and “Doing Nothing”

In some cultures, notably the United States and Western Europe, silence makes people uncomfortable. It may denote nothing is happening or that something is going wrong. The usual response is to say something, to fill the silence or to keep the meeting or conversation going. People in other cultures, including many Asian and Pacific Island nations, are quite comfortable with silence. It is seen as an opportunity to focus inward and gather one's thoughts before you speak. The Japanese emphasize “ma,” which roughly translates as the “space” between things, or the “pause.” It implies that what happens between things, or what doesn't seem to be happening, is as or more important than what is visibly happening. As an extreme example, consider a question people in Brunei often begin their day by asking: “What isn't going to happen today?”

Brislin (2000) has described how cultural misunderstandings and counterproductive decisions often arise from these differences. For example, “Americans will sometimes misinterpret long periods of silence as a signal that they should make a concession. Their negotiating counterparts in Asia know this and will sometimes prolong their silence in the expectation that a concession will be made.”

A related temporal difference concerns what people perceive as “wasted time.” People, cultures, and economies that emphasize the rule that “time is money” may see any time not devoted to tangible production as wasted time. People in other cultures, however, believe that overemphasis on this rule is a waste of one's time in a larger sense, that it is a wasteful way to spend one's life. If something more worthy of one's attention—be it social- or work-related—challenges a planned schedule, it is seen as wasteful to not deviate from the planned schedule. In fact, the term “wasted time” may make little sense. A typical comment may be, “There is no such thing as wasted time. If you are not doing one thing, you are doing something

else" (Levine, 1997).

Norms Concerning Waiting

Cultures differ in their norms for waiting, not only how long it is appropriate to keep a person waiting but how the rules change depending on the situation and the people involved. Levine (1997) describes a number of "rules" to waiting and how these rules differ in various cultures. Some useful questions: Are the rules based on the principle that time is money? Who is expected to wait for whom, under what circumstances, and for how long? Are some individuals—by virtue of their status, power, and/or wealth—exempt from waiting? What is the protocol for waiting in line? Is it an orderly procedure, as in the United Kingdom, or do people just nudge their way through the crowd, pushing the people ahead of them, until they somehow make their way to the front, as in India? Is there a procedure for buying oneself a place in front, or off the line completely? What social message is being sent when the accepted rules are broken?

Temporal Orientation

There are individual and cultural differences in people's orientation toward the past, present, and future. Zimbardo and Boyd (2008) have developed a scale that distinguishes between six types of temporal frames:

1. Past negative—a pessimistic, negative, or aversive orientation toward the past.
2. Past positive—a warm, sentimental, nostalgic, and positive construction of the past.
3. Present hedonistic—hedonistic orientation attitude toward time and life.
4. Present fatalistic—a fatalistic, helpless, and hopeless attitude toward the future and life.
5. Future—planning for, and achievement of, future goals, characterizing a general future orientation.
6. Future transcendental—an orientation to the future beyond one's own death.

Zimbardo and Boyd have found large individual and cultural differences on both the individual subscales and the patterns of the subscales taken together. They describe a wide range of consequences of these differences. Time perspective affects political, economic, personal, social, environmental, and other domains of life and society. One of the paradoxes, they report, is that each particular temporal perspective is associated with numerous personal and social benefits but that, in excess, they are associated with even greater costs. There are both positive



Reflecting on our own lives, we often shift between various temporal orientations. Consider your usual perspective. Do you find yourself with a particular emphasis on the past, present, or future? [Image: Christian, <https://goo.gl/QZzRLB>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

and negative processes associated with each perspective. Individuals who focus on the past, for example, are often described with terms such as happy, grateful, patriotic, high self-esteem, and having strong personal values; on the other hand, past time perspective can be associated with terms such as depressed, guilty, angry and revengeful, and resistant to change. Similarly, a focus on the present may be associated with strong social affiliations, joy, sensuality, sexuality, energy, and improvisation; but it may also be associated with violence, anger, over-fatalism, risk-taking, and addictive behavior. A focus on the future may be associated with achievement, self-efficacy, healthy behaviors, and hope for change; but also

with anxiety, social isolation, competitiveness, and unhealthy physical consequences ranging from coronary artery disease to sexual impotence. The authors argue for the importance of a healthy balance in one's temporal orientation.

The Pace of Life

There are profound differences in the **pace of life** on many levels—individual temperament, cultural norms, between places, at different times, during different activities. Levine and Norenzayan (1999) conducted a series of field experiments measuring walking speed, work speed, and concern with clock time in countries around the world. They found that the characteristic pace of life of a place has consequences—both positive and negative—for the physical, social, economic, and psychological well-being of people who live there. The optimal pace, they argue, requires flexibility and sensitivity to matching individual preferences to the requirements of the situation.

Conclusion

Understanding the values and assumptions a culture places on these temporal dimensions is essential to creating policies that enhance the quality of peoples' lives. The historian Lewis Mumford once observed how "each culture believes that every other space and time is an

approximation to or perversion of the real space and time in which it lives." The truth, however, is there is no single correct way to think about time. There are different ways of thinking, each with their pluses and minuses, and all may be of value in given situations.

Outside Resources

Video: Dealing with Time

<http://www.youtube.com/watch?v=fSvC3i4Sq4>

Video: RSA Animate—The Secret Powers of Time

www.youtube.com/watch?v=A3oliH7BLmg&feature=related

Discussion Questions

1. Can you give an example of Edward Hall's notion of time as a "silent language"?
2. Can you give an example of clock time in your own life? Can you give an example of event time?
3. Are there activities where you might benefit from another culture's approach to time rather than your usual approach? Give an example.
4. What do you think are the consequences, both positive and negative, of a faster pace of life?
5. Is it fair to conclude that some cultural time practices are more advanced than others? That some are healthier than others? Explain.

Vocabulary

Clock time

Scheduling activities according to the time on the clock.

Ma

Japanese way of thinking that emphasizes attention to the spaces between things rather than the things themselves.

Monochronic (M-time)

Monochronic thinking focuses on doing one activity, from beginning to completion, at a time.

Pace of life

The frequency of events per unit of time; also referred to as speed or tempo.

Polychronic (P-time)

Polychronic thinking switches back and forth among multiple activities as the situation demands.

Silent language

Cultural norms of time and time use as they pertain to social communication and interaction.

Social time

Scheduling by the flow of the activity. Events begin and end when, by mutual consensus, participants “feel” the time is right.

Temporal perspective

The extent to which we are oriented toward the past, present, and future.

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Failures of Awareness: The Case of Inattentional Blindness

Daniel Simons

We think important objects and events in our world will automatically grab our attention, but they often don't, particularly when our attention is focused on something else. The failure to notice unexpected objects or events when attention is focused elsewhere is now known as inattentional blindness. The study of such failures of awareness has a long history, but their practical importance has received increasing attention over the past decade. This module describes the history and status of research on inattentional blindness, discusses the reasons why we find these results to be counterintuitive, and the implications of failures of awareness for how we see and act in our world.

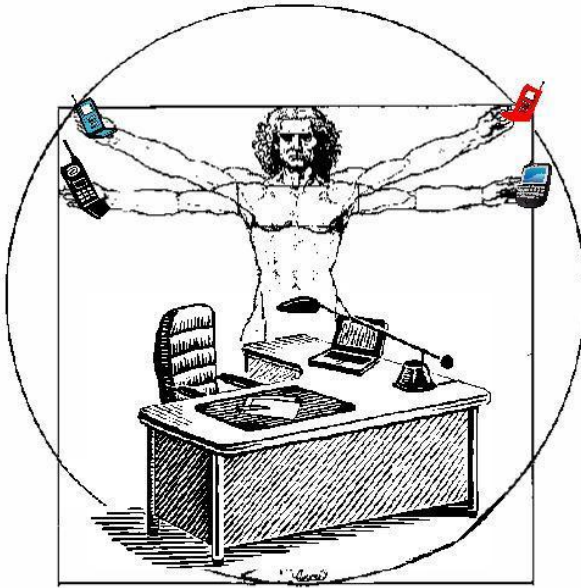
Learning Objectives

- Learn about inattentional blindness and why it occurs.
- Identify ways in which failures of awareness are counterintuitive.
- Better understand the link between focused attention and failures of awareness.

Do you regularly spot editing errors in movies? Can you multitask effectively, texting while talking with your friends or watching television? Are you fully aware of your surroundings? If you answered yes to any of those questions, you're not alone. And, you're most likely wrong.

More than 50 years ago, experimental psychologists began documenting the many ways that our perception of the world is limited, not by our eyes and ears, but by our minds. We appear

able to process only one stream of information at a time, effectively filtering other information from awareness. To a large extent, we perceive only that which receives the focus of our cognitive efforts: our attention.



Some researchers contend that there really is no such thing as multi-tasking. Instead, people are just rapidly switching their attention between tasks, rather than holding those tasks in their attention at the same time. [Image: Mike Licht, <https://goo.gl/z7rkve>, CC BY 2.0, <https://goo.gl/v4Y0Zv>]

Imagine the following task, known as **dichotic listening** (e.g., Cherry, 1953; Moray, 1959; Treisman, 1960): You put on a set of headphones that play two completely different speech streams, one to your left ear and one to your right ear. Your task is to repeat each syllable spoken into your left ear as quickly and accurately as possible, mimicking each sound as you hear it. When performing this attention-demanding task, you won't notice if the speaker in your right ear switches to a different language or is replaced by a different speaker with a similar voice. You won't notice if the content of their speech becomes nonsensical. In effect, you are deaf to the substance of the ignored speech. But, that is not because of the limits of your auditory senses. It is a form of cognitive deafness, due to the nature of

focused, selective attention. Even if the speaker on your right headphone says your name, you will notice it only about one-third of the time (Conway, Cowan, & Bunting, 2001). And, at least by some accounts, you only notice it that often because you still devote some of your limited attention to the ignored speech stream (Holendar, 1986). In this task, you will tend to notice only large physical changes (e.g., a switch from a male to a female speaker), but not substantive ones, except in rare cases.

This **selective listening** task highlights the power of attention to filter extraneous information from awareness while letting in only those elements of our world that we want to hear. Focused attention is crucial to our powers of observation, making it possible for us to zero in on what we want to see or hear while filtering out irrelevant distractions. But, it has consequences as well: We can miss what would otherwise be obvious and important signals.

The same pattern holds for vision. In a groundbreaking series of studies in the 1970s and

early 1980s, Neisser and his colleagues devised a visual analogue of the dichotic listening task (Neisser & Becklen, 1975). Their subjects viewed a video of two distinct, but partially transparent and overlapping, events. For example, one event might involve two people playing a hand-clapping game and the other might show people passing a ball. Because the two events were partially transparent and overlapping, both produced sensory signals on the retina regardless of which event received the participant's attention. When participants were asked to monitor one of the events by counting the number of times the actors performed an action (e.g., hand clapping or completed passes), they often failed to notice unexpected events in the ignored video stream (e.g., the hand-clapping players stopping their game and shaking hands). As for dichotic listening, the participants were unaware of events happening outside the focus of their attention, even when looking right at them. They could tell that other "stuff" was happening on the screen, but many were unaware of the meaning or substance of that stuff.

To test the power of selective attention to induce failures of awareness, Neisser and colleagues (Neisser, 1979) designed a variant of this task in which participants watched a video of two teams of players, one wearing white shirts and one wearing black shirts. Subjects were asked to press a key whenever the players in white successfully passed a ball, but to ignore the players in black. As for the other videos, the teams were filmed separately and then superimposed so that they literally occupied the same space (they were partially transparent). Partway through the video, a person wearing a raincoat and carrying an umbrella strolled through the scene. People were so intently focused on spotting passes that they often missed the "umbrella woman." (Pro tip: If you look closely at the video, you'll see that Ulric Neisser plays on both the black and white teams.)



Have you ever been paying attention to something so closely you missed another event in the background? Or have you ever been so used to seeing something a certain way that when it changed, you didn't even notice it had? [Image: Tilde Ann Thurium, <https://goo.gl/pb8l6Q>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

These surprising findings were well known in the field, but for decades, researchers dismissed their implications because the displays had such an odd, ghostly appearance. Of course, we

would notice if the displays were fully opaque and vivid rather than partly transparent and grainy. Surprisingly, no studies were built on Neisser's method for nearly 20 years. Inspired by these counterintuitive findings and after discussing them with Neisser himself, Christopher Chabris and I revisited them in the late 1990s (Simons & Chabris, 1999). We replicated Neisser's work, again finding that many people missed the umbrella woman when all of the actors in the video were partially transparent and occupying the same space. But, we added another wrinkle: a version of the video in which all of the actions of both teams of players were choreographed and filmed with a single camera. The players moved in and around each other and were fully visible. In the most dramatic version, we had a woman in a gorilla suit walk into the scene, stop to face the camera, thump her chest, and then walk off the other side after nine seconds on screen. Fully half the observers missed the gorilla when counting passes by the team in white.

This phenomenon is now known as **inattentional blindness**, the surprising failure to notice an unexpected object or event when attention is focused on something else (Mack & Rock, 1998). The past 15 years has seen a surge of interest in such failures of awareness, and we now have a better handle on the factors that cause people to miss unexpected events as well as the range of situations in which inattentional blindness occurs. People are much more likely to notice unexpected objects that share features with the attended items in a display (Most et al., 2001). For example, if you count passes by the players wearing black, you are more likely to notice the gorilla than if you count passes by the players wearing white because



The more effort a cognitive task requires the more likely it becomes that you'll miss noticing something significant. [Image: CC0 Public Domain, <https://goo.gl/m25gce>]

the color of the gorilla more closely matches that of the black-shirted players (Simons & Chabris, 1999). However, even unique items can go unnoticed. In one task, people monitored black shapes and ignored white shapes that moved around a computer window (Most et al., 2001). Approximately 30 percent of them failed to detect the bright red cross traversing the display, even though it was the only colored item and was visible for five seconds.

Another crucial influence on noticing is the effort you put into the attention-demanding task. If you have to keep separate counts of bounce passes and aerial passes, you are less likely to notice the gorilla (Simons & Chabris, 1999), and if

you are tracking faster moving objects, you are less likely to notice (Simons & Jensen, 2009). You can even miss unexpected visual objects when you devote your limited cognitive resources to a memory task (Fougnie & Marois, 2007), so the limits are not purely visual. Instead, they appear to reflect limits on the capacity of attention. Without attention to the unexpected event, you are unlikely to become aware of it (Mack & Rock, 1998; Most, Scholl, Clifford, & Simons, 2005).

Inattentional blindness is not just a laboratory curiosity—it also occurs in the real world and under more natural conditions. In a recent study (Chabris, Weinberger, Fontaine, & Simons, 2011), Chabris and colleagues simulated a famous police misconduct case in which a Boston police officer was convicted of lying because he claimed not to have seen a brutal beating (Lehr, 2009). At the time, he had been chasing a murder suspect and ran right past the scene of a brutal assault. In Chabris' simulation, subjects jogged behind an experimenter who ran right past a simulated fight scene. At night, 65 percent missed the fight scene. Even during broad daylight, 44 percent of observers jogged right passed it without noticing, lending some plausibility to the Boston cop's story that he was telling the truth and never saw the beating.

Perhaps more importantly, auditory distractions can induce real-world failures to see. Although people believe they can multitask, few can. And, talking on a phone while driving or walking decreases situation awareness and increases the chances that people will miss something important (Strayer & Johnston, 2001). In a dramatic illustration of cell phone-induced inattentional blindness, Ira Hyman observed that people talking on a cell phone as they walked across a college campus were less likely than other pedestrians to notice a unicycling clown who rode across their path (Hyman, Boss, Wise, McKenzie, & Caggiano, 2010).

Recently, the study of this sort of awareness failure has returned to its roots in studies of listening, with studies documenting inattentional deafness: When listening to a set of spatially localized conversations over headphones, people often fail to notice the voice of a person walking through the scene repeatedly stating "I am a gorilla" (Dalton & Fraenkel, 2012). Under conditions of focused attention, we see and hear far less of the unattended information than we might expect (Macdonald & Lavie, 2011; Wayand, Levin, & Varakin, 2005).

We now have a good understanding of the ways in which focused attention affects the detection of unexpected objects falling outside that focus. The greater the demands on attention, the less likely people are to notice objects falling outside their attention (Macdonald & Lavie, 2011; Simons & Chabris, 1999; Simons & Jensen, 2009). The more like the ignored elements of a scene, the less likely people are to notice. And, the more distracted we are, the less likely we are to be aware of our surroundings. Under conditions of distraction, we effectively develop tunnel vision.

Despite this growing understanding of the limits of attention and the factors that lead to more or less noticing, we have relatively less understanding of individual differences in noticing (Simons & Jensen, 2009). Do some people consistently notice the unexpected while others are obviously unaware of their surroundings? Or, are we all subject to inattentional blindness due to structural limits on the nature of attention? The question remains controversial. A few studies suggest that those people who have a greater working memory capacity are more likely to notice unexpected objects (Hannon & Richards, 2010; Richards, Hannon, & Derakshan, 2010). In effect, those who have more resources available when focusing attention are more likely to spot other aspects of their world. However, other studies find no such relationship: Those with greater working memory capacity are not any more likely to spot an unexpected object or event (Seegmiller, Watson, & Strayer, 2011; Bredemeier & Simons, 2012). There are theoretical reasons to predict each pattern. With more resources available, people should be more likely to notice (see Macdonald & Lavie, 2011). However, people with greater working memory capacity also tend to be better able to maintain their focus on their prescribed task, meaning that they should be less likely to notice. At least one study suggests that the ability to perform a task does not predict the likelihood of noticing (Simons & Jensen, 2009; for a replication, see Bredemeier & Simons, 2012). In a study I conducted with Melinda Jensen, we measured how well people could track moving objects around a display, gradually increasing the speed until people reached a level of 75% accuracy. Tracking ability varied greatly: Some people could track objects at more than twice the speed others could. Yet, the ability to track objects more easily was unrelated to the odds of noticing an unexpected event. Apparently, as long as people try to perform the tracking task, they are relatively unlikely to notice unexpected events.



Now you see me, now you don't! Although the research on attention has only developed over the last few decades, magicians have been taking advantages of our susceptibility to misguided focus for centuries. [Image: ShahanB, <https://goo.gl/p5DYXH>, CC BY-SA 3.0, <https://goo.gl/eLCn2O>]

What makes these findings interesting and important is that they run counter to our intuitions. Most people are confident they would notice the chest-thumping gorilla. In fact, nearly 90% believe they would spot the gorilla (Levin & Angelone, 2008), and in a national survey, 78% agreed with the statement, "People generally notice when something unexpected enters their field of view, even when they're paying attention to something else" (Simons & Chabris, 2010). Similarly, people are convinced that they would spot errors in movies or changes to a

conversation partner (Levin & Angelone, 2008). We think we see and remember far more of our surroundings than we actually do. But why do we have such mistaken intuitions?

One explanation for this mistaken intuition is that our experiences themselves mislead us (Simons & Chabris, 2010). We rarely experience a study situation such as the gorilla experiment in which we are forced to confront something obvious that we just missed. That partly explains why demonstrations such as that one are so powerful: We expect that we would notice the gorilla, and we cannot readily explain away our failure to notice it. Most of the time, we are happily unaware of what we have missed, but we are fully aware of those elements of a scene that we have noticed. Consequently, if we assume our experiences are representative of the state of the world, we will conclude that we notice unexpected events. We don't easily think about what we're missing.

Given the limits on attention coupled with our mistaken impression that important events will capture our attention, how has our species survived? Why weren't our ancestors eaten by unexpected predators? One reason is that our ability to focus attention intently might have been more evolutionarily useful than the ability to notice unexpected events. After all, for an event to be unexpected, it must occur relatively infrequently. Moreover, most events don't require our immediate attention, so if inattentional blindness delays our ability to notice the events, the consequences could well be minimal. In a social context, others might notice that event and call attention to it. Although inattentional blindness might have had minimal consequences over the course of our evolutionary history, it does have consequences now.

At pedestrian speeds and with minimal distraction, inattentional blindness might not matter for survival. But in modern society, we face greater distractions and move at greater speeds, and even a minor delay in noticing something unexpected can mean the difference between a fender-bender and a lethal collision. If talking on a phone increases your odds of missing a unicycling clown, it likely also increases your odds of missing the child who runs into the street or the car that runs a red light. Why, then, do people continue to talk on the phone when driving? The reason might well be the same mistaken intuition that makes inattentional blindness surprising: Drivers simply do not notice how distracted they are when they are talking on a phone, so they believe they can drive just as well when talking on a phone even though they can't (Strayer & Johnston, 2001).

So, what can you do about inattentional blindness? The short answer appears to be, "not much." There is no magical elixir that will overcome the limits on attention, allowing you to notice everything (and that would not be a good outcome anyway). But, there is something you can do to mitigate the consequences of such limits. Now that you know about inattentional blindness, you can take steps to limit its impact by recognizing how your intuitions will lead

you astray.



Even though you may think you can drive, text, listen to music, and drink a smoothie at the same time, really, your focus should be only on the road. Everything else is a potential distraction from what's most important: driving safely! [Image: FMHS The Buzz TV, <https://goo.gl/TSk2RP>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

those limits.

First, maximize the attention you do have available by avoiding distractions, especially under conditions for which an unexpected event might be catastrophic. The ring of a new call or the ding of a new text are hard to resist, so make it impossible to succumb to the temptation by turning your phone off or putting it somewhere out of reach when you are driving. If you know that you will be tempted and you know that using your phone will increase inattention blindness, you must be proactive. Second, pay attention to what others might not notice. If you are a bicyclist, don't assume that the driver sees you, even if they appear to make eye contact. Looking is not the same as seeing. Only by understanding the limits of attention and by recognizing our mistaken beliefs about what we "know" to be true can we avoid the modern-day consequences of

Outside Resources

Article: Scholarpedia article on inattentional blindness

http://www.scholarpedia.org/article/Inattentional_blindness

Video: The original gorilla video

<http://www.youtube.com/watch?v=vjG698U2Mvo>

Video: The sequel to the gorilla video

http://www.youtube.com/watch?v=IGQmdoK_ZfY

Web: Website for Chabris & Simons book, The Invisible Gorilla. Includes links to videos and descriptions of the research on inattentional blindness

<http://www.theinvisiblegorilla.com>

Discussion Questions

1. Many people, upon learning about inattentional blindness, try to think of ways to eliminate it, allowing themselves complete situation awareness. Why might we be far worse off if we were not subject to inattentional blindness?
2. If inattentional blindness cannot be eliminated, what steps might you take to avoid its consequences?
3. Can you think of situations in which inattentional blindness is highly likely to be a problem? Can you think of cases in which inattentional blindness would not have much of an impact?

Vocabulary

Dichotic listening

A task in which different audio streams are presented to each ear. Typically, people are asked to monitor one stream while ignoring the other.

Inattentional blindness

The failure to notice a fully visible, but unexpected, object or event when attention is devoted to something else.

Inattentional deafness

The auditory analog of inattentional blindness. People fail to notice an unexpected sound or voice when attention is devoted to other aspects of a scene.

Selective listening

A method for studying selective attention in which people focus attention on one auditory stream of information while deliberately ignoring other auditory information.

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28

Multi-Modal Perception

Lorin Lachs

Most of the time, we perceive the world as a unified bundle of sensations from multiple sensory modalities. In other words, our perception is multimodal. This module provides an overview of multimodal perception, including information about its neurobiology and its psychological effects.

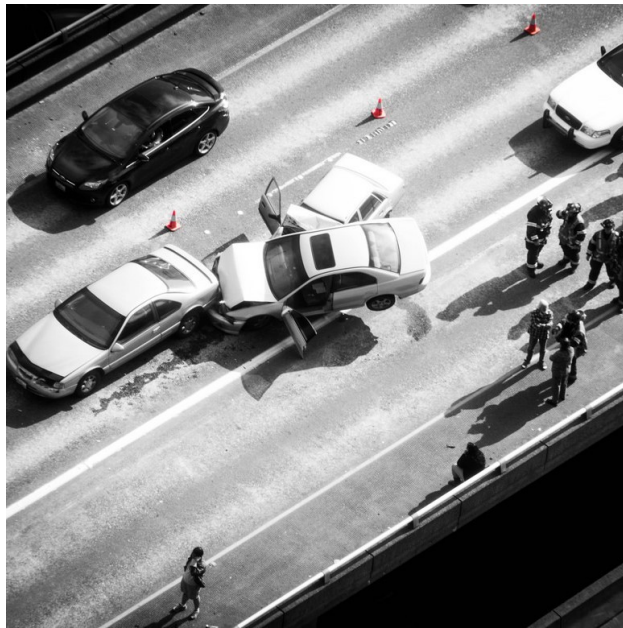
Learning Objectives

- Define the basic terminology and basic principles of multimodal perception.
- Describe the neuroanatomy of multisensory integration and name some of the regions of the cortex and midbrain that have been implicated in multisensory processing.
- Explain the difference between multimodal phenomena and crossmodal phenomena.
- Give examples of multimodal and crossmodal behavioral effects.

Perception: Unified

Although it has been traditional to study the various senses independently, most of the time, perception operates in the context of information supplied by multiple sensory modalities at the same time. For example, imagine if you witnessed a car collision. You could describe the stimulus generated by this event by considering each of the senses independently; that is, as a set of unimodal stimuli. Your eyes would be stimulated with patterns of light energy bouncing off the cars involved. Your ears would be stimulated with patterns of acoustic energy emanating from the collision. Your nose might even be stimulated by the smell of burning

rubber or gasoline.



If you were a witness to this scene you'd be able to describe it using input from many of your senses. Your experience would be multimodal. [Image: Photo Grrrrr, <https://goo.gl/dzfKs8>, CC BY-NC-SA 2.0, <https://goo.gl/Toc0ZF>]

However, all of this information would be relevant to the same thing: your perception of the car collision. Indeed, unless someone was to explicitly ask you to describe your perception in unimodal terms, you would most likely experience the event as a unified bundle of sensations from multiple senses. In other words, your perception would be **multimodal**. The question is whether the various sources of information involved in this multimodal stimulus are processed separately by the perceptual system or not.

For the last few decades, perceptual research has pointed to the importance of **multimodal perception**: the effects on the perception of events and objects in the world that are observed when there is information from more than one sensory

modality. Most of this research indicates that, at some point in perceptual processing, information from the various sensory modalities is **integrated**. In other words, the information is combined and treated as a unitary representation of the world.

Questions About Multimodal Perception

Several theoretical problems are raised by multimodal perception. After all, the world is a “blooming, buzzing world of confusion” that constantly bombards our perceptual system with light, sound, heat, pressure, and so forth. To make matters more complicated, these stimuli come from multiple events spread out over both space and time. To return to our example: Let’s say the car crash you observed happened on Main Street in your town. Your perception during the car crash might include a lot of stimulation that was *not* relevant to the car crash. For example, you might also overhear the conversation of a nearby couple, see a bird flying into a tree, or smell the delicious scent of freshly baked bread from a nearby bakery (or all three!). However, you would most likely not make the mistake of associating any of these stimuli with the car crash. In fact, we rarely combine the auditory stimuli associated with one

event with the visual stimuli associated with another (although, under some unique circumstances—such as ventriloquism—we do). How is the brain able to take the information from separate sensory modalities and match it appropriately, so that stimuli that belong together stay together, while stimuli that do not belong together get treated separately? In other words, how does the perceptual system determine which unimodal stimuli must be integrated, and which must not?

Once unimodal stimuli have been appropriately integrated, we can further ask about the consequences of this integration: What are the effects of multimodal perception that would not be present if perceptual processing were only unimodal? Perhaps the most robust finding in the study of multimodal perception concerns this last question. No matter whether you are looking at the actions of neurons or the behavior of individuals, it has been found that responses to multimodal stimuli are typically greater than the combined response to either modality independently. In other words, if you presented the stimulus in one modality at a time and measured the response to each of these unimodal stimuli, you would find that adding them together would still not equal the response to the multimodal stimulus. This superadditive effect of multisensory integration indicates that there are consequences resulting from the integrated processing of multimodal stimuli.

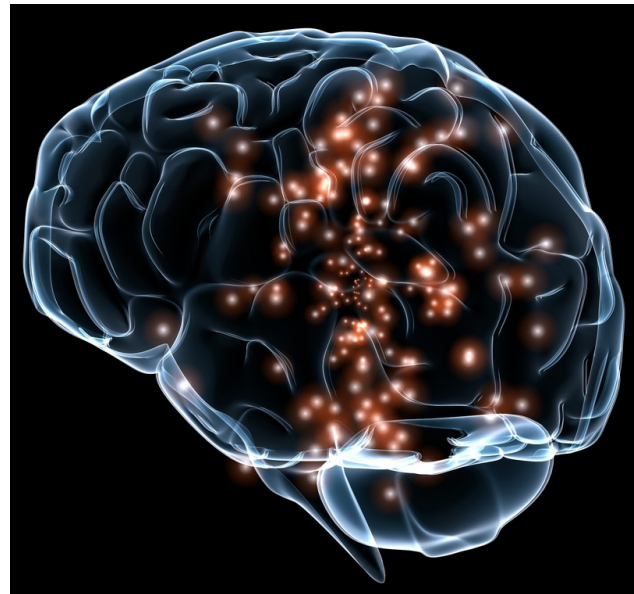
The extent of the superadditive effect (sometimes referred to as multisensory enhancement) is determined by the strength of the response to the single stimulus modality with the biggest effect. To understand this concept, imagine someone speaking to you in a noisy environment (such as a crowded party). When discussing this type of multimodal stimulus, it is often useful to describe it in terms of its unimodal components: In this case, there is an auditory component (the sounds generated by the speech of the person speaking to you) and a visual component (the visual form of the face movements as the person speaks to you). In the crowded party, the auditory component of the person's speech might be difficult to process (because of the surrounding party noise). The potential for visual information about speech—lipreading—to help in understanding the speaker's message is, in this situation, quite large. However, if you were listening to that same person speak in a quiet library, the auditory portion would probably be sufficient for receiving the message, and the visual portion would help very little, if at all (Sumbly & Pollack, 1954). In general, for a stimulus with multimodal components, if the response to each component (on its own) is weak, then the opportunity for multisensory enhancement is very large. However, if one component—by itself—is sufficient to evoke a strong response, then the opportunity for multisensory enhancement is relatively small. This finding is called the Principle of Inverse Effectiveness (Stein & Meredith, 1993) because the effectiveness of multisensory enhancement is inversely related to the unimodal response with the greatest effect.

Another important theoretical question about multimodal perception concerns the neurobiology that supports it. After all, at some point, the information from each sensory modality is definitely separated (e.g., light comes in through the eyes, and sound comes in through the ears). How does the brain take information from different neural systems (optic, auditory, etc.) and combine it? If our experience of the world is multimodal, then it must be the case that at some point during perceptual processing, the unimodal information coming from separate sensory organs—such as the eyes, ears, skin—is combined. A related question asks where in the brain this integration takes place. We turn to these questions in the next section.

Biological Bases of Multimodal Perception

Multisensory Neurons and Neural Convergence

A surprisingly large number of brain regions in the midbrain and cerebral cortex are related to multimodal perception. These regions contain neurons that respond to stimuli from not just one, but multiple sensory modalities. For example, a region called the superior temporal sulcus contains single neurons that respond to both the visual and auditory components of speech (Calvert, 2001; Calvert, Hansen, Iversen, & Brammer, 2001). These multisensory convergence zones are interesting, because they are a kind of neural intersection of information coming from the different senses. That is, neurons that are devoted to the processing of one sense at a time—say vision or touch—send their information to the convergence zones, where it is processed together.



In order for us to perceive the world effectively, neurons from our various senses carry information that is integrated in the brain. [Image: DARPA, <https://goo.gl/kat7ws>, CC0 Public Domain, <https://goo.gl/m25gce>]

One of the most closely studied multisensory convergence zones is the superior colliculus (Stein & Meredith, 1993), which receives inputs from many different areas of the brain, including regions involved in the unimodal processing of visual and auditory stimuli (Edwards, Ginsburgh, Henkel, & Stein, 1979). Interestingly, the superior colliculus is involved in the

“orienting response,” which is the behavior associated with moving one’s eye gaze toward the location of a seen or heard stimulus. Given this function for the superior colliculus, it is hardly surprising that there are multisensory neurons found there (Stein & Stanford, 2008).

Crossmodal Receptive Fields

The details of the anatomy and function of multisensory neurons help to answer the question of how the brain integrates stimuli appropriately. In order to understand the details, we need to discuss a neuron’s receptive field. All over the brain, neurons can be found that respond only to stimuli presented in a very specific region of the space immediately surrounding the perceiver. That region is called the neuron’s **receptive field**. If a stimulus is presented in a neuron’s receptive field, then that neuron responds by increasing or decreasing its firing rate. If a stimulus is presented outside of a neuron’s receptive field, then there is no effect on the neuron’s firing rate. Importantly, when two neurons send their information to a third neuron, the third neuron’s receptive field is the combination of the receptive fields of the two input neurons. This is called neural convergence, because the information from multiple neurons converges on a single neuron. In the case of multisensory neurons, the convergence arrives from different sensory modalities. Thus, the receptive fields of multisensory neurons are the combination of the receptive fields of neurons located in different sensory pathways.

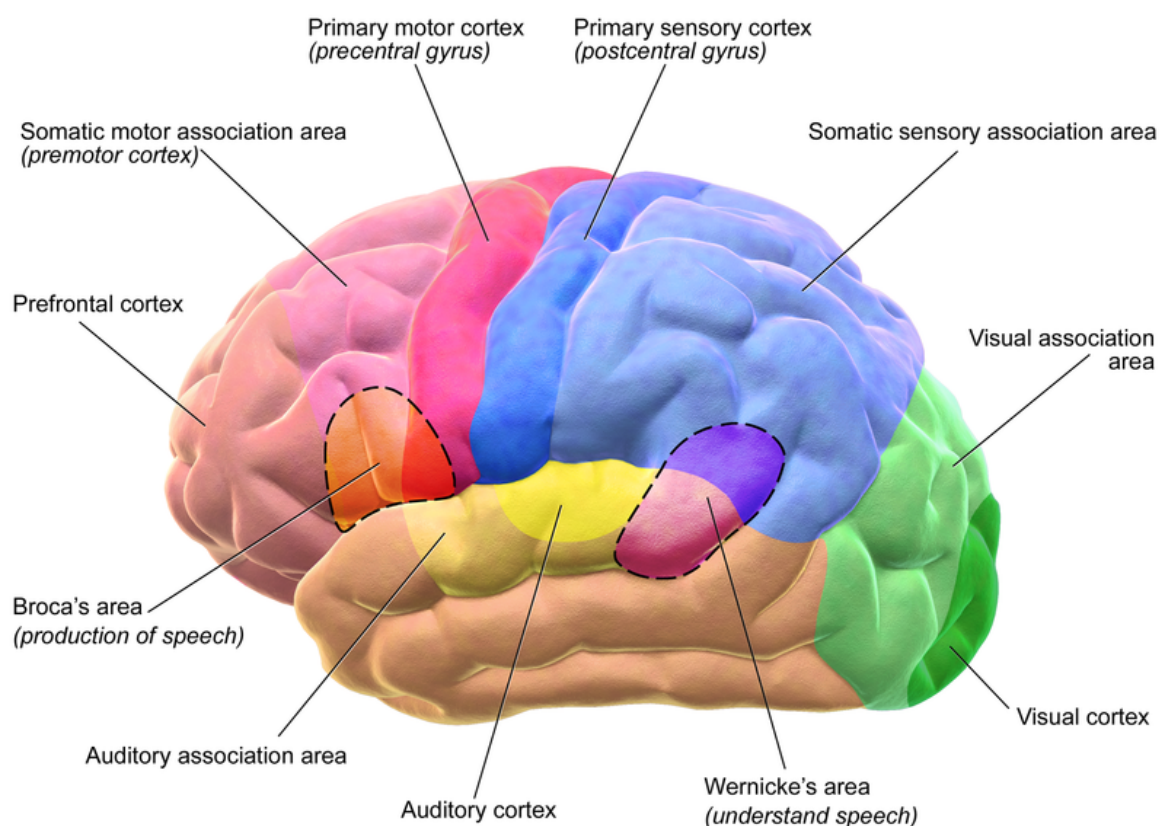
Now, it could be the case that the neural convergence that results in multisensory neurons is set up in a way that ignores the locations of the input neurons’ receptive fields. Amazingly, however, these **crossmodal receptive fields** overlap. For example, a multisensory neuron in the superior colliculus might receive input from two unimodal neurons: one with a visual receptive field and one with an auditory receptive field. It has been found that the unimodal receptive fields refer to the same locations in space—that is, the two unimodal neurons respond to stimuli in the same region of space. Crucially, the overlap in the crossmodal receptive fields plays a vital role in the integration of **crossmodal stimuli**. When the information from the separate modalities is coming from within these overlapping receptive fields, then it is treated as having come from the same location—and the neuron responds with a superadditive (enhanced) response. So, part of the information that is used by the brain to combine multimodal inputs is the location in space from which the stimuli came.

This pattern is common across many multisensory neurons in multiple regions of the brain. Because of this, researchers have defined the **spatial principle of multisensory integration**: Multisensory enhancement is observed when the sources of stimulation are spatially related to one another. A related phenomenon concerns the *timing* of crossmodal stimuli. Enhancement effects are observed in multisensory neurons only when the inputs from

different senses arrive within a short time of one another (e.g., Recanzone, 2003).

Multimodal Processing in Unimodal Cortex

Multisensory neurons have also been observed outside of multisensory convergence zones, in areas of the brain that were once thought to be dedicated to the processing of a single modality (**unimodal cortex**). For example, the **primary visual cortex** was long thought to be devoted to the processing of exclusively visual information. The primary visual cortex is the first stop in the cortex for information arriving from the eyes, so it processes very low-level information like edges. Interestingly, neurons have been found in the primary visual cortex that receives information from the **primary auditory cortex** (where sound information from the auditory pathway is processed) and from the superior temporal sulcus (a multisensory convergence zone mentioned above). This is remarkable because it indicates that the processing of visual information is, from a very early stage, influenced by auditory information.



There are zones in the human brain where sensory information comes together and is integrated such as the Auditory, Visual and Motor Cortices pictured here. [Image: BruceBlaus, <https://goo.gl/UqKB13>, CC BY 3.0, <https://goo.gl/b58TcB>]

There may be two ways for these multimodal interactions to occur. First, it could be that the processing of auditory information in relatively late stages of processing feeds back to influence low-level processing of visual information in unimodal cortex (McDonald, Teder-Sälejärvi, Russo, & Hillyard, 2003). Alternatively, it may be that areas of unimodal cortex contact each other directly (Driver & Noesselt, 2008; Macaluso & Driver, 2005), such that multimodal integration is a fundamental component of all sensory processing.

In fact, the large numbers of multisensory neurons distributed all around the cortex—in multisensory convergence areas and in primary cortices—has led some researchers to propose that a drastic reconceptualization of the brain is necessary (Ghazanfar & Schroeder, 2006). They argue that the cortex should not be considered as being divided into isolated regions that process only one kind of sensory information. Rather, they propose that these areas only *prefer* to process information from specific modalities but engage in low-level multisensory processing whenever it is beneficial to the perceiver (Vasconcelos et al., 2011).

Behavioral Effects of Multimodal Perception

Although neuroscientists tend to study very simple interactions between neurons, the fact that they've found so many crossmodal areas of the cortex seems to hint that the way we experience the world is fundamentally multimodal. As discussed above, our intuitions about perception are consistent with this; it does not seem as though our perception of events is constrained to the perception of each sensory modality independently. Rather, we perceive a unified world, regardless of the sensory modality through which we perceive it.

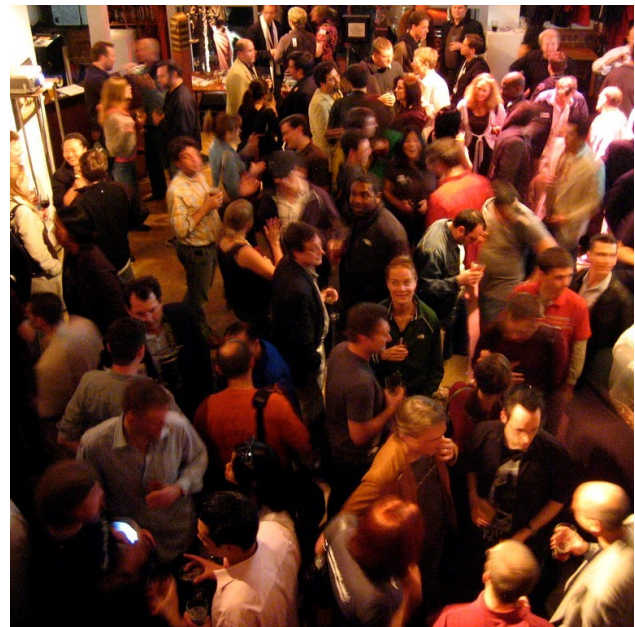
It will probably require many more years of research before neuroscientists uncover all the details of the neural machinery involved in this unified experience. In the meantime, experimental psychologists have contributed to our understanding of multimodal perception through investigations of the behavioral effects associated with it. These effects fall into two broad classes. The first class—**multimodal phenomena**—concerns the binding of inputs from multiple sensory modalities and the effects of this binding on perception. The second class—**crossmodal phenomena**—concerns the influence of one sensory modality on the perception of another (Spence, Senkowski, & Roder, 2009).

Multimodal Phenomena

Audiovisual Speech

Multimodal phenomena concern stimuli that generate simultaneous (or nearly simultaneous) information in more than one sensory modality. As discussed above, speech is a classic example of this kind of stimulus. When an individual speaks, she generates sound waves that carry meaningful information. If the perceiver is also looking at the speaker, then that perceiver also has access to *visual* patterns that carry meaningful information. Of course, as anyone who has ever tried to lipread knows, there are limits on how informative visual speech information is. Even so, the visual speech pattern alone is sufficient for very robust speech perception. Most people assume that deaf individuals are much better at lipreading than individuals with normal hearing. It may come as a surprise to learn, however, that some individuals with normal hearing are also remarkably good at lipreading (sometimes called “speechreading”). In fact, there is a wide range of speechreading ability in both normal hearing and deaf populations (Andersson, Lyxell, Rönnerberg, & Spens, 2001). However, the reasons for this wide range of performance are not well understood (Auer & Bernstein, 2007; Bernstein, 2006; Bernstein, Auer, & Tucker, 2001; Mohammed et al., 2005).

How does visual information about speech interact with auditory information about speech? One of the earliest investigations of this question examined the accuracy of recognizing spoken words presented in a noisy context, much like in the example above about talking at a crowded party. To study this phenomenon experimentally, some irrelevant noise (“white noise”—which sounds like a radio tuned between stations) was presented to participants. Embedded in the white noise were spoken words, and the participants’ task was to identify the words. There were two conditions: one in which only the auditory component of the words was presented (the “auditory-alone” condition), and one in which both the auditory and visual components were presented (the “audiovisual” condition). The noise levels were also varied, so that on some trials, the noise was very loud relative to the loudness of the words, and on other trials, the noise was very soft relative to the words. Sumbly and Pollack (1954) found that the accuracy of identifying the spoken words was much higher for the audiovisual condition than it was in the auditory-alone condition. In addition, the pattern of results was consistent with



In a noisy and poorly lit environment such as a nightclub in order to have a conversation we rely on audiovisual speech to understand others. [Image: Jeremy Keith, <https://goo.gl/18sLfg>, CC BY 2.0, <https://goo.gl/v4Y0Zv>]

the Principle of Inverse Effectiveness: The advantage gained by audiovisual presentation was highest when the auditory-alone condition performance was lowest (i.e., when the noise was loudest). At these noise levels, the audiovisual advantage was considerable: It was estimated that allowing the participant to see the speaker was equivalent to turning the volume of the noise down by over half. Clearly, the audiovisual advantage can have dramatic effects on behavior.

Another phenomenon using audiovisual speech is a very famous illusion called the “McGurk effect” (named after one of its discoverers). In the classic formulation of the illusion, a movie is recorded of a speaker saying the syllables “gaga.” Another movie is made of the same speaker saying the syllables “baba.” Then, the auditory portion of the “baba” movie is dubbed onto the visual portion of the “gaga” movie. This combined stimulus is presented to participants, who are asked to report what the speaker in the movie said. McGurk and MacDonald (1976) reported that 98 percent of their participants reported hearing the syllable “dada”—which was in neither the visual nor the auditory components of the stimulus. These results indicate that when visual and auditory information about speech is integrated, it can have profound effects on perception.

Tactile/Visual Interactions in Body Ownership

Not all multisensory integration phenomena concern speech, however. One particularly compelling multisensory illusion involves the integration of tactile and visual information in the perception of body ownership. In the “rubber hand illusion” (Botvinick & Cohen, 1998), an observer is situated so that one of his hands is not visible. A fake rubber hand is placed near the obscured hand, but in a visible location. The experimenter then uses a light paintbrush to simultaneously stroke the obscured hand and the rubber hand in the same locations. For example, if the middle finger of the obscured hand is being brushed, then the middle finger of the rubber hand will also be brushed. This sets up a correspondence between the tactile sensations (coming from the obscured hand) and the visual sensations (of the rubber hand). After a short time (around 10 minutes), participants report feeling as though the rubber hand “belongs” to them; that is, that the rubber hand is a part of their body. This feeling can be so strong that surprising the participant by hitting the rubber hand with a hammer often leads to a reflexive withdrawing of the obscured hand—even though it is in no danger at all. It appears, then, that our awareness of our own bodies may be the result of multisensory integration.

Crossmodal Phenomena

Crossmodal phenomena are distinguished from multimodal phenomena in that they concern the influence one sensory modality has on the perception of another.

Visual Influence on Auditory Localization



Ventriloquists are able to trick us into believing that what we see and what we hear are the same where, in truth, they are not.

[Image: Amanda Ferrell, CC0 Public Domain, <https://goo.gl/m25gce>]

A famous (and commonly experienced) crossmodal illusion is referred to as “the ventriloquism effect.” When a ventriloquist appears to make a puppet speak, she fools the listener into thinking that the location of the origin of the speech sounds is at the puppet’s mouth. In other words, instead of localizing the auditory signal (coming from the mouth of a ventriloquist) to the correct place, our perceptual system localizes it incorrectly (to the mouth of the puppet).

Why might this happen? Consider the information available to the observer about the location of the two components of the stimulus: the sounds from the ventriloquist’s mouth and the visual movement of the puppet’s mouth. Whereas it is very obvious where the visual stimulus is coming from (because you can

see it), it is much more difficult to pinpoint the location of the sounds. In other words, the very precise visual location of mouth movement apparently overrides the less well-specified location of the auditory information. More generally, it has been found that the location of a wide variety of auditory stimuli can be affected by the simultaneous presentation of a visual stimulus (Vroomen & De Gelder, 2004). In addition, the ventriloquism effect has been demonstrated for objects in motion: The motion of a visual object can influence the perceived direction of motion of a moving sound source (Soto-Faraco, Kingstone, & Spence, 2003).

Auditory Influence on Visual Perception

A related illusion demonstrates the opposite effect: where sounds have an effect on visual perception. In the **double flash illusion**, a participant is asked to stare at a central point on a computer monitor. On the extreme edge of the participant’s vision, a white circle is briefly

flashed one time. There is also a simultaneous auditory event: either one beep or two beeps in rapid succession. Remarkably, participants report seeing *two* visual flashes when the flash is accompanied by two beeps; the same stimulus is seen as a single flash in the context of a single beep or no beep (Shams, Kamitani, & Shimojo, 2000). In other words, the number of heard beeps influences the number of seen flashes!

Another illusion involves the perception of collisions between two circles (called “balls”) moving toward each other and continuing through each other. Such stimuli can be perceived as either two balls moving through each other or as a collision between the two balls that then bounce off each other in opposite directions. Sekuler, Sekuler, and Lau (1997) showed that the presentation of an auditory stimulus at the time of contact between the two balls strongly influenced the perception of a collision event. In this case, the perceived sound influences the interpretation of the ambiguous visual stimulus.

Crossmodal Speech

Several crossmodal phenomena have also been discovered for speech stimuli. These crossmodal speech effects usually show altered perceptual processing of unimodal stimuli (e.g., acoustic patterns) by virtue of prior experience with the alternate unimodal stimulus (e.g., optical patterns). For example, Rosenblum, Miller, and Sanchez (2007) conducted an experiment examining the ability to become familiar with a person's voice. Their first interesting finding was unimodal: Much like what happens when someone repeatedly hears a person speak, perceivers can become familiar with the “visual voice” of a speaker. That is, they can become familiar with the person's speaking style simply by seeing that person speak. Even more astounding was their crossmodal finding:

Familiarity with this *visual* information also led to increased recognition of the speaker's *auditory* speech, to which participants had never had exposure.



Experiments have demonstrated that by simply observing a speaker, with no auditory information, we can gather important clues about the actual sound of their voice. [Ken Whytock, <https://goo.gl/VQJssP>, CC BY-NC 2.0, <https://goo.gl/tgFydH>]

Similarly, it has been shown that when perceivers see a speaking face, they can identify the (auditory-alone) voice of that speaker, and vice versa (Kamachi, Hill, Lander, & Vatikiotis-Bateson, 2003; Lachs & Pisoni, 2004a, 2004b, 2004c; Rosenblum, Smith, Nichols, Lee, & Hale, 2006). In other words, the visual form of a speaker engaged in the act of speaking appears to contain information about what that speaker should sound like. Perhaps more surprisingly, the auditory form of speech seems to contain information about what the speaker should look like.

Conclusion

In this module, we have reviewed some of the main evidence and findings concerning the role of multimodal perception in our experience of the world. It appears that our nervous system (and the cortex in particular) contains considerable architecture for the processing of information arriving from multiple senses. Given this neurobiological setup, and the diversity of behavioral phenomena associated with multimodal stimuli, it is likely that the investigation of multimodal perception will continue to be a topic of interest in the field of experimental perception for many years to come.

Outside Resources

Article: A review of the neuroanatomy and methods associated with multimodal perception:

<http://dx.doi.org/10.1016/j.neubiorev.2011.04.015>

Journal: Experimental Brain Research Special issue: Crossmodal processing

<http://www.springerlink.com/content/0014-4819/198/2-3>

TED Talk: Optical Illusions

http://www.ted.com/talks/beau_lotto_optical_illusions_show_how_we_see

Video: McGurk demo

<http://youtu.be/aFPtc8BVdjk>

Video: The Rubber Hand Illusion

<http://www.youtube.com/watch?v=sxwn1w7MJvk>

Web: Double-flash illusion demo

<http://www.cns.atr.jp/~kmtm/soundInducedIllusoryFlash2/>

Discussion Questions

1. The extensive network of multisensory areas and neurons in the cortex implies that much perceptual processing occurs in the context of multiple inputs. Could the processing of unimodal information ever be useful? Why or why not?
2. Some researchers have argued that the Principle of Inverse Effectiveness (PoIE) results from ceiling effects: Multisensory enhancement cannot take place when one modality is sufficient for processing because in such cases it is not possible for processing to be enhanced (because performance is already at the “ceiling”). On the other hand, other researchers claim that the PoIE stems from the perceptual system’s ability to assess the relative value of stimulus cues, and to use the most reliable sources of information to construct a representation of the outside world. What do you think? Could these two possibilities ever be teased apart? What kinds of experiments might one conduct to try to get at this issue?
3. In the late 17th century, a scientist named William Molyneux asked the famous philosopher

John Locke a question relevant to modern studies of multisensory processing. The question was this: Imagine a person who has been blind since birth, and who is able, by virtue of the sense of touch, to identify three dimensional shapes such as spheres or pyramids. Now imagine that this person suddenly receives the ability to see. Would the person, without using the sense of touch, be able to identify those same shapes visually? Can modern research in multimodal perception help answer this question? Why or why not? How do the studies about crossmodal phenomena inform us about the answer to this question?

Vocabulary

Bouncing balls illusion

The tendency to perceive two circles as bouncing off each other if the moment of their contact is accompanied by an auditory stimulus.

Crossmodal phenomena

Effects that concern the influence of the perception of one sensory modality on the perception of another.

Crossmodal receptive field

A receptive field that can be stimulated by a stimulus from more than one sensory modality.

Crossmodal stimulus

A stimulus with components in multiple sensory modalities that interact with each other.

Double flash illusion

The false perception of two visual flashes when a single flash is accompanied by two auditory beeps.

Integrated

The process by which the perceptual system combines information arising from more than one modality.

McGurk effect

An effect in which conflicting visual and auditory components of a speech stimulus result in an illusory percept.

Multimodal

Of or pertaining to multiple sensory modalities.

Multimodal perception

The effects that concurrent stimulation in more than one sensory modality has on the perception of events and objects in the world.

Multimodal phenomena

Effects that concern the binding of inputs from multiple sensory modalities.

Multisensory convergence zones

Regions in the brain that receive input from multiple unimodal areas processing different sensory modalities.

Multisensory enhancement

See “superadditive effect of multisensory integration.”

Primary auditory cortex

A region of the cortex devoted to the processing of simple auditory information.

Primary visual cortex

A region of the cortex devoted to the processing of simple visual information.

Principle of Inverse Effectiveness

The finding that, in general, for a multimodal stimulus, if the response to each unimodal component (on its own) is weak, then the opportunity for multisensory enhancement is very large. However, if one component—by itself—is sufficient to evoke a strong response, then the effect on the response gained by simultaneously processing the other components of the stimulus will be relatively small.

Receptive field

The portion of the world to which a neuron will respond if an appropriate stimulus is present there.

Rubber hand illusion

The false perception of a fake hand as belonging to a perceiver, due to multimodal sensory information.

Sensory modalities

A type of sense; for example, vision or audition.

Spatial principle of multisensory integration

The finding that the superadditive effects of multisensory integration are observed when the sources of stimulation are spatially related to one another.

Superadditive effect of multisensory integration

The finding that responses to multimodal stimuli are typically greater than the sum of the independent responses to each unimodal component if it were presented on its own.

Unimodal

Of or pertaining to a single sensory modality.

Unimodal components

The parts of a stimulus relevant to one sensory modality at a time.

Unimodal cortex

A region of the brain devoted to the processing of information from a single sensory modality.

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Eyewitness Testimony and Memory Biases

Cara Laney & Elizabeth F. Loftus

Eyewitnesses can provide very compelling legal testimony, but rather than recording experiences flawlessly, their memories are susceptible to a variety of errors and biases. They (like the rest of us) can make errors in remembering specific details and can even remember whole events that did not actually happen. In this module, we discuss several of the common types of errors, and what they can tell us about human memory and its interactions with the legal system.

Learning Objectives

- Describe the kinds of mistakes that eyewitnesses commonly make and some of the ways that this can impede justice.
- Explain some of the errors that are common in human memory.
- Describe some of the important research that has demonstrated human memory errors and their consequences.

What Is Eyewitness Testimony?

Eyewitness testimony is what happens when a person witnesses a crime (or accident, or other legally important event) and later gets up on the stand and recalls for the court all the details of the witnessed event. It involves a more complicated process than might initially be

presumed. It includes what happens during the actual crime to facilitate or hamper witnessing, as well as everything that happens from the time the event is over to the later courtroom appearance. The eyewitness may be interviewed by the police and numerous lawyers, describe the perpetrator to several different people, and make an identification of the perpetrator, among other things.



What can happen to our memory from the time we witness an event to the retelling of that event later? What can influence how we remember, or misremember, highly significant events like a crime or accident? [Image: Robert Couse-Baker, <https://goo.gl/OiPUmz>, CC BY 2.0, <https://goo.gl/BRvSA7>]

Why Is Eyewitness Testimony an Important Area of Psychological Research?

When an eyewitness stands up in front of the court and describes what happened from her own perspective, this testimony can be extremely compelling—it is hard for those hearing this testimony to take it “with a grain of salt,” or otherwise adjust its power. But to what extent is this necessary?

There is now a wealth of evidence, from research conducted over several decades, suggesting that eyewitness testimony is probably the most persuasive form of evidence presented in court, but in many cases, its accuracy is dubious. There is also evidence that mistaken eyewitness evidence can lead to wrongful conviction—sending people to prison for years or decades, even to death row, for crimes they did not commit. Faulty eyewitness testimony has been implicated in at least 75% of DNA exoneration cases—more than any other cause (Garrett, 2011). In a particularly famous case, a man named Ronald Cotton was identified by

a rape victim, Jennifer Thompson, as her rapist, and was found guilty and sentenced to life in prison. After more than 10 years, he was exonerated (and the real rapist identified) based on DNA evidence. For details on this case and other (relatively) lucky individuals whose false convictions were subsequently overturned with DNA evidence, see the Innocence Project website (<http://www.innocenceproject.org/>).

There is also hope, though, that many of the errors may be avoidable if proper precautions are taken during the investigative and judicial processes. Psychological science has taught us what some of those precautions might involve, and we discuss some of that science now.

Misinformation



Misinformation can be introduced into the memory of a witness between the time of seeing an event and reporting it later. Something as straightforward as which sort of traffic sign was in place at an intersection can be confused if subjects are exposed to erroneous information after the initial incident.

In an early study of eyewitness memory, undergraduate subjects first watched a slideshow depicting a small red car driving and then hitting a pedestrian (Loftus, Miller, & Burns, 1978). Some subjects were then asked leading questions about what had happened in the slides. For example, subjects were asked, "How fast was the car traveling when it passed the yield sign?" But this question was actually designed to be misleading, because the original slide included a stop sign rather than a yield sign.

Later, subjects were shown pairs of slides. One of the pair was the original slide containing the stop sign; the other was a replacement slide containing a yield sign. Subjects were asked which of the pair they had previously seen. Subjects who had been asked about the yield sign were likely to pick the slide showing the yield sign, even though they had originally seen the slide with the stop sign. In other words, the misinformation in the leading question led to inaccurate memory.

This phenomenon is called the **misinformation effect**, because the misinformation that subjects were exposed to after the event (here in the form of a misleading question) apparently contaminates subjects' memories of what they witnessed. Hundreds of subsequent studies have demonstrated that memory can be contaminated by erroneous information that people are

exposed to after they witness an event (see Frenda, Nichols, & Loftus, 2011; Loftus, 2005). The misinformation in these studies has led people to incorrectly remember everything from small but crucial details of a perpetrator's appearance to objects as large as a barn that wasn't there at all.

These studies have demonstrated that young adults (the typical research subjects in psychology) are often susceptible to misinformation, but that children and older adults can be even more susceptible (Bartlett & Memon, 2007; Ceci & Bruck, 1995). In addition, misinformation effects can occur easily, and without any intention to deceive (Allan & Gabbert, 2008). Even slight differences in the wording of a question can lead to misinformation effects. Subjects in one study were more likely to say yes when asked "Did you see the broken headlight?" than when asked "Did you see a broken headlight?" (Loftus, 1975).

Other studies have shown that misinformation can corrupt memory even more easily when it is encountered in social situations (Gabbert, Memon, Allan, & Wright, 2004). This is a problem particularly in cases where more than one person witnesses a crime. In these cases, witnesses tend to talk to one another in the immediate aftermath of the crime, including as they wait for police to arrive. But because different witnesses are different people with different perspectives, they are likely to see or notice different things, and thus remember different things, even when they witness the same event. So when they communicate about the crime later, they not only reinforce common memories for the event, they also contaminate each other's memories for the event (Gabbert, Memon, & Allan, 2003; Paterson & Kemp, 2006; Takarangi, Parker, & Garry, 2006).

The misinformation effect has been modeled in the laboratory. Researchers had subjects watch a video in pairs. Both subjects sat in front of the same screen, but because they wore differently polarized glasses, they saw two different versions of a video, projected onto a screen. So, although they were both watching the same screen, and believed (quite reasonably) that they were watching the same video, they were actually watching two different versions of the video (Garry, French, Kinzett, & Mori, 2008).

In the video, Eric the electrician is seen wandering through an unoccupied house and helping himself to the contents thereof. A total of eight details were different between the two videos. After watching the videos, the "co-witnesses" worked together on 12 memory test questions. Four of these questions dealt with details that were different in the two versions of the video, so subjects had the chance to influence one another. Then subjects worked individually on 20 additional memory test questions. Eight of these were for details that were different in the two videos. Subjects' accuracy was highly dependent on whether they had discussed the details previously. Their accuracy for items they had *not* previously discussed with their co-

witness was 79%. But for items that they *had* discussed, their accuracy dropped markedly, to 34%. That is, subjects allowed their co-witnesses to corrupt their memories for what they had seen.

Identifying Perpetrators

In addition to correctly remembering many details of the crimes they witness, eyewitnesses often need to remember the faces and other identifying features of the perpetrators of those crimes. Eyewitnesses are often asked to describe that perpetrator to law enforcement and later to make identifications from books of mug shots or lineups. Here, too, there is a substantial body of research demonstrating that eyewitnesses can make serious, but often understandable and even predictable, errors (Caputo & Dunning, 2007; Cutler & Penrod, 1995).

In most jurisdictions in the United States, lineups are typically conducted with pictures, called **photo spreads**, rather than with actual people standing behind one-way glass (Wells, Memon, & Penrod, 2006). The eyewitness is given a set of small pictures of perhaps six or eight individuals who are dressed similarly and photographed in similar circumstances. One of these individuals is the police suspect, and the remainder are “**foils**” or “fillers” (people known to be innocent of the particular crime under investigation). If the eyewitness identifies the



Mistakes in identifying perpetrators can be influenced by a number of factors including poor viewing conditions, too little time to view the perpetrator, or too much delay from time of witnessing to identification.

suspect, then the investigation of that suspect is likely to progress. If a witness identifies a foil or no one, then the police may choose to move their investigation in another direction.

This process is modeled in laboratory studies of eyewitness identifications. In these studies, research subjects witness a mock crime (often as a short video) and then are asked to make an identification from a photo or a live lineup. Sometimes the lineups are target present, meaning that the perpetrator from the mock crime is actually in the lineup, and sometimes they are target absent, meaning that the lineup is made up entirely of foils. The subjects, or **mock witnesses**, are given some instructions and asked to pick the

perpetrator out of the lineup. The particular details of the witnessing experience, the instructions, and the lineup members can all influence the extent to which the mock witness is likely to pick the perpetrator out of the lineup, or indeed to make any selection at all. Mock witnesses (and indeed real witnesses) can make errors in two different ways. They can fail to pick the perpetrator out of a target present lineup (by picking a foil or by neglecting to make a selection), or they can pick a foil in a target absent lineup (wherein the only correct choice is to not make a selection).

Some factors have been shown to make eyewitness identification errors particularly likely. These include poor vision or viewing conditions during the crime, particularly stressful witnessing experiences, too little time to view the perpetrator or perpetrators, too much delay between witnessing and identifying, and being asked to identify a perpetrator from a race other than one's own (Bornstein, Deffenbacher, Penrod, & McGorty, 2012; Brigham, Bennett, Meissner, & Mitchell, 2007; Burton, Wilson, Cowan, & Bruce, 1999; Deffenbacher, Bornstein, Penrod, & McGorty, 2004).

It is hard for the legal system to do much about most of these problems. But there are some things that the justice system can do to help lineup identifications "go right." For example, investigators can put together high-quality, fair lineups. A fair lineup is one in which the suspect and each of the foils is equally likely to be chosen by someone who has read an eyewitness description of the perpetrator but who did not actually witness the crime (Brigham, Ready, & Spier, 1990). This means that no one in the lineup should "stick out," and that everyone should match the description given by the eyewitness. Other important recommendations that have come out of this research include better ways to conduct lineups, "double blind" lineups, unbiased instructions for witnesses, and conducting lineups in a sequential fashion (see Technical Working Group for Eyewitness Evidence, 1999; Wells et al., 1998; Wells & Olson, 2003).

Kinds of Memory Biases

Memory is also susceptible to a wide variety of other biases and errors. People can forget events that happened to them and people they once knew. They can mix up details across time and place. They can even remember whole complex events that never happened at all. Importantly, these errors, once made, can be very hard to unmake. A memory is no less "memorable" just because it is wrong.

Some small memory errors are commonplace, and you have no doubt experienced many of them. You set down your keys without paying attention, and then cannot find them later when

you go to look for them. You try to come up with a person's name but cannot find it, even though you have the sense that it is right at the tip of your tongue (psychologists actually call this the tip-of-the-tongue effect, or TOT) (Brown, 1991).

Other sorts of memory biases are more complicated and longer lasting. For example, it turns out that our expectations and beliefs about how the world works can have huge influences on our memories. Because many aspects of our everyday lives are full of redundancies, our memory systems take advantage of the recurring patterns by forming and using schemata, or memory templates (Alba & Hasher, 1983; Brewer & Treyens, 1981). Thus, we know to expect that a library will have shelves and tables and librarians, and so we don't have to spend energy noticing these at the time. The result of this lack of attention, however, is that one is likely to remember schema-consistent information (such as tables), and to remember them in a rather generic way, whether or not they were actually present.



For most of our experiences schematas are a benefit and help with information overload. However, they may make it difficult or impossible to recall certain details of a situation later. Do you recall the library as it actually was or the library as approximated by your library schemata? [Dan Kleinman, <https://goo.gl/07xyDD>, CC BY 2.0, <https://goo.gl/BRvSA7>]

False Memory

Some memory errors are so “large” that they almost belong in a class of their own: false memories. Back in the early 1990s a pattern emerged whereby people would go into therapy for depression and other everyday problems, but over the course of the therapy develop memories for violent and horrible victimhood (Loftus & Ketcham, 1994). These patients' therapists claimed that the patients were recovering genuine memories of real childhood abuse, buried deep in their minds for years or even decades. But some experimental psychologists believed that the memories were instead likely to be false—created in therapy. These researchers then set out to see whether it would indeed be possible for wholly false memories to be created by procedures similar to those used in these patients' therapy.

In early false memory studies, undergraduate subjects' family members were recruited to provide events from the students' lives. The student subjects were told that the researchers

had talked to their family members and learned about four different events from their childhoods. The researchers asked if the now undergraduate students remembered each of these four events—introduced via short hints. The subjects were asked to write about each of the four events in a booklet and then were interviewed two separate times. The trick was that one of the events came from the researchers rather than the family (and the family had actually assured the researchers that this event had *not* happened to the subject). In the first such study, this researcher-introduced event was a story about being lost in a shopping mall and rescued by an older adult. In this study, after just being asked whether they remembered these events occurring on three separate occasions, a quarter of subjects came to believe that they had indeed been lost in the mall (Loftus & Pickrell, 1995). In subsequent studies, similar procedures were used to get subjects to believe that they nearly drowned and had been rescued by a lifeguard, or that they had spilled punch on the bride's parents at a family wedding, or that they had been attacked by a vicious animal as a child, among other events (Heaps & Nash, 1999; Hyman, Husband, & Billings, 1995; Porter, Yuille, & Lehman, 1999).

More recent false memory studies have used a variety of different manipulations to produce false memories in substantial minorities and even occasional majorities of manipulated subjects (Braun, Ellis, & Loftus, 2002; Lindsay, Hagen, Read, Wade, & Garry, 2004; Mazzoni, Loftus, Seitz, & Lynn, 1999; Seamon, Philbin, & Harrison, 2006; Wade, Garry, Read, & Lindsay, 2002). For example, one group of researchers used a mock-advertising study, wherein subjects were asked to review (fake) advertisements for Disney vacations, to convince subjects that they had once met the character Bugs Bunny at Disneyland—an impossible false memory because Bugs is a Warner Brothers character (Braun et al., 2002). Another group of researchers photoshopped childhood photographs of their subjects into a hot air balloon picture and then asked the subjects to try to remember and describe their hot air balloon experience (Wade et al., 2002). Other researchers gave subjects unmanipulated class photographs from their childhoods along with a fake story about a class prank, and thus enhanced the likelihood that subjects would falsely remember the prank (Lindsay et al., 2004).

Using a false feedback manipulation, we have been able to persuade subjects to falsely remember having a variety of childhood experiences. In these studies, subjects are told (falsely) that a powerful computer system has analyzed questionnaires that they completed previously and has concluded that they had a particular experience years earlier. Subjects apparently believe what the computer says about them and adjust their memories to match this new information. A variety of different false memories have been implanted in this way. In some studies, subjects are told they once got sick on a particular food (Bernstein, Laney, Morris, & Loftus, 2005). These memories can then spill out into other aspects of subjects' lives, such that they often become less interested in eating that food in the future (Bernstein & Loftus, 2009b). Other false memories implanted with this methodology include having an

unpleasant experience with the character Pluto at Disneyland and witnessing physical violence between one's parents (Berkowitz, Laney, Morris, Garry, & Loftus, 2008; Laney & Loftus, 2008).

Importantly, once these false memories are implanted—whether through complex methods or simple ones—it is extremely difficult to tell them apart from true memories (Bernstein & Loftus, 2009a; Laney & Loftus, 2008).

Conclusion

To conclude, eyewitness testimony is very powerful and convincing to jurors, even though it is not particularly reliable. Identification errors occur, and these errors can lead to people being falsely accused and even convicted. Likewise, eyewitness memory can be corrupted by leading questions, misinterpretations of events, conversations with co-witnesses, and their own expectations for what should have happened. People can even come to remember whole events that never occurred.

The problems with memory in the legal system are real. But what can we do to start to fix them? A number of specific recommendations have already been made, and many of these are in the process of being implemented (e.g., Steblay & Loftus, 2012; Technical Working Group for Eyewitness Evidence, 1999; Wells et al., 1998). Some of these recommendations are aimed at specific legal procedures, including when and how witnesses should be interviewed, and how lineups should be constructed and conducted. Other recommendations call for appropriate education (often in the form of expert witness testimony) to be provided to jury members and others tasked with assessing eyewitness memory. Eyewitness testimony can be of great value to the legal system, but decades of research now argues that this testimony is often given far more weight than its accuracy justifies.

Outside Resources

Video 1: Eureka Foong's - The Misinformation Effect. This is a student-made video illustrating this phenomenon of altered memory. It was one of the winning entries in the 2014 Noba Student Video Award.

<https://www.youtube.com/watch?v=iMPIWkFtd88>

Video 2: Ang Rui Xia & Ong Jun Hao's - The Misinformation Effect. Another student-made video exploring the misinformation effect. Also an award winner from 2014.

<https://www.youtube.com/watch?v=gsn9iKmOJLQ>

Discussion Questions

1. Imagine that you are a juror in a murder case where an eyewitness testifies. In what ways might your knowledge of memory errors affect your use of this testimony?
2. How true to life do you think television shows such as CSI or Law & Order are in their portrayals of eyewitnesses?
3. Many jurisdictions in the United States use "show-ups," where an eyewitness is brought to a suspect (who may be standing on the street or in handcuffs in the back of a police car) and asked, "Is this the perpetrator?" Is this a good or bad idea, from a psychological perspective? Why?

Vocabulary

False memories

Memory for an event that never actually occurred, implanted by experimental manipulation or other means.

Foils

Any member of a lineup (whether live or photograph) other than the suspect.

Misinformation effect

A memory error caused by exposure to incorrect information between the original event (e.g., a crime) and later memory test (e.g., an interview, lineup, or day in court).

Mock witnesses

A research subject who plays the part of a witness in a study.

Photo spreads

A selection of normally small photographs of faces given to a witness for the purpose of identifying a perpetrator.

Schema (plural: schemata)

A memory template, created through repeated exposure to a particular class of objects or events.

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